SIDE TRACK-2 PROGRAM

AZADEGAN OIL FIELD

SIDE TRACK-2 PROGRAM

AZNS – 127 (PAD #54), TYPE: IIF

Drilling Department

APRIL 2026

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1. General Information

1.1 Field Introduction

Azadegan South Oil Field is situated in West Karoun province on a large long-axis anticline with N-S trending with approximate dimensions about 20 km width and 60 km length near to Iran-Iraq border, to the east of Yaran Oil Field.

The trap area of the south Azadegan is limited to 43.2x17.2 km. The Azadegan Oil field was discovered in 1999, when Well AZN-1 was drilled vertically down to the Cretaceous Fahliyan Formation with total depth of 4,643m. The results of this well, revealed the reservoir characteristics of the field. Following to the result of the well no. AZN-1, another 5 exploration wells was drilled in different parts of this huge field down to Fahliyan Formation. In the later stage the well No.10 assumed as exploration well for evaluation of deeper formations. This well was drilled in 2008, down to the depth of 5340m (Neyriz Fm.), mainly to evaluate the Jurassic formations.

Based on the exploration wells results, carbonate reservoir of Sarvak Formation is broadly developed in Azadegan Field. This formation was penetrated in all of seven exploration wells (AZN-1, AZN-2, AZN-3, AZN-4, AZN-5, AZN-6 and AZN-10). Clastic oil-bearing zones of Kazhdumi Formation developed in main anticline of Azadegan Field, extending from north to center of South Azadegan. The Clastic oil-bearing zones of Gadvan Formation developed only in crest area and south parts of South Azadegan Filed, which was discovered by AZN-1 in crest and AZN-4 in south parts. The carbonates of Fahliyan Formation also developed in Azadegan field, but the oil reservoir of this formation developed in South part of South Azadegan.

The Azadegan Field is an elongated anticline, trending in N-S direction, belonging to "Arabian Trend". The stratigraphy sequence of the field includes AghaJari, Gachsaran, Fahliyan, Pabdeh, Gurpi, Ilam, Laffan, Sarvak, Kazhdumi, Dariyan, Gadvan and Fahliyan Formations, from top to bottom. This sequence in well no AZN-10, the deepest up to now drilled well of the field, continues with Garau, Gotnia, Najmah, Sargalu, Alan, Mus and Neyriz formations.

The Dominant lithofacies of Sarvak Formation are Rudist-bearing packstone and partially grainstone, which contains abundant bioclasts. Some other facies like matrix-dominated wackestone and mudstone are also found.

The AZADEGAN Oilfield is Located in Khuzestan Province and is Approximately Located 80 Kilometers West of Ahwaz City, near to Iran-Iraq Border. It is in The Northwest of The YADAVARAN Field and in The West of JUFEYR Field. The Average Monthly Temperature is 45ºc in summer, and Highest Temperature is 52ºc. The Lowest Temperature is -4.4ºc in winter. Maximum Rainfall is 78 mm in February, Winter Season.

This Document covers the work over of well AZNS-122. Main headlines of program are de-complete the well and run new completion string.

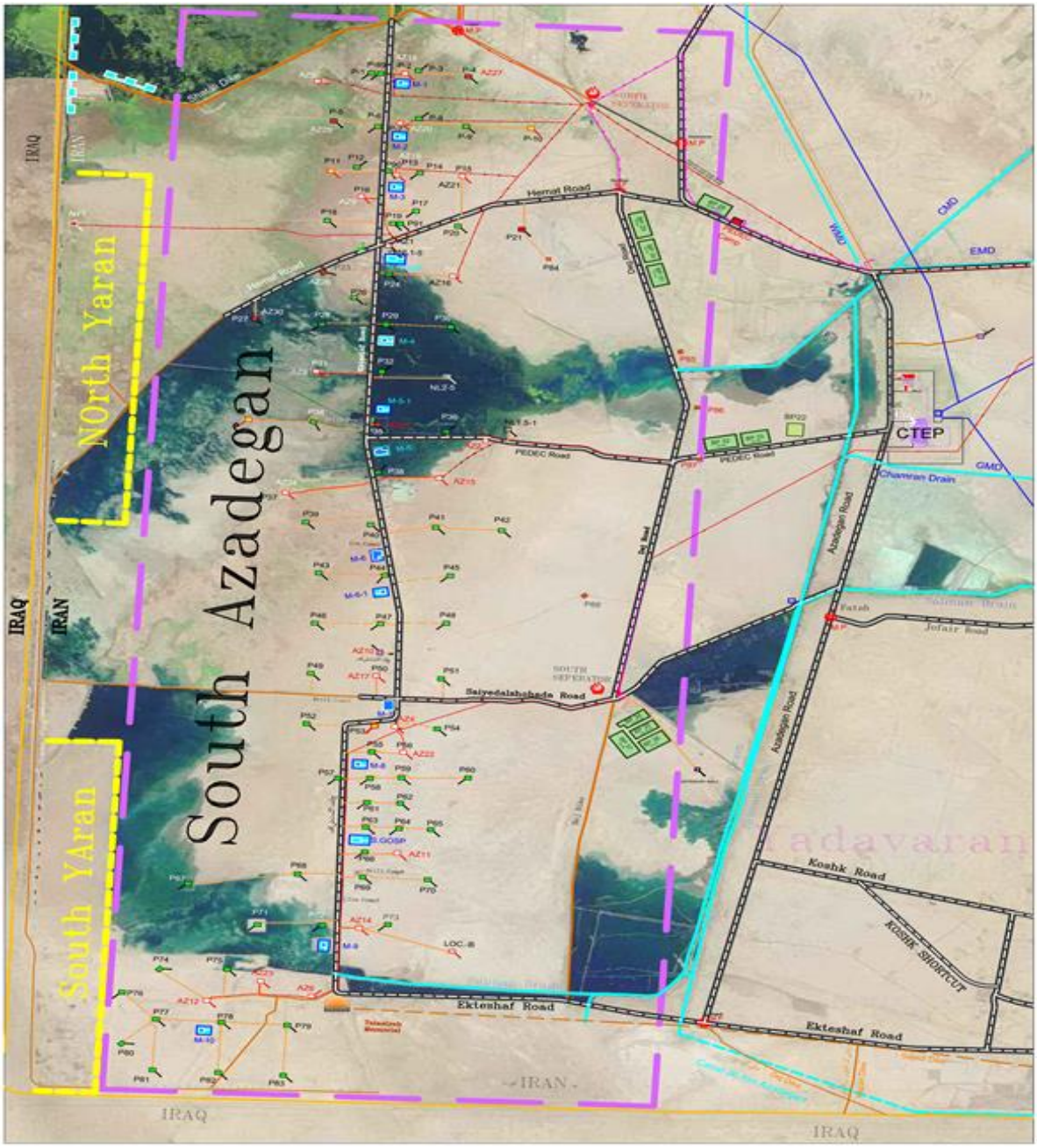


Figure 1-1: AZNS Oilfield Location Map

1.2 Well Information

Country / Area Name:	Iran / Khuzestan
Client	Petro Pars Limited
Field Name:	Azadegan
Type Of Operation:	Side Track-2
Well Name / Type :	AZNS – 127, ST-2 / Fahliyan well in Block II
Pad Number:	54
Well Type:	Side Track
Wellhead Coordinate:	N: 3,448,561 E: 196,167
Existing TD Coordinate:	N: 3,448,543.57 E: 196,187.32
New TD Coordinate:	N: 3,448,563.24 E: 196,166.98
Well Objective:	Side Track The Well from 3055 mMD, Drill 8 – 1/2" & 6 – 1/8" Hole Section & Complete the well.
RTE:	12.95^{m}
GLE:	3.95^{m}
Well Duration :	115.4 Days (Main Plan) + 6.0 Days (Contingency)
Total Depth (MD):	4150^{m}
Producer Formation:	Fahliyan Formation
Formation TD :	Fahliyan
Reservoir :	Fahliyan

Table 1-1 AZNS-127 information

1.3 Area Distribution

According to the drilling history, the CONTRACT AREA is distributed into three blocks as per the geological features with regard to DRILLING OPERATION. This distribution may differ according to knowledge accumulated in the future WORK.
Contract area is as per below:

X (m)	Y (m)
185640	3434275
188020	3477361
205224	3476408
202842	3433327

Table 1-2 Area Distribution

Block I is located on the north part of the CONTRACT AREA in north of well AZN-010. Gachsaran formation is characterized by abnormal high pressure in this block.

Block II is located between well AZN-010 and AZN-011. In this block, Gachsaran formation is characterized by pressure tending to normal

Block III is located on the south of line well AZN-011. In this block, Gachsaran formation is characterized by normal pressure and all drilled wells in this block suffered from complete loss at Pabdeh formation.

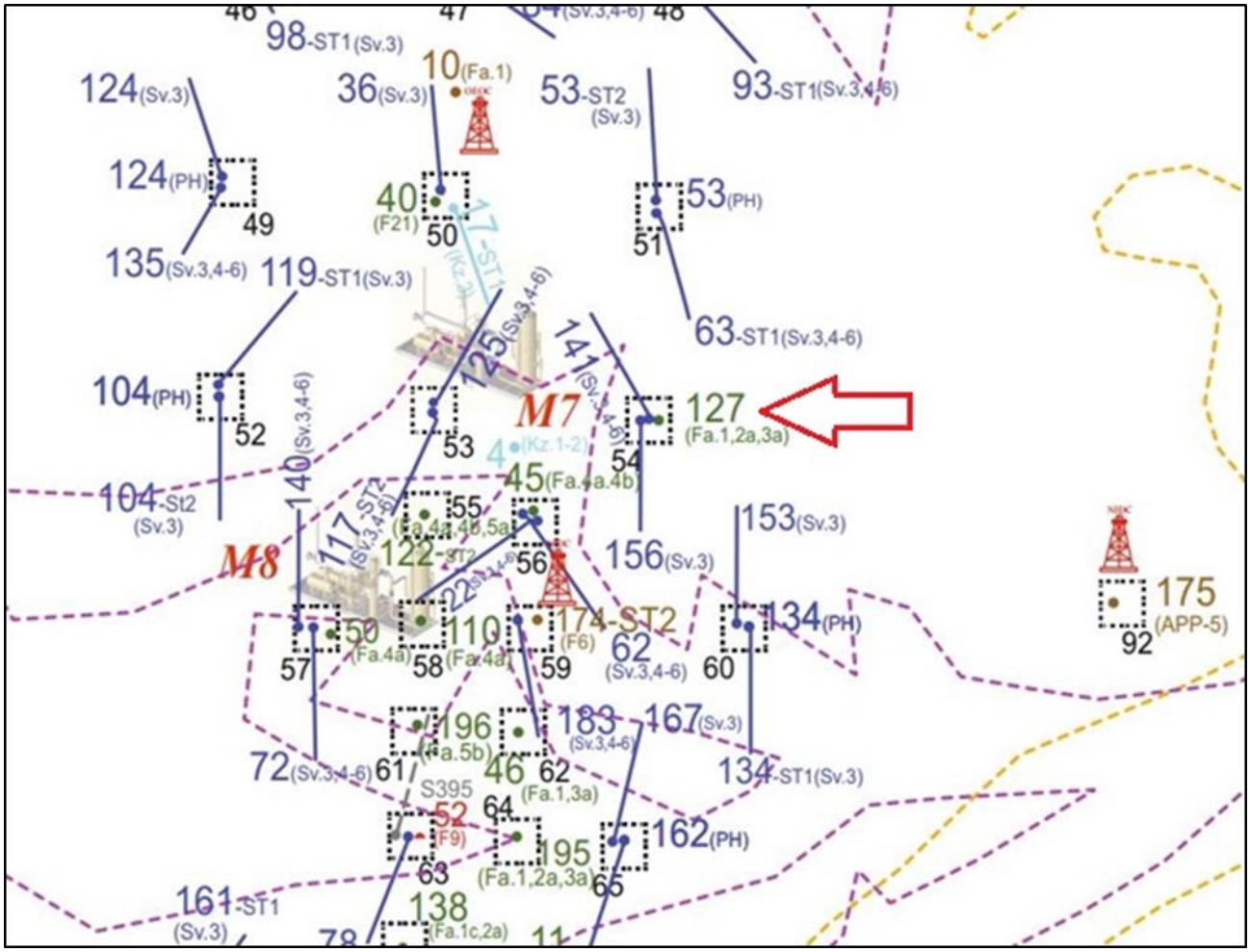


Figure 1-2 AZNS-127 Area Distribution

1.4 Well Pad Specification

Well Pad Type	Well Pad Dimension (m)	Mud Pit Dimension (m)	Cellar Dimension (m)	Conductor Size
Single Well	130 x 100	25 x 20 x 2	7.5L x 3.80W x 3.40D	30"
Double Well	145 x 100	25 x 20 x 2	7.5L x 3.80W x 3.40D + 4.0L x 3.8W x 3.4D	30"
Triple Well	160 x 100	25 x 20 x 2	7.5L x 3.80W x 3.40D + 4.0L x 3.8W x 3.4D + 4.0L x 3.8W x 3.4D	30'

Table 1-3 Well Pad Specification

Compressive Strength for reinforced concrete after 28 days: 280 kg/cm^2

• Fuel tank concrete pond dimension: Length 20^m ×Width 12^m ×Height 0.5^m
• Drainage: Width 30^{cm} × Depth 30^{cm}
• Burning pit: 15^m×15^m square, located on South-East or South-West with at least 350^m distance from well head
• Cellar to cellar distance in multi-well well pad: 15^m
• Conductor pipe depth under cellar bottom is 5.5^m. The conductor pipe will be driven while constructing the well pad.
• Mud material dimension is 25^m*35^m +granular surface
• Mud drying dimension is 25^m*35^m + concrete surface
• Well pad surface is gravel. Access road granular surface. The level of well pads is 3.95 MSL

1. Types of well pads:

• One well in one pad.
• Two wells in one pad. 2 Sarvak horizontal wells or 1 Sarvak horizontal well plus 1 vertical/directional well).
• Three wells in one pad: 2 Sarvak horizontal wells and 1 vertical/directional wells.

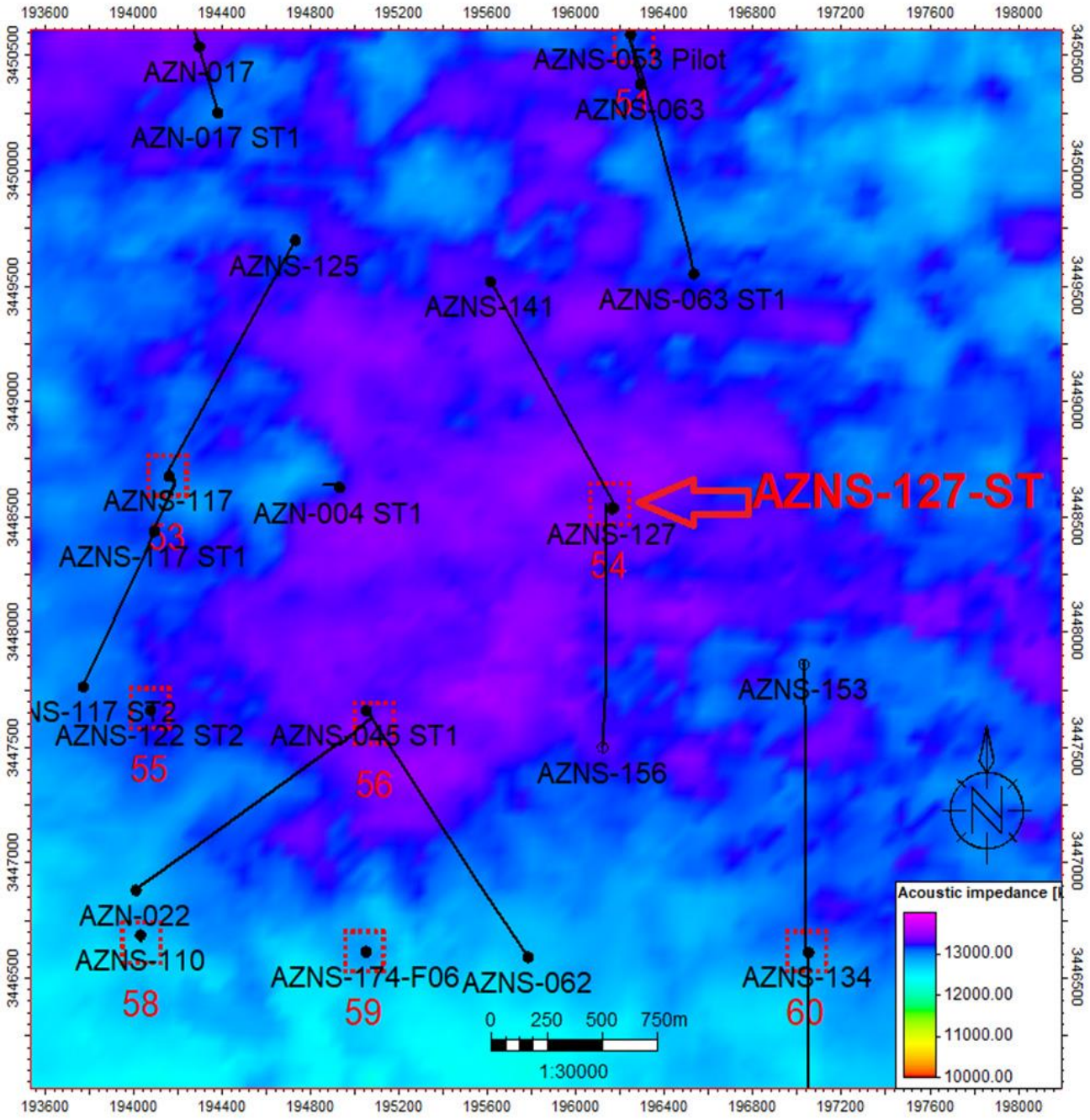


Figure 1-3 The Location of Pad 54 of AZNS - 127

1.5 Reservoir Data

Reservoir Name:	Fahliyan	Unit
Fluids:	Oil	….
Datum Depth:	-4000	M,TVDss
Datum Pressure:	Fah-4a= max 6800 psi Fah-5a= max 7400 psi	Psi
Pressure Gradient:	0.28	Psi/ft
Temperature:	132	C°
Temperature Gradient:	N/A	1.32 F/100ft 0.73 C/100ft 2.4 C/100m
H2S:	H2S in Gas at Atm condition 3000 PPM	PPM
CO2:	Co2 in Gas at Atm Condition 50000 PPM	PPM
Salt Salinity :	143000	ppm
Calcium Content:	8750	(mg/lit)

Table 1-4 Reservoir Data

1.6 Formation Pressure Prediction in Block II

Predicted Formation Pressure in Block II				
Formation	**Interval**	**Pore Pressure**	**Collapse Pressure (Break-Out Pressure)**	**Fracture Pressure (Break-Down Pressure)**
	m	*SG*	*SG*	*SG*
Aghajari	0-1095.9	0.98-1.18	0.06-1.39	1.01-1.89
Gachsaran	1095.9-1461	1.13-1.31	0.78-1.36	1.89-2.01
Asmari	1461-1728	1.14-1.26	0.6-1.28	1.43-1.79
Pabdeh	1728-2196	1.20-1.26	0.82-1.36	1.40-1.78
Gurpi	2196-2536	1.21-1.26	0.66-1.02	1.40-1.76
Ilam	2536-2626	1.18-1.24	0.20-0.9	1.41-1.72
Laffan	2626-2633	1.13-1.18	0.24-1.11	1.41-1.72
Sarvak	2633-3278	1.14-1.26	0.24-1.23	1.44-1.81
Kazhdumi	3278-3481	1.1-1.14	0.46-0.96	1.93-1.94
Dariyan	3481-3647	1.18-1.25	0.55-1.1	1.96-2.06
Gadvan	3647-3937	1.1-1.25	0.55-1.44	1.97-2.11
Fahliyan	3937-T.D.	1.60-1.3	1.20-1.75	2.08-2.12

Table 1-5 Formation Pressure Prediction

Reservoir	Sub-layer	Mid of Sub-Layer Depth in AZNS-127 (mss)	Temp (°F)	Temperature Gradient (°F/100ft)	Current Reservoir Pressure (psia)	Basis for Reservoir Pressure
Fahliyan	Fah-1a	-3966.7	256.3	1.32	9100	MDT 195
	Fah-1b	-3983	257.0		9100	MDT 195
	Fah-1c	-4009	258.0		9200	MDT 195
	Fah-2a	-4035.7	259.3		8912	Dynamic Model
	Fah-2b	-4049.8	259.9		8924	Pressure Gradient
	Fah-3a	-4058	260.3		8440	Dynamic Model
	Fah-3b	-4067	260.6		8447	Pressure Gradient
	Fah-4a	-4077	261.1		7551	Dynamic Model
	Fah-4b	-4091	261.7		8100	MDT 195
	Fah-5a	-4108.6	262.4		8100	MDT 195
	Fah-5b	-4163.7	264.8		8100	MDT 195

Table 1-6 Predicted Pressure and Temperature and TTZ Contact

1.7 Drilling History

The AZNS – 127 well was drilled & completed by TDDC on date 1395.10.08 at 12:00 as describe below:

- 26" hole size drilled to 150^{m} w/ Mw 66 – 68^{pcf} & 20" casing ran to 146^{m} & cemented as program by 170^{bbl} cement.
- 17 – 1/2" hole size drilled to 1441^{m}, while increased Mw from 68 to 90^{pcf}, 13 – 3/8" casing ran & cemented to 1441^{m} by 780^{bb} cement.
- 12 – 1/4" hole size drilled to 3910^{m} w/ performance drlg & 3 – 15BPH loss. At depth 2354^{m} suddenly pressure dropped f/ 2850 to 1950^{psi} & string twist off from float sub & 9.18^{m} fish left in hole. Fish recovered by overshot 10 – 1/8". At depth 3117^{m}due to pressure dropped f/ 2560 to 2320^{psi} POOH string but string got stuck at 3040^{m} & free by applied 50^{klbs} overpull, POOH to surface & observed wash out on float sub above 8" PDM. Took MSFL/DLL/SONIC/CALIPER/GR/CDR & ZDL/CNC/DSL/WTS/TTRM logs, 9 – 5/8" casing ran & cemented to 3909^{m} with 1133^{bbl} cement.
- 8 – 1/2" hole size drilled to 4226^{m} with no major problem, took VDL/SBT/CCL/GR logs, string held up at 3905^{m}, tried to pass this point several times with no success, logged up from this pont to 100^{m} above 13 – 3/8" casing shoe. Due to tight hole Mw increased f/ 104 to 105^{pcf}. Took first full suite log as DLL/SONIC/MSFL/CDR from TD to 3810^{m} & second full suite log as CNT/LDT/NGS/GR from 3910^{m} to TD. 7" liner ran to 4226^{m}. Due to first dry test results, decided was made to tie back 7" liner to 3777^{m}. The well suspended with cement plug & completion string & rig released @1396.06.27.
- Re – entry the well at 1396.11.03 by retrieving completion string & ran new completion & rig released at 24:00 on 1396.12.23
- Rig less operation was performed on date 17.05.1398. tried to RIH with coiled tubing to unload the well. There was an obstruction in the well at depth 3918^{m}. Unable to pass the depth. Several runs were performed by coiled tubing and mill to remove the obstruction.
- Rig less operation for unloading and well flowing was performed on date 30.01.1400.
- Pressure monitoring was performed on tubing head spool vent port, it was 4,000^{psi}. Metal seal of tubing hanger is damaged.
- Try to run coiled tubing to TD and wash the wellbore with xylene. Run CCL/GR/MFC logging BHA in the well. It was observed 7" liner ID has been decreased to about 3inches in interval 3889 – 3911^{m}.
- Flow the well, an abnormal annulus pressure was observed.
- The other Rig less monitoring well annulus pressure on date 18.09.1400. It was annulus pressure is 5,300^{psi}. Sampling was performed from annulus fluid. It was contained with gas and hydrocarbon.

AZNS – 127 side track #1 was drilled & completed by OEOC on 1403 – 02 – 19 and describe as below:

- The completion string was pulled out & the well was side track by directional drilling 8 – 1/2" hole section from 3338.6m (K.O.P) and 7" liner was ran from 3147 to 3891m.
- The 6 – 1/8" hole section was drilled & 4 – ½" liner ran from 3780.41 to 4139.5^{m} and the well completed by 4 – 1/2" * 3 – 1/2" completion string on 1403 – 07 – 04.

1.8 Well Montage

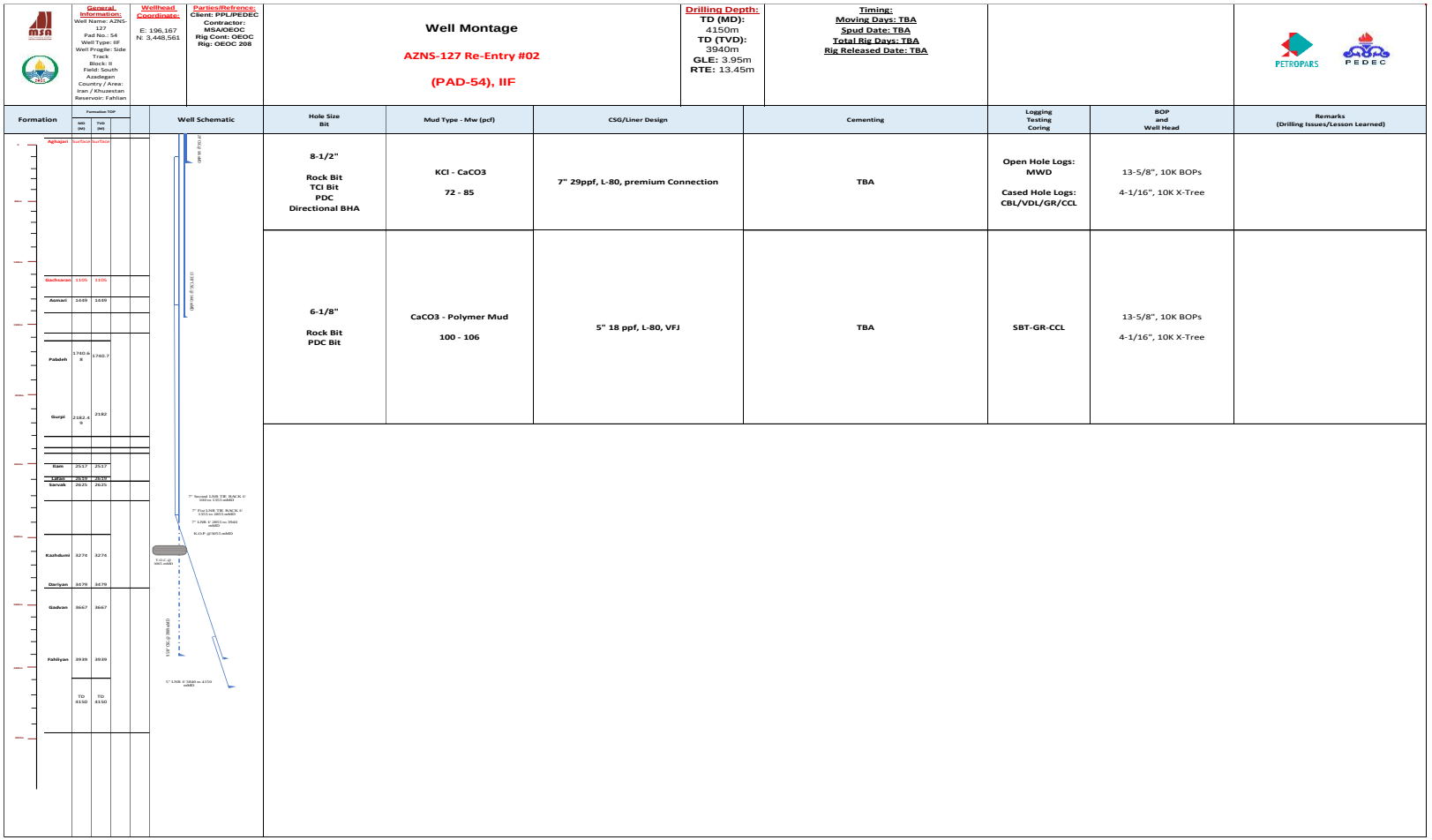


Figure 1-4 AZNS-127 Well Montage

1.9 Casing & Liner Depth before Side Track-2

Casing Spec.	20"	K – 55, 94#, BTC	0 – 146^{m}
	13 – 3/8"	N – 80, 68#, BTC	0 – 1441^{m}
	9 – 5/8"	103 joint 53.5ppf, T – 95, TSH Blue, 226 joints 53.5ppf, T – 95, TPCQ.	0 – 3909^{m}
	7" (S.T #1)	110 SS, 29#, TPCQ	3147 – 3891^{m}
	4 – 1/2" (S.T #1)	110 SS, 13.5#, TPG2	3780.41 – 4139.5^{m}

Table 1-7 Casing & Liner Data

1.10 Well Schematic before Side Track-2

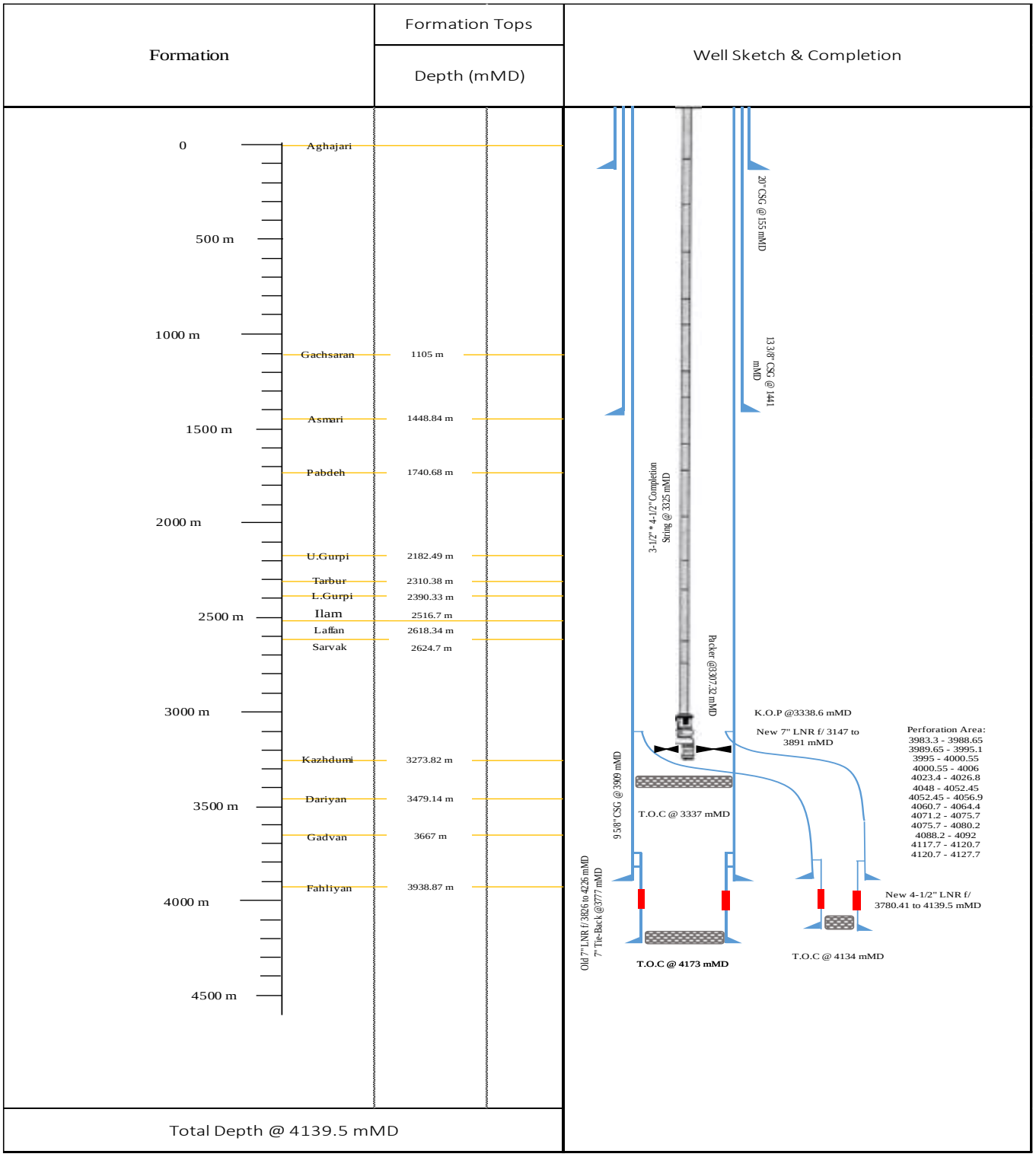


Figure 1-5 - Well Schematic before Side Track #2

1.11 Well Schematic after Side Track-2

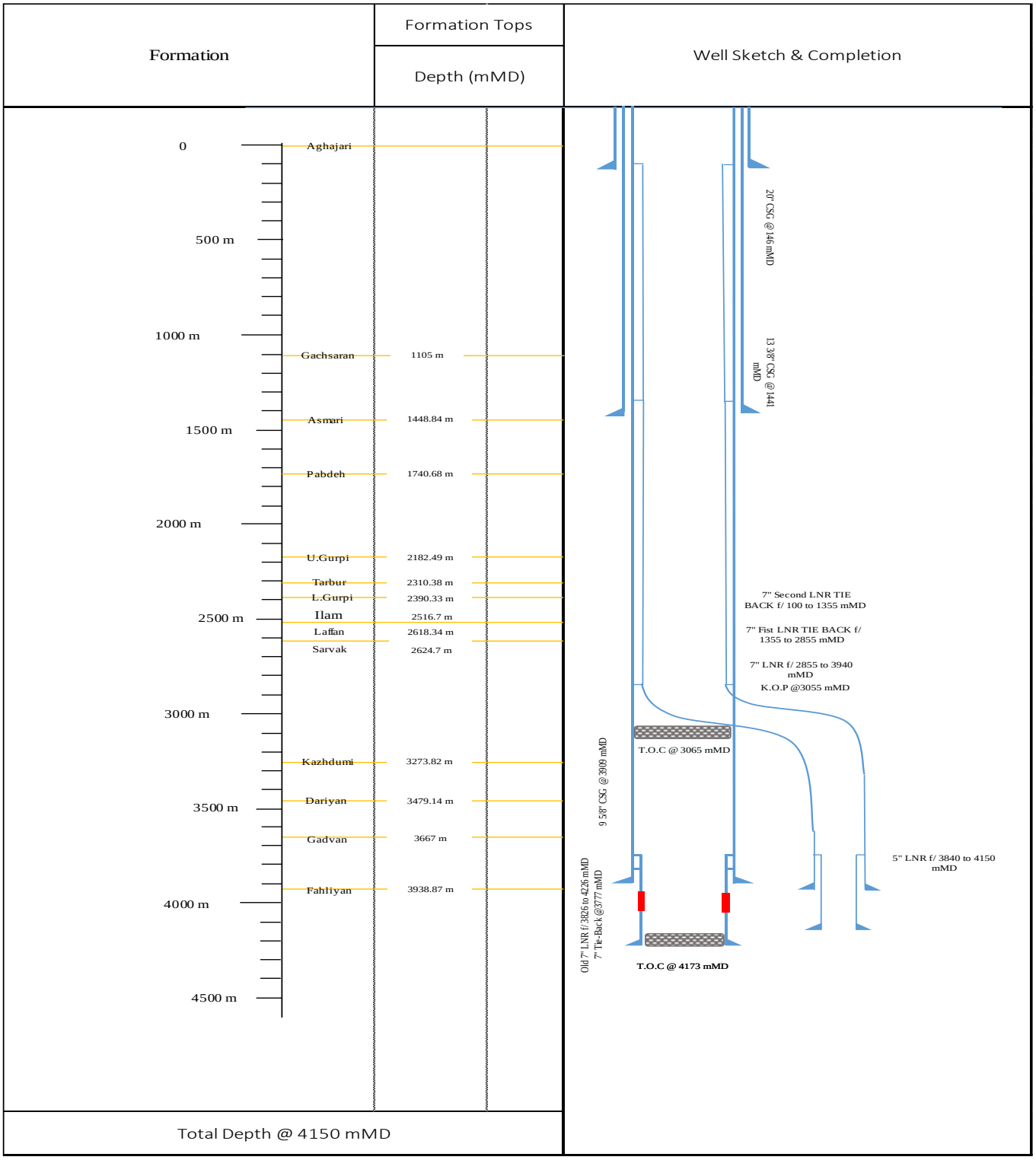


Figure 1-6 - Well Schematic after Side Track #2

2. Geological Information

2.1 Formation Stratigraphy and Lithology

Stage	Group	Formation	Lithology	Lithology Description	Block I Top (mTVD)	Block II Top (mTVD)	Block III Top (mTVD)
Pliocene Miocene	Fars	Agha Jari		Marl, Sandstone, Anhydrite	Surface	Surface	Surface
Miocene	Fars	Gachsaran		Marl, Anhydrite, Claystone	1380	1150	961
E. Miocene Oligocene		Asmari		Limestone, Sandstone, Claystone	1776	1503	1318
E. Oligocene L. Paleocene		Pabdeh		Limestone, Argillaceous Limestone, Claystone, Dolomite	2079	1786	1525
Maastrichtian Campanian		U. Gurpi Tarbur L. Gurpi		Limestone, Argillaceous Limestone	2352	2303	2153
Santonian Coniacian	Bangestan	Ilam		Limestone, Claystone	2606	2534	2549
	Bangestan	Laffan		Claystone	2691	2642	2625
Turonian Cenomanian	Bangestan	Sarvak		Limestone, Claystone	2697	2648	2632
Cenomanian Albian	Bangestan	Kazhdumi		Sandstone, Argillaceous Limestone, Claystone, Shale	3335	3300	3269
Aptian	Khami	Dariyan		Limestone	3538	3499	3471
Aptian Barremian	Khami	Gadvan		Argillaceous Limestone, Claystone, Sandstone	3718	3696	3635
Neocomian	Khami	Fahliyan		Limestone, Marl	4092	3989	3974

Figure 2-1 - Azadegan Stratigraphic Columne

Lithostratigraphic units	Formation Top(m)				Vertical Thickness	Difference (m)	Casing Data	
	Forecast (m)	MD (Sample) (m)	MD (log) (m)	Subsea (msl)	(m)	(Forecast-MD)	Size (Inch)	Depth (m)
Aghajari	Surface	Surface	0	14.45	1059.17		20"	140m
Gachsaran	1107	1105	1084	-1069.67	364.53	-23		
Asmari	1470	1451	1448.84	-1434.2	291.8	-21.16	13 3/8"	1441m
Pabdeh	1747	1724	1740.68	-1726	441.8	-6.32		
U-Gurpi	2203	2189	2182.49	-2167.8	127.9	-20.51		
Tarbur	2335	2314	2310.38	-2295.7	79.9	-24.62		
L-Gurpi	2407	2390	2390.33	-2375.6	126.4	-16.67		
Ilam	2537	2519	2516.7	-2502	101.6	-20.3		
Laffan	2635	2617	2618.34	-2603.6	0.4	-16.66		
Sarvak-1	2602	2595	2624.7	-2610	19.3	22.7		
Sarvak-2			2644.06	-2629.3	8.8			
Sarvak-3			2652.83	-2638.1	31.5			
Sarvak-4			2684.29	-2669.6	94.3			
Sarvak-7			2778.6	-2763.9	21.9			
Sarvak-8			2800.52	-2785.8	59.5			
Sarvak-9			2859.98	-2845.3	37.5			
Sarvak-10			2897.55	-2882.8	22.4			
Sarvak-11			2919.95	-2905.2	32			
Sarvak-12			2951.9	-2937.2	93.6			
Intra Sarvak			3045.5	-3030.8	228.3			
Kazhdumi	3242	3241	3273.82	-3259.1	150.9	31.82		
Kazhdumi-1			3424.78	-3410	23.1			
Kazhdumi-2			3447.89	-3433.1	10.4			
Kazhdumi-3			3458.28	-3443.5	7.7			
Kazhdumi-4			3465.98	-3451.2	7.9			
Kazhdumi-5			3473.83	-3459.1	4.8			
Dariyan	3441	3448	3478.62	-3463.9	171.9	37.62		
Gadvan-1	3596	3605	3650.57	-3635.8	8.8	54.57		
Gadvan-2u			3659.36	-3644.6	54.7			
Khalij Mbr.			3714.04	-3699.3	10.4			
Gadvan-2l			3724.44	-3709.7	4			
Gadvan-3			3728.49	-3713.7	8.2			
Gadvan-4			3736.69	-3721.9	16.3			
Gadvan-5a			3753	-3738.2	5.3			
Gadvan-5b			3758.31	-3743.5	5.4			
Gadvan-5c			3763.62	-3748.9	7.4			
Gadvan-6			3771.11	-3756.3	20.9			
Gadvan-7			3792	-3777.2	24.7			
Gadvan-8			3816.67	-3801.9	121.1		9 5/8"	3909m
Fahliyan	3950	3934	3937.73	-3923	35	-12.27		
Fahliyan-1a			3972.82	-3958	17.5			
Fahliyan-1b			3990.29	-3975.5	15.4			
Fahliyan-1c			4005.63	-3990.9	36.6			
Fahliyan-2a			4042.32	-4027.5	16.4			
Fahliyan-2b			4058.69	-4043.9	11.8			
Fahliyan-3a			4070.51	-4055.7	6.2			
Fahliyan-3b			4076.69	-4061.9	10.4			
Fahliyan-4a			4087.09	-4072.3	10.4			
Fahliyan-4b			4097.52	-4082.7	17.1			
Fahliyan-5a			4114.61	-4099.8	17.8			
Fahliyan-5b			4132.35	-4117.6	93.9			
TD	4226.5	4226.5	4226.5	-4211.5	-	0	7"L	4225.5m

Figure 2-2 Actual Formation Tops of AZNS-127 original well

2.2 Planned Trajectory Data of Side Track

Directional Drilling Plan of AZNS-127 S.T.-2											
RTE: 12.95m								GLE: 3.95m			
Vertical Section was calculted toward Azimuth 340deg.											
MD (m)	Inc (°)	Azi (°)	TVD (m)	TVDss (m)	N/S (m)	E/W (m)	VSEC (m)	DLS (°/30m)	Map N (m)	Map E (m)	Remarks
3045.45	0.93	267.99	3045.25	-3032.30	-21.24	8.95	-23.02	0.22	3448539.76	196175.95	Intra Sarvak
3055.00	1.00	269.09	3054.79	-3041.84	-21.24	8.79	-22.97	0.00	3448539.76	196175.79	K.O.P.
3085.00	3.17	340.00	3084.78	-3071.83	-20.47	8.24	-22.05	3.00	3448540.53	196175.24	
3115.00	6.17	340.00	3114.67	-3101.72	-18.17	7.40	-19.61	3.00	3448542.83	196174.40	
3145.00	9.17	340.00	3144.40	-3131.45	-14.40	6.03	-15.60	3.00	3448546.60	196173.03	
3173.25	12.00	340.00	3172.17	-3159.22	-9.53	4.26	-10.41	3.00	3448551.47	196171.26	M.B.U.A.P.
3175.00	11.83	340.00	3173.88	-3160.93	-9.19	4.14	-10.05	3.00	3448551.81	196171.14	
3205.00	8.83	340.00	3203.39	-3190.44	-4.14	2.30	-4.67	3.00	3448556.86	196169.30	
3235.00	5.83	340.00	3233.14	-3220.19	-0.54	0.99	-0.85	3.00	3448560.46	196167.99	
3265.00	2.83	340.00	3263.05	-3250.10	1.58	0.22	1.41	3.00	3448562.58	196167.22	
3275.53	1.77	340.00	3273.57	-3260.62	1.98	0.07	1.84	3.00	3448562.98	196167.07	Kazhdumi Fm.
3293.25	0.00	340.00	3291.29	-3278.34	2.24	-0.02	2.11	3.00	3448563.24	196166.98	Drop Point
3295.00	0.00	0.00	3293.04	-3280.09	2.24	-0.02	2.11	0.00	3448563.24	196166.98	
3325.00	0.00	0.00	3323.04	-3310.09	2.24	-0.02	2.11	0.00	3448563.24	196166.98	
3355.00	0.00	0.00	3353.04	-3340.09	2.24	-0.02	2.11	0.00	3448563.24	196166.98	
3385.00	0.00	0.00	3383.04	-3370.09	2.24	-0.02	2.11	0.00	3448563.24	196166.98	
3415.00	0.00	0.00	3413.04	-3400.09	2.24	-0.02	2.11	0.00	3448563.24	196166.98	
3445.00	0.00	0.00	3443.04	-3430.09	2.24	-0.02	2.11	0.00	3448563.24	196166.98	
3475.00	0.00	0.00	3473.04	-3460.09	2.24	-0.02	2.11	0.00	3448563.24	196166.98	
3480.32	0.00	0.00	3478.36	-3465.41	2.24	-0.02	2.11	0.00	3448563.24	196166.98	Dariyan Fm.
3505.00	0.00	0.00	3503.04	-3490.09	2.24	-0.02	2.11	0.00	3448563.24	196166.98	
3535.00	0.00	0.00	3533.04	-3520.09	2.24	-0.02	2.11	0.00	3448563.24	196166.98	
3565.00	0.00	0.00	3563.04	-3550.09	2.24	-0.02	2.11	0.00	3448563.24	196166.98	
3595.00	0.00	0.00	3593.04	-3580.09	2.24	-0.02	2.11	0.00	3448563.24	196166.98	
3625.00	0.00	0.00	3623.04	-3610.09	2.24	-0.02	2.11	0.00	3448563.24	196166.98	
3652.27	0.00	0.00	3650.31	-3637.36	2.24	-0.02	2.11	0.00	3448563.24	196166.98	Gadvan Fm.
3655.00	0.00	0.00	3653.04	-3640.09	2.24	-0.02	2.11	0.00	3448563.24	196166.98	
3685.00	0.00	0.00	3683.04	-3670.09	2.24	-0.02	2.11	0.00	3448563.24	196166.98	
3715.00	0.00	0.00	3713.04	-3700.09	2.24	-0.02	2.11	0.00	3448563.24	196166.98	
3745.00	0.00	0.00	3743.04	-3730.09	2.24	-0.02	2.11	0.00	3448563.24	196166.98	
3775.00	0.00	0.00	3773.04	-3760.09	2.24	-0.02	2.11	0.00	3448563.24	196166.98	
3805.00	0.00	0.00	3803.04	-3790.09	2.24	-0.02	2.11	0.00	3448563.24	196166.98	
3835.00	0.00	0.00	3833.04	-3820.09	2.24	-0.02	2.11	0.00	3448563.24	196166.98	
3865.00	0.00	0.00	3863.04	-3850.09	2.24	-0.02	2.11	0.00	3448563.24	196166.98	
3895.00	0.00	0.00	3893.04	-3880.09	2.24	-0.02	2.11	0.00	3448563.24	196166.98	
3925.00	0.00	0.00	3923.04	-3910.09	2.24	-0.02	2.11	0.00	3448563.24	196166.98	
3939.41	0.00	0.00	3937.45	-3924.50	2.24	-0.02	2.11	0.00	3448563.24	196166.98	Fahliyan Fm. & 7" L. Shoe
3955.00	0.00	0.00	3953.04	-3940.09	2.24	-0.02	2.11	0.00	3448563.24	196166.98	
3985.00	0.00	0.00	3983.04	-3970.09	2.24	-0.02	2.11	0.00	3448563.24	196166.98	
4015.00	0.00	0.00	4013.04	-4000.09	2.24	-0.02	2.11	0.00	3448563.24	196166.98	
4045.00	0.00	0.00	4043.04	-4030.09	2.24	-0.02	2.11	0.00	3448563.24	196166.98	
4075.00	0.00	0.00	4073.04	-4060.09	2.24	-0.02	2.11	0.00	3448563.24	196166.98	
4105.00	0.00	0.00	4103.04	-4090.09	2.24	-0.02	2.11	0.00	3448563.24	196166.98	
4135.00	0.00	0.00	4133.04	-4120.09	2.24	-0.02	2.11	0.00	3448563.24	196166.98	
4150.00	0.00	0.00	4148.04	-4135.09	2.24	-0.02	2.11	0.00	3448563.24	196166.98	T.D. & 5" Liner Shoe

3. Logging Program

Hole Size	Casing / Liner	Formation	Cased Hole Wire Line Logging
8 – 1/2”	7” Liner	Sarvak, Kazhdumi, Dariyan, Gadvan	CBL/VDL – GR – CCL
6 – 1/8”	5” Liner	Fahliyan	RBT /CCL/GR Perforation

Table 3-1 AZNS-127 Side Track-2 Logging Program

4. Program Outline

4.1 Contractors Involved

Table 4-1 AZNS-127 Contractor's Involve

Item	Service Item	Contractor
1	Drilling Rig, Mobilization and Demobilization the Rig, Rig Move Between Well Pads	OEOC 208
2	Drilling Fluids and Completion Fluids	VIRA
3	Waste Management	Pars Hafaran Arvand
4	Electrical Logging and Perforation	SEALAND
5	Mud Logging	MAPSA KISH
6	H_2S Monitoring and Control	BONYAN IMEN
7	Casing and Tubing Running	OEOC
8	Cementing	SPAD ENERGY
9	Dry Test, DST and RTTS	OEOC
10	Slick Line	OEOC
11	Stimulation (Coiled Tubing and Acidizing and Nitrogen Lifting)	OEOC
12	Surface Well Testing	OEOC
13	Liner Hanger	PARS POOLAD
14	Well Head	OEOC
15	Directional Drilling	Hana Energy

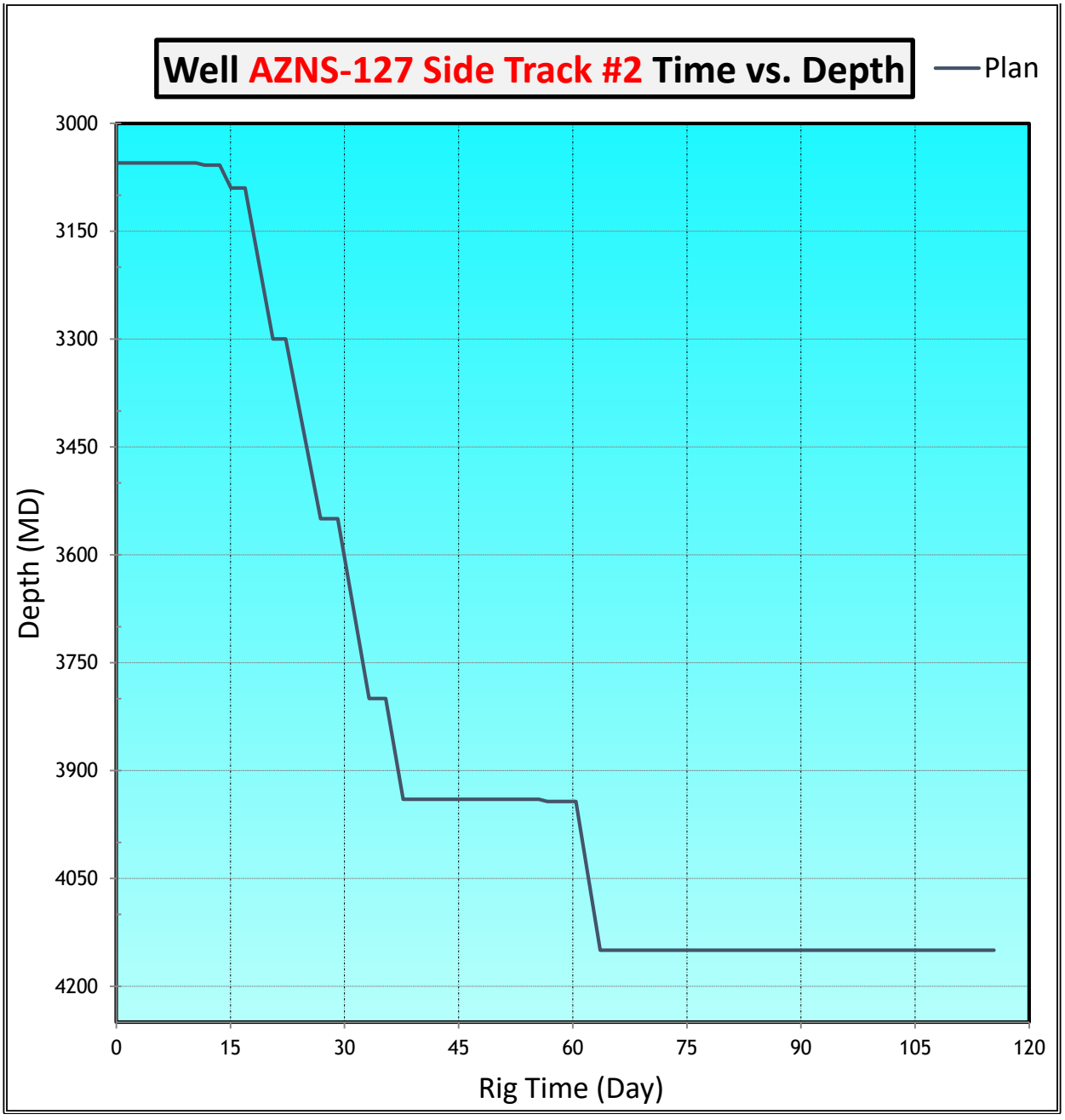
4.2 Time Breakdown

Side Track Operation TBD of Well: AZNS-127 Work-over (Fahlian Producer)	IADC Code	Interval (meters)	Formation	Depth (meters)	ROP	Time Duration (Hours)	Time Duration (Days)	Total Duration (Days)	Contingency (%) 10	Time Duration w/ contingency (Hours)	Time Duration w/ contingency (Days)	Total Duration w/ contingency (Days)
Work-Over												
Open The well, Set Plug Tester in T.H.S & Perform BOP Test as NIOC Policy , Retrieve Plug Tester.	Trip		sarvak	3055		7	0.29	0.3	0.3	7.7	0.32	0.3
P/U & M/U 6 1/8" C. Cut Bit & RIH to 3100 m for, while fill up string with 104 PCF fresh mud due to mud in hole is gelled up & break Circ. Every 3 Std.	Trip		sarvak	3055		24	1.00	1.3	1.1	26.4	1.10	1.4
Perform CBU in 2 stage to balance mud weight.	Circulation		sarvak	3055		10	0.42	1.7	0.5	11.0	0.46	1.9
L/U & pressure test CMT line, perform 20 bbl CMT balance plug job, POOH 10 Std, flush string, Cont. POOH to surface.	CMT		sarvak	3055		20	0.83	2.5	0.9	22.0	0.92	2.8
P/U & M/U 8 1/2" R.Bit & Related BHA, RIH to top of CMT estimated @ 3011m, tag CMT, D/O & Dress CMT to 3065 m, Perform CBU, POOH to surface.	D/O		sarvak	3055		52	2.17	4.7	2.4	57.2	2.38	5.2
P/U & M/U 8 1/2" R. Bit & 9 5/8" scraper BHA & RIH to BTM while Scrap inside 9 5/8" Casing (whipstock packer depth), perform CBU, POOH to surface & L/D Scraper.	Trip		sarvak	3055		28	1.17	5.9	1.3	30.8	1.28	6.5
P/U & M/U 9 5/8" Hydraulic set whipstock including UBHO Sub, align whipface to UBHO (gyro sub), RIH to setting depth +/-3060 m (KOP will be @ 3055 m).	Trip		sarvak	3055		28	1.17	7.0	1.3	30.8	1.28	7.7
R/U wireline, Run gyro, Orient whipface and adjust toolface (Azi: 340 Deg.), POOH gyro, L/D wireline.	Trip		sarvak	3055		18	0.75	7.8	0.8	19.8	0.83	8.6
Set whipstock packer (as per procedure) and check. Pull string upward and shear mill pins, Displace the well with proper mud (82-84 PCF) & Cont. Circ. To balance mud weight	Trip		sarvak	3055		12	0.50	8.3	0.6	13.2	0.55	9.1
Start opening window in 9-5/8" Casing, Pump Hi-Vis pill, sweep the well, in order to change Mill, POOH to surface.	Drlg		sarvak	3055		30	1.25	9.5	1.4	33.0	1.38	10.5
P/U & M/U New Mill & RIH to BTM, Cont. opening window and drill to 3m inside formation. Check window condition.	Drlg	3	sarvak	3058	0.2	24	1.00	10.5	1.1	26.4	1.10	11.6
POOH to surface & L/D Mill.	Trip		sarvak	3058		10	0.42	11.0	0.5	11.0	0.46	12.1
Perform BOP Test as NIOC Policy.	BOP Test		sarvak	3058		6	0.25	11.2	0.3	6.6	0.28	12.3
P/U & M/U 8-1/2" Tri-cone Bit & Dir. BHA (Including MWD & Gyro), Perform shallow hole test, RIH to BTM, R/U wireline, Run gyro, adjust toolface. POOH gyro, L/D wireline.	Trip		sarvak	3058		28	1.17	12.4	1.3	30.8	1.28	13.6
Commence Dir.Drilling 8 1/2" Hole Section (slide & Rotary Mode) from 3058 to 3090 m, while pump Hi-Vis Pill periodically.	Drlg	32	sarvak	3090	1	32	1.33	13.7	1.5	35.2	1.47	15.1
In order to change the bit, POOH to surface.	Trip		sarvak	3090		12	0.50	14.2	0.6	13.2	0.55	15.6
P/U & M/U 8 1/2" PDC Bit & Dir. BHA (Including MWD & Gyro Sub), Perform shallow hole test, RIH to BTM, R/U wireline, Run gyro, adjust toolface. POOH gyro, L/D wireline.	Trip		sarvak	3090		28	1.17	15.4	1.3	30.8	1.28	16.9
Profile the bit, Cont. Dir. Drlg 8 1/2" H.S F/3090 to 3300 m, Perform CBU,	Drlg	210	Kzh.	3300	2.6	80	3.33	18.7	3.7	88.0	3.67	20.6
In order to change BHA, POOH to surface.	Trip		Kzh.	3300		15	0.63	19.3	0.7	16.5	0.69	21.3
P/U & M/U 8 1/2" PDC Bit & Dir. BHA (including MWD), RIH to BTM, perform CBU to uniform mud weight	Circulation		Kzh.	3300		22	0.92	20.3	1.0	24.2	1.01	22.3
Cont. Dir. Drlg 8 1/2" H.S F/3300 to 3550 m, while Pump hi-Vis pill periodically,	Drlg	250	Dariyan	3550	2.5	100	4.17	24.4	4.6	110.0	4.58	26.9
In order to change BHA, perform CBU, POOH to surface, L/D Bit.	Trip		Dariyan	3550		18	0.75	25.2	0.8	19.8	0.83	27.7
Performe BOP Test as per Policy	BOP Test		Dariyan	3550		6	0.25	25.4	0.3	6.6	0.28	28.0
P/U & M/U 8 1/2" PDC Bit & Dir. BHA (including MWD), RIH to BTM, perform CBU to uniform mud weight	Circulation		Dariyan	3550		25	1.04	26.2	1.1	27.5	1.15	29.1
Cont. Drlg 8 1/2" H.S F/3550 to 3800 m, while Pump hi-Vis pill periodically,	Drlg	250	Gad.	3800	2.7	90	3.75	30.0	4.1	99.0	4.13	33.2
Perform CBU, In order to change BHA, perform CBU, POOH to surface, L/D Bit.	Trip		Gad.	3800		21	0.88	30.8	1.0	23.1	0.96	34.2
P/U & M/U 8 1/2" PDC Bit & Dir. BHA (including MWD), RIH to BTM, perform CBU to uniform mud weight	Trip		Gad.	3800		27	1.13	32.0	1.2	29.7	1.24	35.4
Cont. Drlg 8 1/2" H.S to F/3800 to 3940 m (Section TD) while Pump Hi-Vis pill Periodically.	Drlg	140	Fah.	3940	2.8	50	2.08	34.0	2.3	55.0	2.29	37.7
In order to perform condition Trip prior to run 7" LNR, Perform CBU, POOH to 3055 m (KOP), Run back to BTM as condition trip, Pump Hi-Vis Pill & sweep the well, POOH to surface while SLM 5" DPs Std.	Trip		Fah.	3940		40	1.67	35.7	1.8	44.0	1.83	39.6
Perform BOP Test as NIOC Policy.	BOP Test		Fah.	3940		6	0.25	36.0	0.3	6.6	0.28	39.8
Held PJSM, R/U 7" LNR handling equipment, M/U 7" F.Shoe & shoe track, M/U 7" F. Collar, check floating function, Cont. Run about 110 Jts 7" LNR while fill up string every 10 Jt.	Casing Running		Fah.	3940		23	0.96	36.9	1.1	25.3	1.05	40.9
P/U & M/U 7" LNR HGR with top PKR, check POB sealing, L/D LNR running Eq., RIH 7" LNR on 5" DPs to 3055 m while fill up string every 10 Std. Perform Circ. As a 1.5 cap of string.	Casing Running		Fah.	3940		20	0.83	37.8	0.9	22.0	0.92	41.8
Cont. RIH 7" LNR on 5" DPs Std to 3940 m (7" L.Lap will be +/- 2855 m) while fill up string every 5 Std, Perform Circulation as 2 cap of string meanwhile prepare solution for CMT Job.	Casing Running		Fah.	3940		20	0.83	38.6	0.9	22.0	0.92	42.7
Perform 7" LNR CMT Job, release setting tools, POOH to desired TOC, Flush String, POOH 10 Std, Perform CBU, POOH to surface & L/D setting tools.	CMT		Fah.	3940		22	0.92	39.5	1.0	24.2	1.01	43.7
P/U & M/U 8 1/2" Bit & RIH to BTM, Tag CMT, D/O CMT to 7" L.Lap, Perform CBU, POOH to surface.	Trip		Fah.	3940		50	2.08	41.6	2.3	55.0	2.29	46.0
P/U & M/U 6 1/8" R. Bit & 3 1/2" DPs as Jt by Jt(+/-66 Jt), Cont. RIH on 3 1/2" & 5" DPs Std, to 7" L.Lap, D/O POB & dress same, Cont. RIH & D/O inside 7" LNR to LDC depth. Perform CBU, POOH to 7" L.lap, perform positive pressure test of 7" L.Lap, Cont. POOH to surface.	D/O		Fah.	3940		60	2.50	44.1	2.8	66.0	2.75	48.8
P/U & M/U 8 1/2" Bit & 9 5/8" Scraper BHA, RIH to 7" L.Lap while scrap Dry test PKR setting interval for several times, Pump Hi-Vis Pill & Sweel the well, POOH to surface.	Trip		Fah.	3940		32	1.33	45.4	1.5	35.2	1.47	50.2
P/U & M/U 9 5/8" Dry Test Assy & RIH to +/- 30 m above 7" L.Lap, while fill up string with with water cushion, Set PKR, Perform 7" LNR Inflow test as per procedure.	Dry Test		Fah.	3940		48	2.00	47.4	2.2	52.8	2.20	52.4
POOH 9 5/8" Dry Test Assy to surface & L/D same.○	Trip		Fah.	3940		24	1.00	48.4	1.1	26.4	1.10	53.5
Perform BOP Test as NIOC Policy.	BOP Test		Fah.	3940		6	0.25	48.7	0.3	6.6	0.28	53.8
P/U & M/U 6 1/8" R.Bit & 3 1/2" DPs, RIH to 7" LDC depth, D/O plugs, 7" LDC and 7" Liner Shoe Track to 5 m above 7" LNR shoe, Pump Hi-Vis pill & sweep the well, Displace the well with proper mud weight (102-104 PCF), Cont. Drlg shoe track to 7" LNR shoe	D/O		Fah.	3940		38	1.58	50.3	1.7	41.8	1.74	55.6
Total Duration of 8-1/2" Hole Section:								*50.3*				*55.6*

6 1/8" Hole Section													
Commence Drilling 3m, new Formation, CBU till hole Clean, POOH to Surface due to L/D Bit and change BHA.	Drlg	3	Fah.	3943	1	24	1.00	51.3	1.1	26.4	1.10	56.7
P/U & M/U 6 1/8" R.Bit & 7" Scrapper BHA, RIH to 3940 m (7" LNR shoe) while Scrape Inside 7" Liner prior to take 7" Liner CMT quality Log, CBU, POOH to Surface and L/D bit & BHA.	Trip		Fah.	3943		34	1.42	52.7	1.6	37.4	1.56	58.2
Change Rams & Test same, R/U shooting nipple, Wire line equipment & BOP, pressure test same, M/U CBL/VDL-GR-CCL logging tools, Perform 7"LNR cement quality log, POOH to surface & L/D logging tools & wire line equipment	Log		Fah.	3943		24	1.00	53.7	1.1	26.4	1.10	59.3
P/U & M/U 6 1/8" PDC Bit with rotary BHA (P/U & M/U required 3 1/2" DPs as Jt by Jt), RIH to BTM while fill up string every 10 Std.	Trip		Fah.	3943		24	1.00	54.7	1.1	26.4	1.10	60.4
Cont. Drilling 6 1/8" H.S from 3943m to 4150 m, (Well TD) Note: Take full suite loge base on client decision (Contingency #1)	Drlg	207	Fah.	4150	2.9	70	2.92	57.6	3.2	77.0	3.21	63.6
Perform CBU, POOH to 3940 m (7" LNR shoe) & run back to BTM as condition trip, Perform CBU, POOH to surface.	Circulation		Fah.	4150		24	1.00	58.6	1.1	26.4	1.10	64.7
Perform BOP Test as NIOC Policy.	BOP Test		Fah.	4150		6	0.25	58.8	0.3	6.6	0.28	65.0
R/U 5" LNR running Eq., M/U 5" F.Shoe & shoe track, M/U 5" F. Collar, check floating function, Cont. Run about 35 Jts 5" LNR while fill up string every 10 Jt.	Casing Running		Fah.	4150		12	0.50	59.3	0.6	13.2	0.55	65.5
P/U & M/U 5" LNR HGR with top PKR, check POB sealing, L/D LNR running Eq., RIH 5" LNR on 3 1/2" & 5" DPs to 3940 m (7" LNR shoe), while fill up string every 10 Std. Perform Circ. As a 1.5 cap of string.	Casing Running		Fah.	4150		25	1.04	60.4	1.1	27.5	1.15	66.7
Cont. RIH 5" LNR on 5" DPs Std to 4150 m (5" L.Lap will be +/- 3840 m) while fill up string every 5 Std, Perform Circulation as 1.5 cap of string meanwhile prepare solution for CMT Job.	Casing Running		Fah.	4150		6	0.25	60.6	0.3	6.6	0.28	67.0
Perform 5" LNR CMT Job, release setting tools, POOH to desired TOC, Flush String, POOH 10 Std, Perform CBU, POOH to surface while L/D excess 5" DPs as Jt by Jt, & L/D setting tools.	CMT		Fah.	4150		35	1.46	62.1	1.6	38.5	1.60	68.6
P/U & M/U 6 1/8" Bit & D/O BHA, RIH to BTM, Tag CMT, D/O CMT to 5" L.Lap, Perform CBU, POOH to surface.	D/O		Fah.	4150		52	2.17	64.3	2.4	57.2	2.38	71.0
R/U 2 7/8" DPs handling Eq., P/U & M/U 3 1/2" J.Mill & 2 7/8" DPs Jt by Jt(+/- 57 Jts), RIH to 5" L.Lap, D/O POB & dress same, Cont. RIH & D/O inside 5" LNR to LDC depth. Perform CBU, POOH to 5" L.lap, perform positive pressure test of 5"L.Lap, POOH to Surface while L/D 2 7/8" DPs as Jt by Jt	D/O		Fah.	4150		68	2.83	67.1	3.1	74.8	3.12	74.1
P/U & M/U 7" Dress Mill, RIH to 2855 m (7" L.Lap) while dress TBS, Perform CBU, POOH to surface.	Trip		Fah.	4150		28	1.17	68.3	1.3	30.8	1.28	75.4
P/U & M/U 7 5 1/16" polish Mill RIH to 7"L.Lap, Dress inside 7" TBS for several times, Pump Hi-Vis Pill & Sweep the well, POOH to surface.	Trip		Fah.	4150		28	1.17	69.4	1.3	30.8	1.28	76.6
R/U 7" LNR handling Eq., P/U & M/U 7" Tie back Assy & Approx. 150 Jt 7" LNR tie back string, P/U & M/U TBS, check POB sealing meanwhile fill up string every 10 Jts.	Casing Running		Fah.	4150		24	1.00	70.4	1.1	26.4	1.10	77.7
R/D 7" LNR handling Eq., Cont. RIH 7" LNR tie back string on 5" DPs to 2855 m (7" L.Lap) while fill up string every 10 Std, Perform Circ. As 1.5 cap of string while prepare CMT solution.	Casing Running		Fah.	4150		10	0.42	70.8	0.5	11.0	0.46	78.2
Perform 7" LNR tie back CMT Job, sting In, release setting tools (new 7" L.Lap +/- 1355 m) & POOH to 1200 m, Flush String, POOH to surface while L/D excess 5" DPs as Jt by Jt..	CMT		Fah.	4150		48	2.00	72.8	2.2	52.8	2.20	80.4
Perform BOP Test as NIOC Policy.	BOP Test		Fah.	4150		6	0.25	73.1	0.3	6.6	0.28	80.7
P/U & M/U 8 3/8" Bit & D/O BHA, RIH to TOC, D/O CMT to new 7" L.Lap (+/- 1355 m), Pumo Hi-Vis pill & swep the well, POOH to surface.	D/O		Fah.	4150		24	1.00	74.1	1.1	26.4	1.10	81.8
P/U & M/U 6 1/8" Bit & D/O BHA, RIH to 1355 m (7" L.Lap), D/O POB & chased same to 100 m below 7" L.Lap, Perform CBU, POOH to surface.	D/O		Fah.	4150		36	1.50	76.4	1.7	39.6	1.65	83.4
P/U & M/U 7" dress mill , RIH to 1355 m (7" L.Lap) while dress TBS several times, Perform CBU, POOH to surface.	Trip		Fah.	4150		20	0.83	74.9	0.9	22.0	0.92	84.3
P/U & M/U 7 5 1/16" polish Mill RIH to 7"L.Lap, Dress inside 7" TBS for several times, Pump Hi-Vis Pill & Sweep the well, POOH to surface.meanwhile L/D 5" DPs	Trip		Fah.	4150		28	1.17	77.6	1.3	30.8	1.28	85.6
R/U 7" LNR handling Eq., P/U & M/U 7" Tie back Assy & Approx. 130 Jt 7" LNR tie back string, P/U & M/U TBS, check POB sealing while fill up string every 10 Jts.	Casing Running		Fah.	4150		24	1.00	78.6	1.1	26.4	1.10	86.7
R/D 7" LNR handling Eq., Cont. RIH 7" LNR tie back string on 5" DPs to 1355 m (7" L.Lap) fill up string, Perform Circ. As 1.5 cap of string while prepare CMT solution.	Casing Running		Fah.	4150		5	0.21	78.8	0.2	5.5	0.23	86.9
Perform 7" LNR tie back CMT Job, sting In, release setting tools (new 7" L.Lap +/- 100 m) Flush String, POOH to surface.	CMT		Fah.	4150		8	0.33	79.1	0.4	8.8	0.37	87.3
P/U & M/U 8 3/8" Bit & RIH to TOC, D/O CMT to new 7" L.Lap @ 100 m, POOH to surface.	D/O		Fah.	4150		6	0.25	79.4	0.3	6.6	0.28	87.6
P/U & M/U 6 1/8" Bit & D/O BHA, (P/U & M/U required 3 1/2" DPs as Jt by Jt RIH to 100 m (7" L.Lap), D/O POB, RIH to LDC depth of 7" tie back string while clean out inside 7" tie back string, D/O & LDC & plug, Perform CBU to sweep the well, Cont. RIH to LDc depth of first 7" tie-back string, while C/O inside 7" tie back string, D/O LDC & Plugs & chased same to 5" L.Lap, Perform CBU, POOH to surface.	D/O		Fah.	4150		82	3.42	82.8	3.8	90.2	3.76	91.3
P/U & M/U 4 1/8" Mill and D/O BHA and RIH with 2 7/8" DP's Jt by Jt and 3 1/2" DP's to 5" L.Liner, D/O POB & dress same, Cont, RIH to LDC depth while C/O any CMT, Perform CBU, POOH to surafce.	Circulation		Fah.	4150		48	2.00	84.8	2.2	52.8	2.20	93.5
Change Rams & Test same, R/U shooting nipple, Wire line equipment & BOP, pressure test same, M/U RBT-GR-CCL logging tools, Perform 5"LNR cement quality log, POOH to surface & L/D logging tools & wire line equipment	Log		Fah.	4150		36	1.50	86.3	1.7	39.6	1.65	95.2
Perform BOP Test as NIOC Policy.	BOP Test		Fah.	4150		6	0.25	86.5	0.3	6.6	0.28	95.5
HPJSM, Rig Up WL & Pressure Test wire line BOP, M/U CCL & Gummy gun & RIH CCLwith same, POOH & Rig Down.	Log		Fah.	4150		16	0.67	87.2	0.7	17.6	0.73	96.2
Rig Up for Perforation, RIH String (Firing head & shooting ,CCL+Guns), Correlate Depth, Shooting the gun,POOH String & RIH again to Perforate all the Depths then Rig Down (8 Run Had been considered)	perforation		Fah.	4150		96	4.00	91.2	4.4	105.6	4.40	100.6
P/U & M/U 6 1/8" Bit & 7" Scraper BHA, RIH to 5" L.Lap while and scrape 7" production PKR setting area, Dispalce the well with completion fluid, POOH to surface meanwhile L/D all tubulars.	perforation		Fah.	4150		120	5.00	96.2	5.5	132.0	5.50	106.1
Perform BOP Test as NIOC Policy.	BOP Test		Fah.	4150		6	0.25	96.5	0.3	6.6	0.28	106.4
R/U Completion running Eq, Run 3-1/2" * 4-1/2", 10K Completion string as per program, R/D completion running Eq	Completion		Fah.	4150		140	5.83	102.3	6.4	154.0	6.42	112.8
Install TWCV in TBG Hanger, N/D BOP stack, N/U X.M.T, perform required test as per program.	Completion		Fah.	4150		48	2.00	104.3	2.2	52.8	2.20	115.0
Secure the well by set BPV in TBG Hanger, close all X-mas Tree valves except BMV, and Release the Rig.	Completion		Fah.	4150		8	0.33	104.6	0.4	8.8	0.37	115.4
Retrieve BPV,R/U Slick Line, Set protection sleeve, retrieve plug & prong from NO GO landing nipple, POOH.	Completion		Fah.	4150		Rig Less	0.00	104.6	0.0	0.0	0.00	115.4
RIH with CT to bottom, displace the well with diesel or nitrogen & observ well, POOH.	Completion		Fah.	4150		Rig Less	0.00	104.6	0.0	0.0	0.00	115.4
R/U SL, RIH & retrieve protection sleeve, POOH, R/D SL.	Completion		Fah.	4150		Rig Less	0.00	104.6	0.0	0.0	0.00	115.4
Flow the well as pre-acid clean up till acceptable BSW & reading stable flow rate on 2 different choke size for 72 hrs.	Completion		Fah.	4150		Rig Less	0.00	104.6	0.0	0.0	0.00	115.4
R/U & RIH S.L, Set protection sleeve, POOH & R/D S.L RIH with C/T & perform acid wash as per program, POOH to surface. RIH w/SL and retrieve protection sleeve	Completion		Fah.	4150		Rig Less	0.00	104.6	0.0	0.0	0.00	115.4
Flow the well as post-acid clean-up till acceptable BSW & reading stable flow rate on 2 different choke size for min 96 hrs. Note 1: The time is only for flowing duration (R/U, R/D & preparation time is excluded). Note 2: Acidizing & post-acid clean-up maybe perform as rig less job upon contractor request & client approval but in case of any failure in well integrity the contractor should provide drilling rig & required services in own responsibility & fix the problem.	Completion		Fah.	4150		Rig Less	0.00	104.6	0.0	0.0	0.00	115.4
RIH with S/L, set protection Sleeve, POOH & R/D SL.	Completion		Fah.	4150		Rig Less	0.00	104.6	0.0	0.0	0.00	115.4
RIH with C/T to bottom & displace the well with diesel, POOH.	Completion		Fah.	4150		Rig Less	0.00	104.6	0.0	0.0	0.00	115.4
RIH with S/L, install plug & prong into NO GO landing nipple, POOH.	Completion		Fah.	4150		Rig Less	0.00	104.6	0.0	0.0	0.00	115.4
RIH with S/L, retrieve protection sleeve.	Completion		Fah.	4150		Rig Less	0.00	104.6	0.0	0.0	0.00	115.4
Secure the well by set BPV in TBG Hanger, close all X-mas Tree valves except BMV.	Completion		Fah.	4150		Rig Less	0.00	104.6	0.0	0.0	0.00	115.4
Total Duration :							***53.4***	***104.6***			***59.8***	***115.4***

NOTE:												
Total Duration for Cut & Slip DL in all hole section.	Cut & Slip			4150		0	0.00	104.6	0.0	0.0	0.00	115.4
Total Duration for Rig Service (Consider 0.5hr per day) in all hole section.	Rig Service			4150		0	0.00	104.6	0.0	0.0	0.00	115.4
Total Duration of AZNS-127 Work-over Well:								104.6				115.4
Contingency Plans TBD of Well: AZNS-127 Work-over	IADC Code	Interval (meters)	Formation	Depth (meters)	ROP	Time Duration (Hours)	Time Duration (Days)	Total Duration (Days)	Contingency (%) 10	Time Duration w/ contingency (Hours)	Time Duration w/ contingency (Days)	Total Duration w/ contingency (Days)
Based on operation situation in each step, the following contingency plans might be followed												
Contigency Plan#1: Change Ram, Install shooting nipple, R/U WL Logging equipment & BOP, Pressure test same, Perform Logging (Full Suite/CDR/GR Logs,), R/D Logging Eq., P/U & M/U 6 1/8" Bit & BHA, RIH to BTM as condition trip, Pump Hi-Vis Pill & sweep the well, spot 100 BBl Hi-Vis Pill, POOH to surface.	Log		Kzh.	4150		144	6.00	6.00	0.0	144.0	6.00	6.0
Total Duration of Contingency Plan AZNS-127 Work-over Well:								6.0				6.0

4.3 Time vs. Depth Chart



5. Sequence of Operation

5.1 Open Window & Drilling 8 – 1/2" Hole Section

Casing / Liner Point	7" Liner shoe @3940^{m} (7" L.Lap @ ±2855^{m})	
Formation	Asmari, Pabdeh, Gurpi, Ilam, Laffan, Sarvak, Kazhdumi, Dariyan, Gadvan	
Risks & Hole Problems	Losses, Tight Hole & Pipe Stuck, Oil/Gas Flow, Shale Instability,Hole Deviation , Complete Loss	
Deviation Control	MWD	
Bit Type & Nozzle	Tri Cone, 517 & PDC 1665. Matrix Body	
Logging	**Cased Hole:** SBT or CBL/VDL – GR/CCL from 7" LNR	
Testing	Dry Test	
Drilling Fluid Parameters	**Salt Saturated Kcl/Glycol/Polymer/$CaCO_3$ WBM**	
	MW (PCF)	80 – 85
	FV (Seq/qt)	32 – 52
	PV (CP)	ALAP
	YP (Lbs./100ft^2)	12 – 20
	HPHT FL (CC/30min)	<3
	Gel 10 sec/10 min (Lbs./100ft^2)	2 - 5 / 3 - 7
	PH	9 – 10
Drilling Parameters	GPM	350-500
	WOB (Klbs)	10-20
	RPM	100-170

Table 5-1 AZNS-127 8 – ½" Hole Section General Data

Formation Sampling

Sample would be taken from Gachsaran to Gadvan every 2-5 meters

NOTE: In addition to this general rule, geologist may decide to take more samples.

- Ensure that drilling tools & equipment are on wellsite & are in serviceable condition.
- Ensure that adequate fishing tools for this section of hole are available.
- Ensure that enough mud material, LCM & CMT are available.

1. Open the well, set plug tester in T.H.S & perform BOP test as NIOC policy, retrieve plug tester.
2. P/U & M/U 6 – 1/8" centre cut bit and RIH to 3100^{m} Whipstock, while fill up string with 104^{pcf} fresh mud due to mud in hole is galled up & break Circ. every 3 stands.
3. Perform CBU in 2 stages to balance mud weight.
4. L/U & pressure test cement line, perform 20^{bbl} cement balance plug job, POOH 10 stands, flush string, Cont. POOH to surface.
5. P/U & M/U 8 – 1/2" rock bit & related BHA, RIH to top of cement estimated @ 3011^{m},tag cement, D/O & dress cement to 3065^{m}, perform CBU, POOH to surface.
6. P/U & M/U 8 – 1/2" rock bit & 9 – 5/8" scraper BHA & RIH to bottom while scrap inside 9 – 5/8" casing (Whipstock packer depth), perform CBU, POOH to surface & L/D scraper.
7. P/U & M/U 8 – 1/2" 9 – 5/8" hydraulic set Whipstock including UBHO sub, align Whipface to UBHO (Gyro sub), RIH to setting depth @ ±3060^{m} (KOP will be @3055^{m}).
8. R/U wireline, run Gyro, orient Whipface & adjust toolface (Azi: 340 Deg), POOH Gyro, L/D wireline.
9. Set Whipstock packer as per procedure & check, pull string upward & shear mill pins, displace the well with proper mud (82 – 84^{pcf}) & Cont. Circ. to balance mud weight.
10. Start opening window in 9 – 5/8" casing, pump Hi-Vis pill, sweep the well, in order to change mill, POOH to surface.
11. P/U & M/U new mill RIH to bottom, Cont. opening window & drill to 3^{m} inside formation. Check window condition.
12. POOH to surface & L/D mill.
13. Perform BOP test as NIOC policy.
14. P/U & M/U 8 – 1/2" Tri – Cone bit & directional BHA (including MWD & Gyro sub), perform shallow hole test, RIH to bottom.
15. R/U wireline, run Gyro, adjust toolface, POOH Gyro & L/D wireline.
16. Profile the bit, Cont. directional drilling hole section from 3090^{m} to 3300^{m}, perform CBU.
17. In order to change the BHA, POOH to surface.
18. P/U & M/U 8 – 1/2" PDC bit & directional BHA (including MWD), RIH to bottom, perform CBU to uniform mud weight.
19. Cont. directional drilling hole section to 3550^{m} while pump Hi-Vis pill periodically.

20. In order to chang th BHA, perform CBU, POOH to surface & L/D bit.
21. Perform BOP test as NIOC policy.
22. P/U & M/U 8 – 1/2" PDC bit & directional BHA (including MWD), RIH to bottom, perform CBU to uniform mud weight.
23. Cont. directional drilling hole section to 3800^{m} while pump Hi-Vis pill periodically.
24. In order to chang th BHA, perform CBU, POOH to surface & L/D bit.
25. P/U & M/U 8 – 1/2" PDC bit & directional BHA (including MWD), RIH to bottom, perform CBU to uniform mud weight.
26. Cont. directional drilling hole section to 3940^{m} (section TD) while pump Hi-Vis pill periodically.
27. In order to perform condition trip prior to run 7" liner, perform CBU, POOH to 3055^{m} (K.O.P).
28. Run back to bottom as condition trip, pump Hi-Vis pill & sweep the well, POOH to surface while SLM 5" DP's stands.
29. Perform BOP test as NIOC policy.
30. Held PJSM, R/U 7" liner handling equipmet, M/U 7" float shoe & shoe track, M/U 7" float collar, check floating function, Cont. run about 110 joints 7" liner while fill up string every 10 joints.
31. P/U & M/U 7" liner hanger with top packer, check POB sealing, L/D liner running equipment, RIH 7" liner on 5" DP's to 3055^{m} while fill up string every 10 joints. Perform Circ. as 1.5 cap of string.
32. Cont. RIH 7" liner on 5" DP's stand to 3940^{m} (7" liner lap will be ±2855^{m}) while fill up string every 5 stands, perform Circ. as 2 cap of string meanwhile prepare solution for cement job.
33. Perform 7" liner cement job, release seting tools, POOH to desired top ofcement, flush string, POOH 10 stands, perform CBU, POOH to surface & L/D setting tools.
34. P/U & M/U 8 – 1/2" bit & RIH to bottom, tag cement, D/O cement to 7" liner lap, perform CBU & POOH to surface.
35. P/U & M/U 6 – 1/8" rock bit & 3 – 1/2" DP's as joint by joint (±66 joints), Cont. RIH on 3 – 1/2" & 5" DP's to 7" liner lap, D/O POB & dress same, Cont. D/O inside 7" liner to LDC depth. Perform CBU, POOH to 7" liner lap, perform positive pressure test of 7" liner lap, Cont. POOH to surface.
36. P/U & M/U 8 – 1/2" rock bit & 9 – 5/8" scraper BHA, RIH to 7" liner lap whil scrap dry test packer setting interval several times, pump Hi-Vis pill & sweep the well, POOH to surface.
37. P/U & M/U 9 – 5/8" dry test assy. & RIH to ±30^{m} above 7" liner lap while fill up string with water cushion, set packer, perform 7" liner inflow test as per procedure.
38. POOH 9 – 5/8" dry test assy. to surface & L/D same.
39. Perform BOP test as NIOC policy.

40. P/U & M/U 6 – 1/8" rock bit & 3 – 1/2" DP's, RIH to 7" LDC depth, D/O plugs, 7" LDC & liner shoe track to 5^{m} above 7" liner shoe, pump Hi-Vis pill & sweep the well, displace the well with proper mud weight (102 – 104^{pcf}), Cont. drilling shoe track to 7" liner shoe.

6 – 1/8" Hole Section

Casing/Liner Point	5" Liner shoe @ 4150^{m} (5" L.Lap @ 3840^{m})	
Formation	Fahliyan	
Risks & Hole Problem	Oil Flow, Tight Hole and Differential Sticking, H2S,Complete Loss, Hole Deviation	
Deviation Control	Single/Multi shot Survey	
Bit Type & Nozzle	Tri Cone 537 & PDC 1665 Matrix body	
Logging	**Cased Hole:** RBT/CCL/GR from 5" Liner	
Testing		
Drilling Fluid Parameters	**Salt Saturated Kcl/Polymer/$CaCO_3$ WBM**	
	MW (PCF)	102 – 104
	FV (Seq/qt)	40 – 56
	PV (CP)	ALAP
	YP (Lbs./100ft^2)	14 – 24
	API FL(CC/30min)	< 3
	Gel 10 sec/10 min (Lbs./100ft^2)	3-7 /5-10
Drilling Parameters	GPM	180-400
	WOB	5 – 15
	RPM	110-170

Table 5-2 AZNS-127 6 – 1/8" Hole Section General Data

- **Formation Sampling**

Sample would be taken from Fahliyan every 2 meters.

NOTE: In addition to this general rule, geologist may decide to take more samples.

- Ensure that drilling tools & equipment are on wellsite & are in serviceable condition.
- Ensure that adequate fishing tools for this section of hole are available.
- Ensure that enough mud material, LCM & CMT are available.

1. Commence drilling 3m new formation, CBU till hole clean, POOH to surface due to L/D bit & chabge BHA.
2. P/U & M/U 6 – 1/8" rock bit & 7" scraper BHA, RIH to 3940^{m} (7" liner shoe) while scrap inside 7" liner prior to take 7" liner cement quality log, CBU, POOH to surface & L/D bit & BHA.
3. Change rams & test same, R/U shooting nipple, wirelie equipment & BOP, pressure test same, M/U CBL/VDL – GR – CCL logging tools, perform 7" liner cement quality log, POOH to surface & L/D wireline logging tools & equipment.
4. P/U & M/U 6 – 1/8" PDC bit with rotary BHA, RIH to bottom while fill up string every 10 stands.
5. Cont. drilling hole section from 3940m TD section @ 4150^{m}.

NOTE: take full suite log base on client's decision (contingency plan #1).

6. Perform CBU, POOH to 3940^{m} (7" liner shoe) & run back to bottom as condition trip, perform CBU, POOH to surface.
7. Perform BOP test as NIOC policy.
8. R/U 5" liner running equipment, M/U 5" float shoe & track, M/U 5" float collar, check floating function, Cont. run about 35 joints 5" liner while fill up string every 10 joints.
9. P/U & M/U 5" liner hanger with top packer, check POB sealing, L/D liner running equipment, RIH 5" liner on 3 – 1/2" & 5" DP's to 7" liner lap @ 3940^{m} while fill up string every 10 stands, perform Circ. as a 1.5 cap of string.
10. Cont. RIH 5" liner on 5" DP's stand to 4150^{m} (5" liner lap will be ±3840^{m}) while fill up string every 5 stands, perform Circ. as 1.5 cap of string meanwhile prepare solution for cement job.
11. Perform 5" liner cement job, release setting tools, POOH to desire top of cement, flush string, POOH 10 stands, perform CBU, POOH to surface while L/D excess 5" DP's as joint by joint & L/D setting tools.
12. P/U & M/U 6 – 1/8" bit & D/O BHA, RIH to bottom, tag cement, D/O cement to 5" liner lap, perform CBU, POOH to surface.
13. R/U 2 – 7/8" DP's handling equipment, P/U & M/U 3 – 1/2" junk mill & 2 – 7/8" DP's joint by joint (±57 joints), RIH to 5" liner lap, D/O POB & dress same, Cont. RIH & D/O inside 5" liner to LDC depth.
14. Perform CBU, POOH to 5" liner lap, perform positive pressure test of 5" liner lap, POOH to surface while L/D 2 – 7/8" DP's joint by joint.
15. P/U & M/U 7" dress mill, RIH to 7" liner lap @ 2855^{m} while dress TBS several times, perform CBU, POOH to surface.
16. P/U & M/U 7 – 5/16" polish mill & RIH to 7" liner lap, dress inside 7" TBS several times, pump Hi-Vis pill & sewwp the well, POOH to surface.

17. R/U 7” liner handling equipment, P/U & M/U 7” tie – back assy. & approximately 150 joints 7” liner tie – back string, P/U & M/U TBS, check POB sealing meanwhile fill up string every 10 joints.

18. R/D 7” liner handling equipment, Cont. RIH 7” liner tie – back string on 5” DP’s to 2855^{m} (7” liner lap) while fill up string every 10 stands, perform Circ. as 1.5 cap of string while prepare cement solution.

19. Perform 7” liner tie – back cement job, sting in, release setting tools (new 7” liner lap @ ±1355^{m}) & POOH to 1200^{m}, flush string, POOH to surface while L/D excess 5” DP’s joint by joint.

20. Perform BOP test as NIOC policy.

21. P/U & M/U 8 – 3/8” bit & D/O BHA, RIH to top of cement to new 7” liner lap @ ±1355m, pump Hi-Vis pill & sweep the well, POOH to surface.

22. P/U & M/U 6 – 1/8” bit & D/O BHA, RIH to 1355^{m}, D/O POB & chase same to 100^{m} below 7” liner lap, perform CBU & POOH to surface.

23. P/U & M/U 7” dress mill, RIH to 1355^{m} while dress TBS several times, perform CBU & POOH to surface.

24. P/U & M/U 7 – 5/16” polish mill & RIH to 7” liner lap, dress inside 7” TBS several times, pump Hi-Vis pill & seewp the well, POOH to surface meanwhile L/D 5” DP’s.

25. R/U 7” liner handling equipment, P/U & M/U 7” tie – back assy. & approximately 130 joints 7” liner tie – back string, P/U & M/U TBS, check POB sealing meanwhile fill up string every 10 joints.

26. R/D 7” liner handling equipment, Cont. RIH 7” liner tie – back string on 5” DP’s to 1355^{m} (7” liner lap) while fill up, perform Circ. as 1.5 cap of string while prepare cement solution.

27. Perform 7” liner tie – back cement job, sting in, release setting tools (new 7” liner lap @ ±100^{m}), flush string, POOH to surface.

28. P/U & M/U 8 – 3/8” bit & RIH to top of cement, D/O cement to new 7” liner lap @±100^{m}, POOH to surface.

29. P/U & M/U 6 – 1/8" bit & D/O BHA, (P/U & M/U required 3 – 1/2" DP’s Joint by Joint) RIH to 100^{m} (7" liner lap), D/O POB, RIH to LDC depth of 7" tie back string while clean out inside 7" tie back string, D/O & LDC & plug, perform CBU to sweep the well, Cont. RIH to LDC depth of first 7" tie – back string, while C/O inside 7" tie – back string, D/O LDC & plugs & chased same to 5" liner lap, perform CBU, POOH to surface.

30. P/U & M/U 4 – 1/8" mill and D/O BHA and RIH with 2 – 7/8" DP's Jt by Jt and 3 – 1/2" DP's to 5" liner lap, D/O POB & dress same, Cont. RIH to LDC depth while C/O any cement, perform CBU, POOH to surafce.

31. Change rams & test same, R/U shooting nipple, wireline equipment & BOP, pressure test same, M/U RBT – GR – CCL logging tools, perform 5" liner cement quality log, POOH to surface & L/D logging tools & wireline equipment.

32. Perform BOP test as NIOC policy.

33. HPJSM, R/U wireline & pressure test wireline BOP, M/U CCL & Gummy gun & RIH CCLwith same, POOH & R/D.

34. R/U for perforation, RIH string (Firing head & shooting ,CCL+Guns), correlate depth, shooting the gun, POOH string & RIH again to perforate all the depths then R/D (8 runs had been considered).

35. P/U & M/U 6 – 1/8" bit & 7" scraper BHA, RIH to 5" liner lap while scrape 7" production packer setting area, dispalce the well with completion fluid, POOH to surface meanwhile L/D all tubulars.

36. Perform BOP test as NIOC policy.

Completion Section

1. R/U completion running equipment, run 3 – 1/2" * 4 – 1/2", 10K completion string as per program, R/D completion running.

2. Install TWCV in tubing hanger, N/D BOP stack, N/U X.M.T, perform required test as per program.

3. Secure the well by set BPV in tubing hanger, close all X – mas tree valves except BMV, and **Release the Rig**.

4. Retrieve BPV, R/U slickline, set protection sleeve, retrieve prong & plug from NO GO landing nipple, POOH.

5. RIH with CT to bottom, displace the well with diesel or nitrogen & observ well, POOH.

6. R/U SL, RIH & retrieve protection sleeve, POOH, R/D SL.

7. Flow the well as pre – acid clean up till acceptable BSW & reading stable flow rate on 2 different choke size for 72 hrs.

8. R/U & RIH S.L, set protection sleeve, POOH & R/D S.L

9. RIH with C/T & perform acid wash as per program, POOH to surface. RIH w/SL and retrieve protection sleeve

10. Flow the well as post – acid clean up till acceptable BSW & reading stable flow rate on 2 different choke size for min 96 hrs.

Note 1: The time is only for flowing duration (R/U, R/D & preparation time is excluded).

Note 2: Acidizing & post-acid clean-up maybe perform as rig less job upon contractor request & client approval but in case of any failure in well integrity the contractor should provide drilling rig & required services in own responsibility & fix the problem.

11. RIH with S/L, set protection Sleeve, POOH & R/D SL.

12. RIH with C/T to bottom & displace the well with diesel, POOH.

13. RIH with S/L, install plug & prong into NO GO landing nipple, POOH.

14. RIH with S/L, retrieve protection sleeve.

15. Secure the well by set BPV in TBG Hanger, close all X-mas Tree valves except BMV.

Contingency Plan:

Contingency plan #1: Change ram, install shooting nipple, R/U wireline logging equipment & BOP, pressure test same, perform logging (full suite/CDR/GR logs), R/D logging equipment. P/U & M/U 6 – 1/8" bit & BHA, RIH to bottom as condition trip, pump Hi-Vis pill & sweep the well, spot 100^{bbl} Hi-Vis pill, POOH to surface.

6. Directional Drilling Operation

6.1 Hydraulic Set Whipstock System Operation Procedure

SCOPE

The following procedures are intended to provide concise but definitive steps to follow in operations involving the milling of lateral hole sections from a parent well using the Hydraulic Set Whipstock System.

The Hydraulic Set Whipstock System is a hydraulically set whipstock that requires a hydraulic force to activate the packer assembly, setting itself in the casing.

Milling a window in casing using a whipstock usually consists of one run for setting the Whipstock in place and milling and completing the window through the casing wall and into the formation, while dressing the window to eliminate drag.

EQUIPMENT REQUIRED

Hole Preparation Run Equipment

- A Full Gauge Drift Mill
- Casing Scrapper
- Cross Over Sub
- String Stabilizer
- Pony Drill Collar or 1 Joint of Drill collar
- String Stabilizer
- Drill Collars
- Drilling Jar
- Drill Collars
- HWDP's
- Balance Of Drill Pipes To Surface

The stiffness of this assembly will serve to identify any problem doglegs or restrictions in the hole that may hamper running of the whipstock assembly.

Whipstock Run Equipment

- Hydraulic Set Whipstock
- Tri-Mill (Optional Bi-Mill or a combination of Rugby/Back/Window Mill)
- Running Tools
- Nuzzle plug Assembly
- Flex Joint (1 joint of HWDP)

- Universal Bottom Hole Orientation Sub (UBHO sub) or MWD
- Pup joints (for space-out if necessary)
- Tool kit

PRE-OPERATIONAL PROCEDURES

Pre-Run Checklist

As part of the Pre-operational Procedures, all whipstock jobs should include a completed Pre-Run Preparation Checklist.

Pre-Run Preparation Checklist

† The Packer slip is in place, is in good condition, and is the correct size for the casing size and weight and the Allen screws are tight.

† The shear pins are properly installed and in good condition and the Number of Shear pins are enough.

† The shear screw is the proper size, and it will make up into the Whipface.

† The shear screw will fit through the Rugby/Lead Mill hole.

† Running Tools, Nuzzle plug Assembly and Hydraulic Lines and Fittings are in good condition and properly installed.

† All mills are on location and the OD's are correct for the casing's size/weight. (Check mill OD's with a gauge ring that is OD + 1/32").

† All downhole equipment has been properly measured and OD's, ID's, and lengths have been properly recorded.

† The slots on the gyro match and are compatible with the lug assembly on the orientation sub.

† The survey tools will pass through all drill string tool joints and drill collar ID's.

† All rotary connections are compatible.

† All rotary shouldered connections are made up to the recommended torque.

† Enough Pump Pressure is available on Site for setting the Packer.

Making the Hole Preparation Run

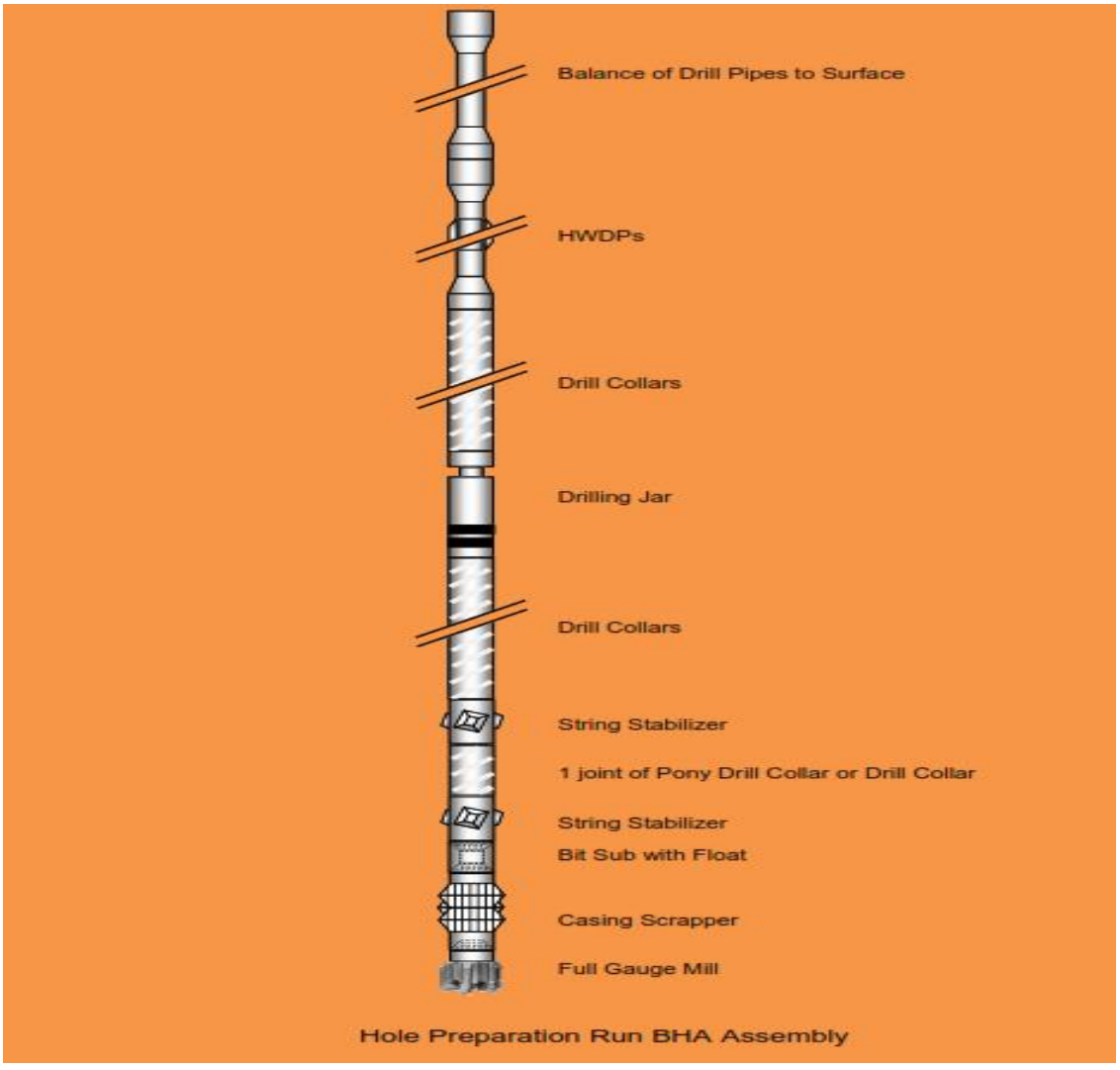
1. Make a run to ensure that the casing is clean and there is smooth and clean place for setting the packer. The run should include the following (from the bottom):

- A Full Gauge Drift Mill
- Casing Scrapper (Do not run casing scrapers across open perforations.)
- Cross Over Sub (Bit Sub with Float)
- String Stabilizer
- Pony Drill Collar or 1 Joint of Drill collar
- String Stabilizer
- Drill Collars
- Drilling Jar
- Drill Collars
- HWDP's
- Balance of Drill Pipes To Surface

2. Run this assembly in the well to the Packer setting depth.
3. Tally the pipe while running in or out of the well to get an accurate measurement to the Packer setting depth.
4. Record the measurements to ensure that:
   - the whipstock can be run and set at the desired depth
   - the depth measurement to the Packer setting depth is accurate
   - the casing wall is clean where the whipstock is to be set

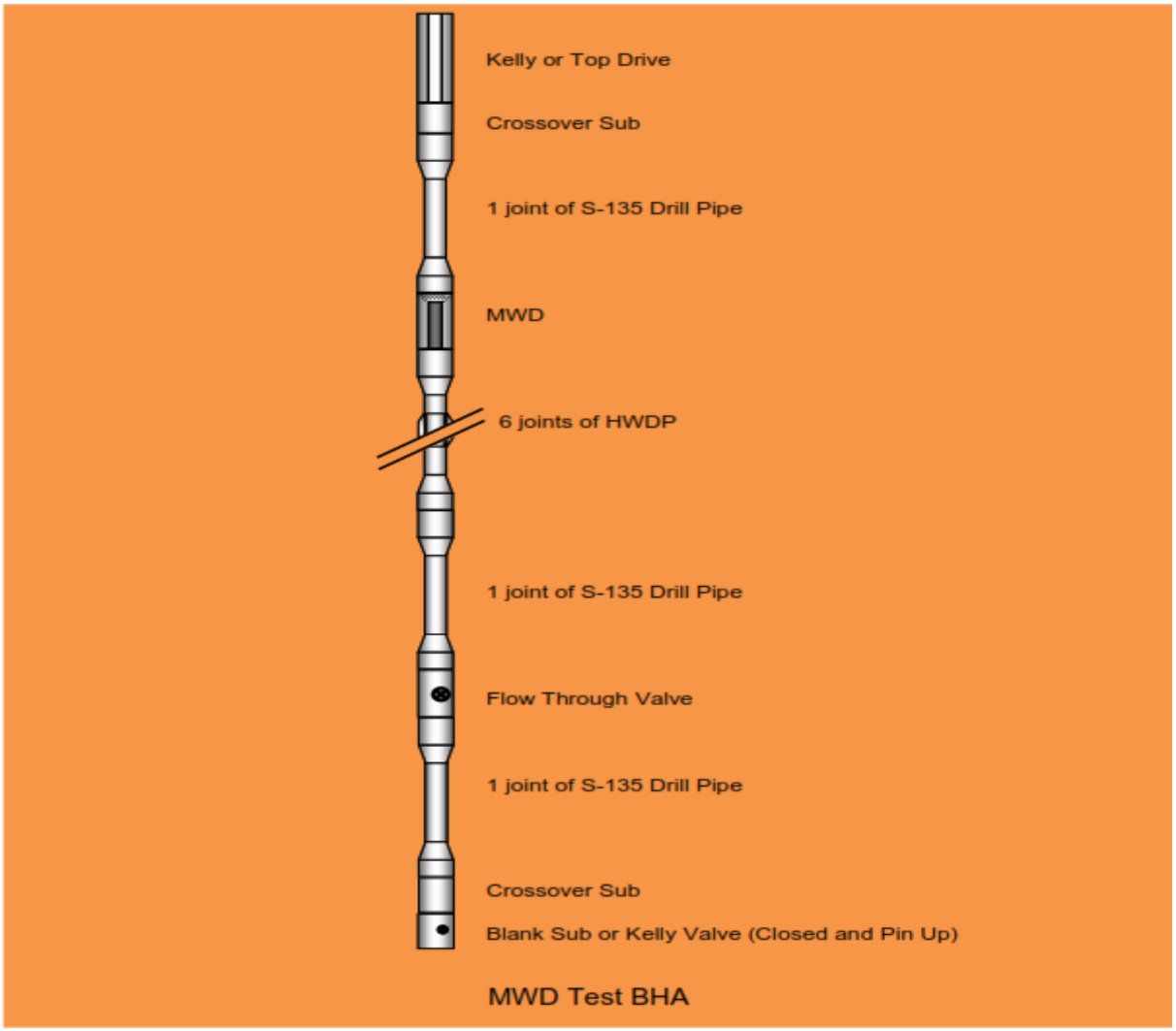
NOTE: Scrape Packer setting depth twice.

Note: It is very important to get an accurate measurement for the packer setting depth so the Whipstock will be oriented as to setting depth as possible. Using this measurement facilitates proper placement of the Whipstock.



Test the MWD tool
prior, to running the assembly in hole.

1. Pick up the following assembly:
   - o Blank Sub or Kelly Valve (valve to be closed and run pin up)
   - o Crossover Sub
   - o One joint of S-135 Drill Pipe
   - o Flow Through Valve
   - o One joint of S-135 Drill Pipe
   - o 6 joints of HWDP
   - o MWD
   - o One joint of S-135 Drill Pipe
   - o Crossover Sub
   - o Kelly or Top Drive
2. RIH to top of Drill Pipe. With top drive or Kelly connected, record the flow rate required to obtain a survey. The Flow Through Valve will remain open.



Preparing the Milling Equipment

1. Lay the whipstock on the catwalk, face up, with the top of the whipstock pointing toward the rig.

WARNING
Never use a nylon sling to lift or move a Whipstock because nylon may be cut. Always us a wire sling or wire rope to lift or move a Whipstock.

2. Complete the whipstock assembly, checking all OD's, ID's, and descriptions.

3. Pick up a joint of S-135 Drill Pipe (FLEXJOINT) and make up the **Flow Through Valve** to the top of the drill pipe and make up the Mill Assembly covered with Running Tool to the bottom of Drill Pipe.

Note: If the window is exiting through two or more strings of casing or into a known hard formation, exclude the Watermelon Mills (in the case of using combination mills), using only the Rugby Mill below the single joint of DP. Depending on the casing size and grade, more than one Rugby Mill may be required to make multiple exits.

Once the casing string(s) has been exited into the formation, pull out of the hole. Make up the Back Mill (and Window Mill) in the assembly as described above and run in the hole, reaming through the window with the Watermelon Mills.

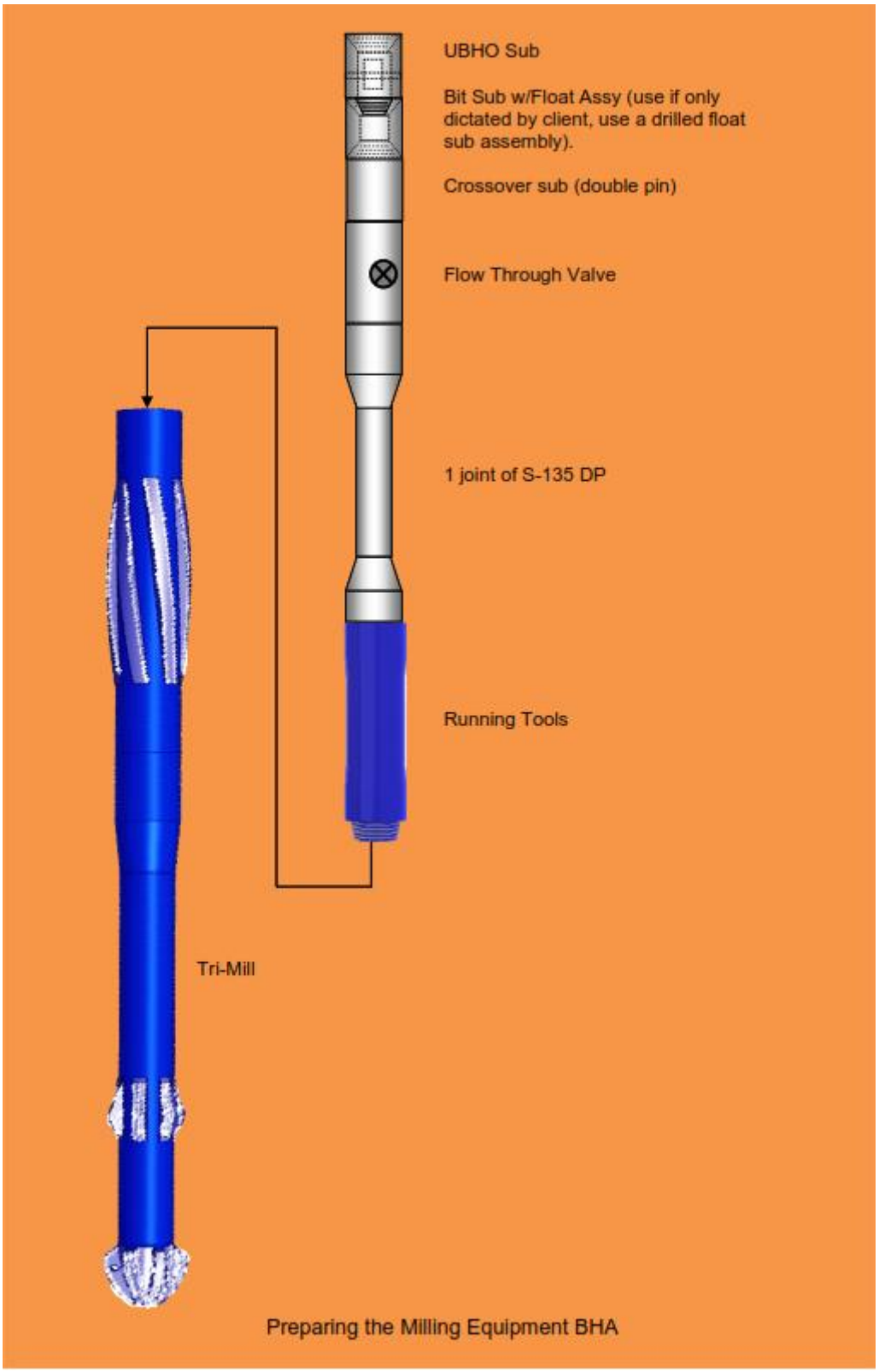
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4. Make up one joint of S-135 Drill Pipe (with cone type DP strainer installed) to the top of Flow Through Valve and make Up MWD to the top of Drill Pipe. (or Make up UBHO Sub to the top of Flow Through Valve).

5. If a bit float have to be used, place the bit float sub above the Flow Through Valve and beneath the UBHO sub.

6. Make up all connections to the recommended torque.

7. Ensure that everything is properly made up before continuing (see Figure).



Preparing the Milling Equipment BHA

There is a hole in a one of Rugby/Lead Mill blades for screwing the shear screw and attaching Rugby Mill to the Whipface. Mark the opposite blade of the blade with shear

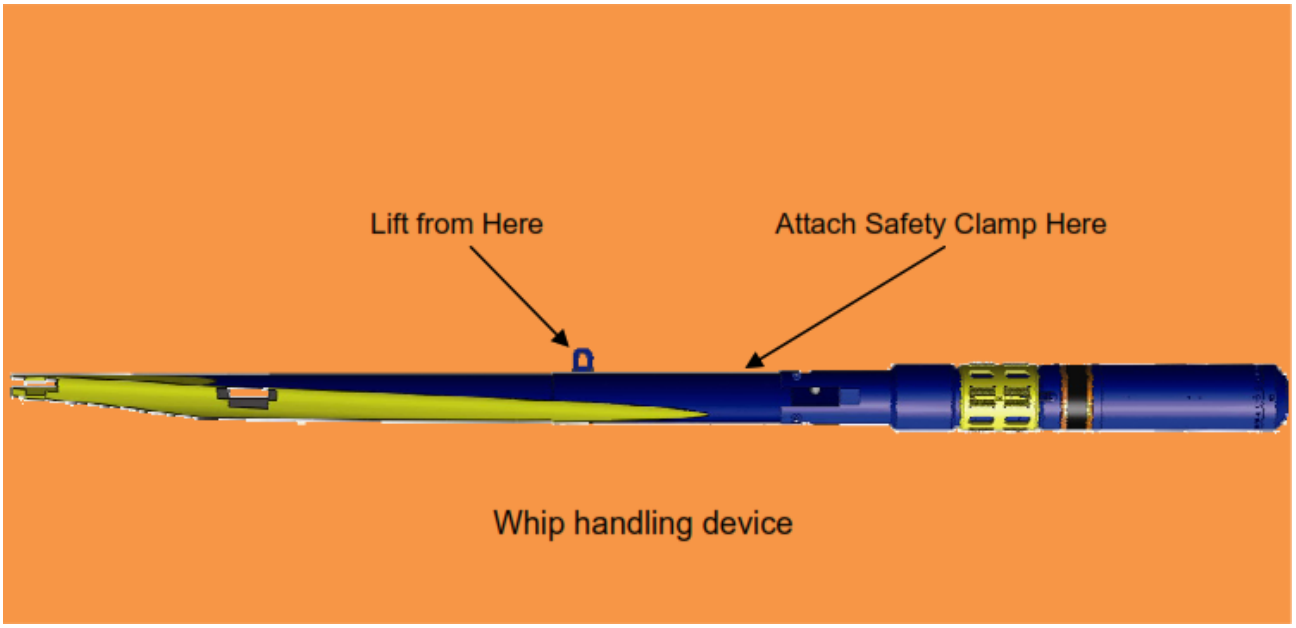
screw hole on the Rugby Mill and scribe a line from this blade, up the Back Mill (and Window Mill), Bit Sub, and the single joint of HWDP to the top of the UBHO Sub.

9. Have the survey company representative back out the set screws that hold the internal key in the UBHO Sub, and rotate the key until it is aligned with the scribed line. After it is in position, tighten the setscrews to lock the key in alignment with the scribed line.

10. Run the Mill assembly through the BOP stack and into the casing liner/hanger (being careful not to damage the wear bushings or strike any closed BOP rams) to ensure that the whipstock's OD is compatible with the drift diameter of the casing. Remove wear bushing if necessary.

11. Pull the assembly out of hole.

12. Secure a safety clamp around the whipstock body about halfway down the face of the Whipface while the whipstock is lying on the pipe deck/catwalk.



13. Pick up the Whipstock with tugger from the catwalk. A whip handling device is provided which is bolted onto the back of whipstock. This is to be used as a false rotary to support the assembly in the Rotary Table when making up the mill to whip via the shear screw. Assist with a crane as necessary.

14. Set Whipstock in the rotary table (rams shut).

15. Lower the whipstock into the mouse hole (or other suitable work hole on the drill floor), allowing both the safety clamp and the winch/tugger line to support the whipstock's weight.

Note: *Larger Whipstocks (9⅝ to 20 inches) are usually too big for the mouse hole and must be set in the rotary to attach to the Mill assembly. Properly sized rotary slips must be used. The 13⅜" whipstock will not fit through the standard rotary (10⅝" ID) and the rotary bushing, in this event must be pulled. Use of an API bowl and slips to hold the whipstock in the rotary is required.*

CAUTION

- **Close the BOP Shear/Blind ram when working on Whipstock.**
- **Do not drop anything in the wellbore when working over the wellbore.**
- **Use extreme caution when handling the whipstock, setting drill collar slips, safety clamps, etc. to prevent dropping the whipstock in the hole.**
- **Run the draw works in low. Pick up and set down the whipstock easily in the slips to prevent damage or premature setting.**
- **Ensure the lead tongs have most of the slack out of the snub line to prevent jerking against the back-up tongs.**

16. Put a light amount of thread compound on the shear screw, adjust Rugby Mill hole and Whipface and connect the whipstock assembly to the mill assembly with shear screw.

Shear screws with a **higher** value should be considered when:

- running the whipstock in a highly deviated hole
- heavy mud is in the well
- other wellbore restrictions such as a liner hanger may be encountered

Shear screws with a **lower** shear value should be considered when:

- whipstock is being set shallow (less than 1500 m)
- running and setting Whipstock with coiled tubing

17. Attach the Hydraulic Line to the connection on the bottom of the Milling Assembly and back of whipstock.

18. Fill Whipstock and Mill Assembly / Running Tool with Hydraulic Oil, allow air to bleed out, fit piston ensuring the top of piston is below the top running tool box connection.

19. Run the Whipstock & Mill Assembly below the rotary table to the top of UBHO sub, Break the connection and top-up with Hydraulic Oil.

20. Pick up the Milling / whipstock assembly.

21. Install the rotary table cover.

22. Remove the safety clamp and whip handling device which is bolted onto the back of whipstock.

23. Remove the rotary table cover.

24. Lower the whipstock slowly into the well, setting the rotary slips on the DP.

25. Check the orientation of the key in the UBHO sub, make any adjustments necessary, and tighten the setscrews again.

Note: For safety purposes, install a drill pipe wiper to prevent dropping the Allen wrench in the hole.

CAUTION

Run the draw works in low. ***Pick up and set down the whipstock gently in the slips to prevent damage or premature setting. Make sure the lead tongs have most of the slack out of the snub line to prevent jerking against the back-up tongs.***

RUNNING PROCEDURE

Running and Orienting the Whipstock

1. Lower the whipstock assembly slowly into the wellbore and **do not pump** anything in the well to prevent premature setting of the whipstock.

   Do not alter the orientation of the UBHO sub to the Whipface's face when making the first connection. Do not let the UBHO sub turn or move, or the direction of the milled window could be affected.
2. Refer to dogleg checklist for doglegs greater than 4 degrees. Watch the weight indicator carefully and record up-weights regularly.
3. Monitor trip-tank for proper fill up.
4. Run the whipstock slowly, like a packer assembly, at a rate of 2 to 3 minutes per stand.
5. Do not turn string.
6. Do not Pump in the Well (unless an open Flow Throgh Valve is Run above the Running Tools).
7. Fill the drill string every 5 strands.

Proceed slowly and with caution while running the whipstock assembly through the BOPs, wellhead, liner top, and any other restrictive sections to prevent damage or premature setting. The difference between the whipstock OD and the casing ID is minimal, making it similar to a piston, so lower the assembly very slowly in these small bypass areas. DO NOT PUMP IN THE WELL.

8. Run slowly into the hole until approximately 30 feet off the bottom.
9. Slowly move the whipstock assembly up and down from 5 to 20 feet to remove all trapped torque from the drill string.
10. Record up and down free weights.
11. Run survey tools on wireline to the UBHO sub if using a survey.
12. Take survey readings
13. To orient the whipstock, rotate the drill pipe to the right in ¼ turn increments (no more that ½ turn at a time) until the whipstock assembly is oriented in the correct position.
14. Take another survey, making additional corrections as needed and lowering 4 to 5 feet after each increment.

CAUTION
A crooked hole may require several readings and corrections to remove all of the trapped torque from the drill string. Work slowly and carefully in order to achieve proper orientation.

CAUTION
This Model of Whipstock, because it is hinged, must be set within 30° on either side of the low side to prevent the whipstock's whipface from falling across the wellbore.

15. Continue taking readings until 3 sequential reading are within a 5° range before assuming the direction is correct.
16. After reaching and verifying the proper direction, call a meeting with the customer, driller, and survey/MWD company representative to review the orientation of the whipstock's Whipface once more.
17. Continue the procedure if everyone is in agreement.
18. POOH the survey tools and make a mark on the rotary table and the Drill Pipe, make up the Kelly (or Top Drive), DO NOT ALTER THE ORIENTATION WHILE MAKING UP THE KELLY.
19. Lower the whipstock assembly slowly to the setting depth.
20. Start pump and increase pressure until the packer shear pins are sheared and the packer sets in place (at 1200-1400 psi), hold pressure for 10 minutes to confirm the setting).

Note: In case of Using MWD and Flow Through Valve, increase pump rate until the Flow Through Valve shear pins are sheared and the valve closes, then continue pumping until the packer shear pins are sheared and the packer sets in place.

21. Slack off 10-15klbs and over pull 10-15k lbs to prove packer is set.
22. Pick-up 35,000-55,000 lbs (Depend on whipstock size) over pull, continue over pull on the screw as required to initiate the shear. A sharp drop in the pipe occurs when the shear screw shears.
23. Raise the drill string approximately 1 to 2 feet.
24. Set the drill string back down to make sure the shear screw has sheared. Repeat if necessary.
25. Mark the depth on the pipe each time. It should read the same each time.

26. Set the pipe back down with approximately 20,000 lb of pressure to ensure that the Whipface is over against the casing wall.

27. Raise the drill string so that the Rugby Mill is 5 to 10 feet above the top of the whipstock.

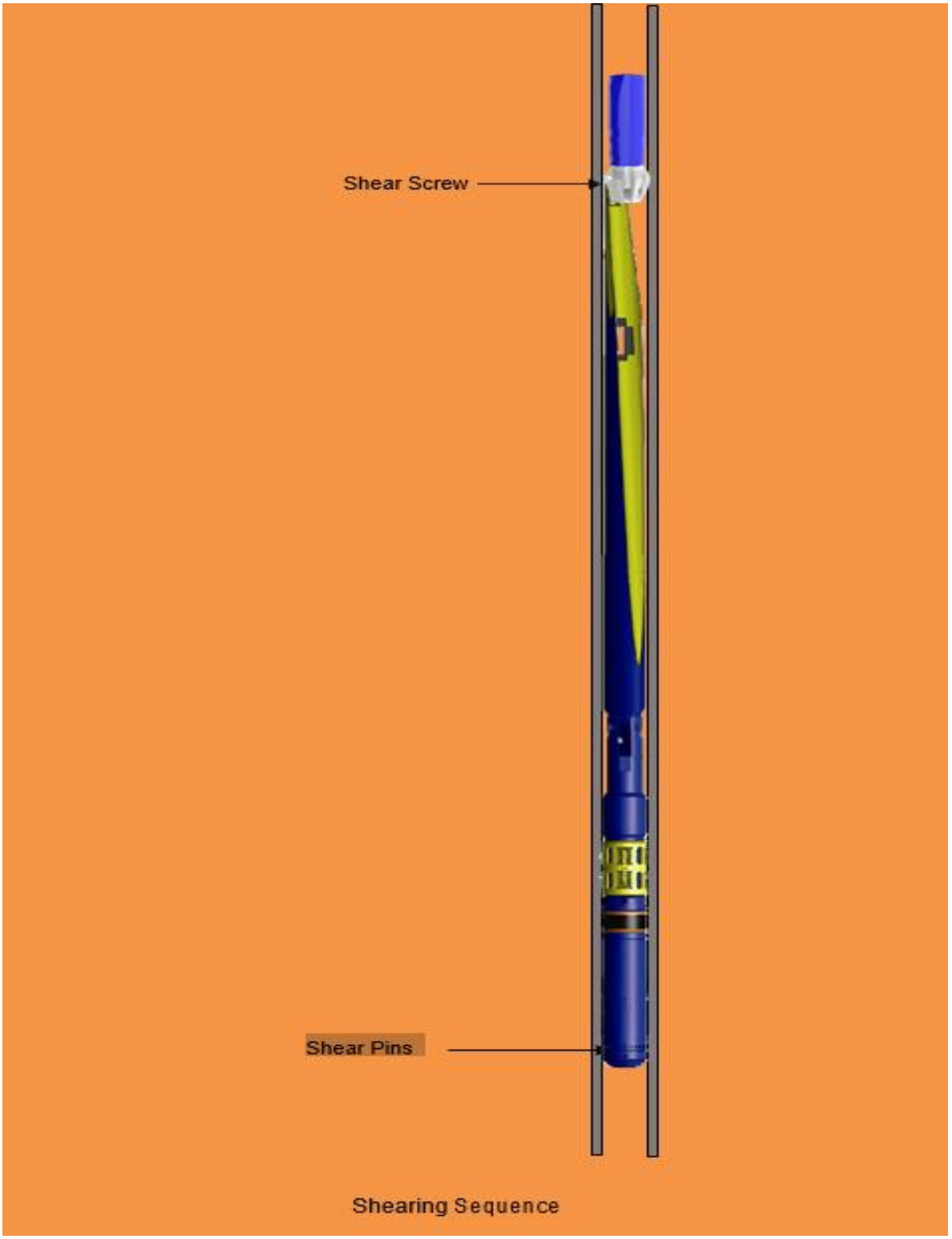
28. Lay down the necessary drill pipe to pick up the kelly.

29. Start pump with maximum allowable GPM (According to the standpipe pressure). Engage with Whipface with Reduced RPM (20-30). Monitor pressure drop due to breaking Nozzle plugs.

Note: Do not Mill and work on Nozzle plugs more than One min with low GPM which leads to Mill burning.

30. Displace the Mud in the well (if needed) and check mud properties for proper rheology for steel removal.

31. The milling operation can now begin.



Shearing Sequence

Running Milling Assembly and Opening the Window

This section assumes that the Rugby Mill has been sheared from the whipstock and raised 5 to 10 feet above the whipstock.

NOTE: Install Ditch Magnets

1. Make up the kelly to the drill string.

2. Record string weight up and down.

3. Engage the pumps, slowly increasing to the planned circulation rate, and record pump pressure and rate.

4. Engage the rotary and slowly increase the speed between 20 and 40 RPM.

5. After the rotary has stabilized, slowly increase between 50 and 75 RPM (depending on the mill size).

6. Record the following information:
   - Free rotary torque
   - Pump pressure
   - Pump rate
   - Free weight – up and down (string weight with zero weight on mill)

7. While rotating, slowly lower the drill string until the Rugby Mill contacts the whipstock and the casing wall, causing an immediate and substantial increase in rotary torque.

8. Mark the kelly. This is the starting point for the Rugby Mill run.

9. After making the initial mark on the kelly, raise the Mill 3 to 5 feet.

10. While rotating, slowly lower the drill string again to see if the increase in rotary torque is indicated at the same place. Work slowly and carefully; repeat if necessary.

11. After marking the kelly with the correct starting point for the Mill, begin milling, using light weights on the Mill (1,000 to 3,000 lb depending on the size of the mill and grade of the casing). As the window milling process begins, an increase in torque and decrease in weight will be indicated.

12. As the torque increases, increase the rotary speed to the recommended RPM (based on the mill size).

13. Continue milling with a constant rotary torque and weight (1,000 to 3,000 lb) on the mill. Additional weight may be required if the rate of penetration (ROP) begins to slow.

Use recommended parameters as follows.

Casing Size (in)	Weight on Mill (klb)	Milling RPM
9 ⅝”	2 – 15	60 – 120
7”	2 – 12	60 – 120
5”	1 – 7	60 – 120

14. Monitor returns for cuttings during milling operations and **pump high viscosity sweeps** as needed to remove cuttings from the well. This practice is especially important if the window is being milled in a lateral section.

15. The Window Mill run is completed after milling down the Whipface’s face into the formation AND when the Dress Mill is centered at the bottom of the window.

16. Ream back and forth through the window until no signs of additional torque or drag are indicated.

17. Stop the rotary and slide the Mill assembly through the window several times. If any weight or drag is indicated, start the rotary and ream back and forth through the window until weight or drag is eliminated.

18. Stop the rotary and check for weight or drag again. Repeat this process as necessary.

19. Once the window is complete, stop the rotary, **clean the hole of cuttings with high viscosity sweeps**, and pull out of the hole without rotating.

20. Closely inspect and gauge all mills for wear using a gauge ring of OD + 1/32"

21. Determine the estimated size of the window.

22. Review the window requirements with the customer to determine if the Elongation Run is required.

Note: The Elongation run is recommended if:

- *You have run a bi-Mill Assembly for opening the Window and:*
- *The customer requests it.*
- *PDC bit will be used through the window for additional drilling*
- *Drill collars and stabilizers will be used through the window for additional drilling*
- *Liner will be installed through the window*
- *Double bent sub and PDM assembly will be used through the window for additional drilling*
- *Unusual wear is indicated on the Rugby Mill, resulting in an under-gauge hole*

CAUTION

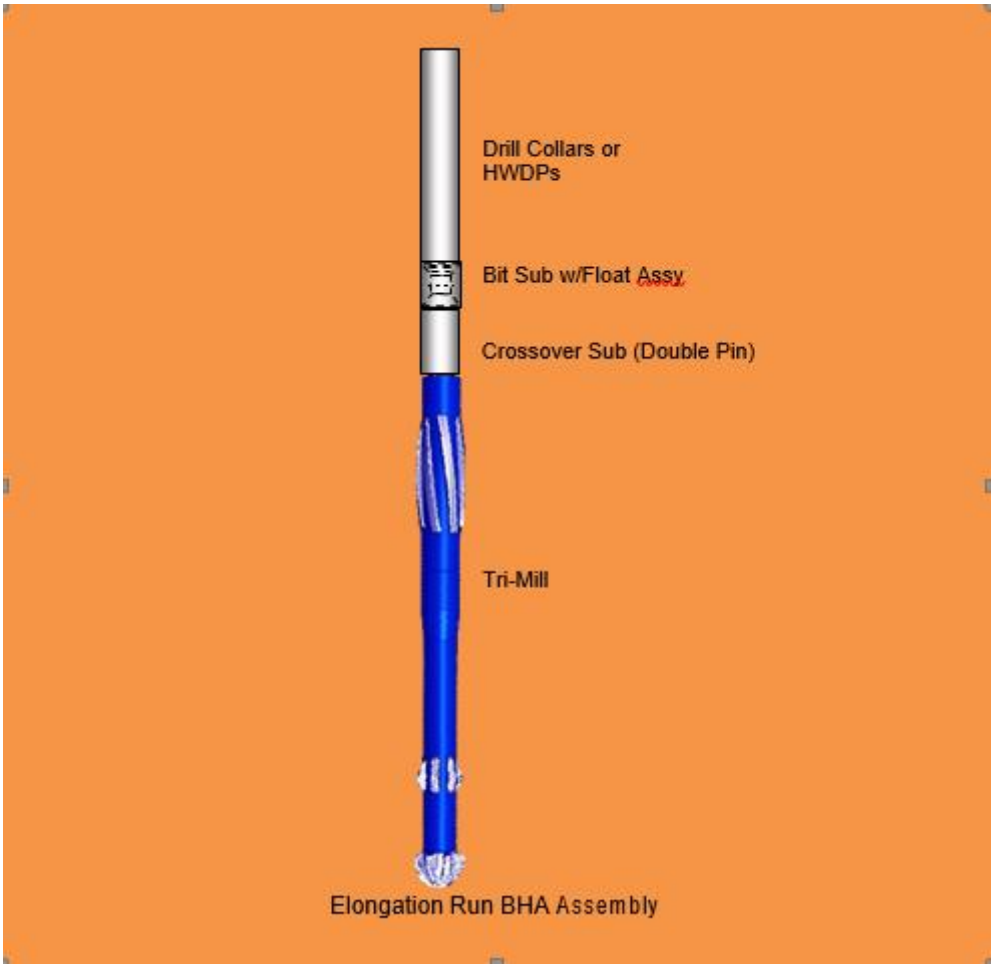
When using a 3° whipface whipstock, do not run the Elongation assembly until AFTER the casing is fully exited. Otherwise, excessive torque will be encountered as the two Watermelon Mills begin elongating the top of the window.



Elongation run

1. Assemble the Elongation run components
   - Balance of drill string
   - One or more drill collars or heavy weight
   - Bit Sub (minimum length, bored for float assembly)
   - Crossover sub (double pin, if Bit sub is used)
   - Tri-Mill

Note: Be sure to eliminate the one joint of drill pipe above the Bit Sub and replace it with one or more drill collars/heavy weight to provide "stiffness" to the milling assembly.



Elongation Run BHA Assembly

2. Lower the Elongation run assembly into the hole, stopping 5 to 10 feet above the whipstock.

3. Make up the kelly to the drill string.

4. Engage the pumps and slowly increase to the desired circulation rate.

5. Engage the rotary and slowly increase to between 50 and 75 RPM (depending on the mill size).

6. Record the following information:
   - Free rotary torque
   - Free weight – up and down (string weight with zero weight on mill)
   - Pump pressure
   - Pump rate

7. While rotating, slowly lower the drill pipe string and ream back into the window.

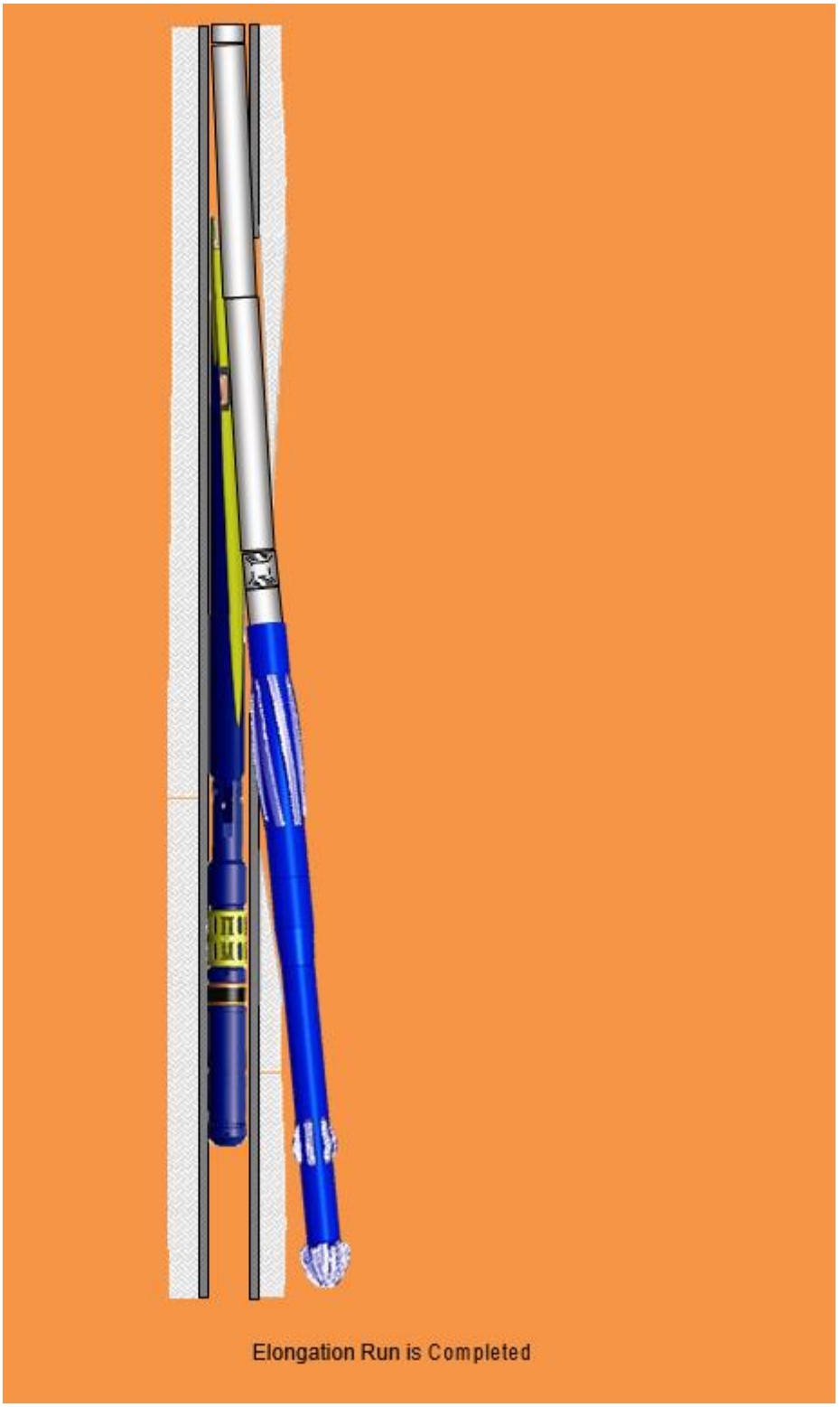
8. As the torque increases, increase the rotary speed to the recommended RPM (based on the mill size).

9. Continue milling with a constant rotary torque and weight on the mill.

10. Monitor returns for cuttings during milling operations. Pump high viscosity sweeps as needed to remove cuttings from the well.

11. Continue milling down the Whipface, elongating the milled window at the top with the Dress Mill.

12. The Elongation run is complete (unless directed otherwise by the customer) when the Dress Mill is centered at the bottom of the window.



13. While rotating, ream back and forth through the window until no additional weight or drag is indicated.

Note: If using stabilizers for additional drilling through the window, it is recommended to drill enough hole into the formation to accommodate the stabilizers. The additional rathole should allow the stabilizers to be in the formation BEFORE drilling commences. Rotating the stabilizers or drill bits while on the Whipface can damage the Whipface and the window.

14. Stop the rotary and slide the Mill assembly through the window several times. If any weight or drag is indicated, start the rotary and ream back and forth through the window until weight or drag is eliminated.

15. Circulate the hole clean with **one or more high viscosity sweeps** to remove all cuttings from the well.

16. When circulation is complete, stop the rotary and pull out of the hole without rotating.

17. Inspect and gauge all mills using a gauge ring of OD + 1/32", to determine wear to the mill and to estimate window/hole size. Review signs of wear with the customer. Repeat Run #3 if the hole appears to be undersized.

18. Lay down the Elongation Run assembly.

Abnormal Conditions

Whipstock stuck

- Try to release the whipstock with 20000klb overpull in 7" hole and 30000klb over pull in 9-5/8" hole.
- Close an annular preventor hydrill and apply 500-1000 psi pressure in annulus to remove casing collapse.
- Apply weight on bit.
- If no success, shear screw and pull out the whipstock out of hole with fishing tool later.

No success in passing the whipstock through liner lab

- Come back to neutral point and turn string 1/4 round in clockwise direction and try to pass through liner lab again.

The packer does not set with designed setting pressure

- Release the pressure and apply the pressure again.
- Pull out of hole to surface and check the tools.

Pressure drop during the milling operation

- Twisting off the string or mill:
- In this case the torque will be constant and progress will be stop.
- Pressure drop due to string washing out:
- In this case the standpipe pressure will decrease gradually.
- Pressure drop due to breaking the nozzle plugs:
- In this case the pressure decreases suddenly and no change in other parameters will happen.
- Pressure drop due to washing out the nozzles:
- In this case the standpipe pressure will decrease gradually.

Note: In all the above items pulling the string out of hole to surface and checking the tools will be necessary.

No progress during milling

- Change the drilling parameters, for example increase or decrease RPM, increase WOB, decrease GPM.

- If you observe no progress in 2 hours, pulling the string out of hole to surface will be necessary.

Observing complete loss during milling

- Don’t continue milling till curing the complete loss.
- If the client insists in continuing the operation, due to working with low GPM, stop the milling operation for 30 minutes after each hour of milling to cool the mill and prevent from thermal cracking.

POH string due to failure of whipstock:

- Do not shock the string while put and pick up slips
- Do not shock the string while wrenching
- Ensure that the rotary table is locked for prevent of rotating string
- Pick up each stand in 5 to 7 minutes
- Disconnect the connections with wrench and air wrench.
- Be careful for 5” and 7” whipstocks , because it will has more swing in larger casing size

Caution: do not use rotary table for disconnect the connections

7. BHA Program

7.1 8 – 1/2" Hole Section

BIT SIZE (inch)	8-1/2"	Directional BHA Configuration to Drill 8-1/2" Hole Section (PDC Bit)				DC	OD
Depth In/Out (m)	3055 - 3940					DC-1	6-1/2"
HOLE INCLINATION (deg)	0						
MUD WEIGHT (pcf)	85						
B.FACTOR	0.827						
Weight in Air (lb)	21,393						
Bouyed Weight (lb)	17,682						
Bouyed Weight Below Jar (lb)	15,815						

SCHEMATIC	DESCRIPTION	QUANTITY (JT)	OD (in)	ID (in)	CONNECTION	LENGTH (m)	
	5" Drill Pipe	to surface	5	4 1/2	4-1/2" IF Pin x Box	-	
	5" Heavy Weight Drill Pipe	3	5	3	4-1/2" IF Pin x Box	27.94	271.89
	6 -1/4" D. Jar	1	6-1/4	2-1/4	4-1/2" IF Pin x Box	5.83	243.95
	5" Heavy Weight Drill Pipe	24	5	3	4-1/2" IF Pin x Box	219.36	238.12
	6-3/4" NMDC	1	6-3/4	3	4-1/2" IF Pin x Box	9.14	18.76
	6-3/4" MWD Sub	1	6-3/4	2-1/2	4-1/2" IF Pin x Box	0.93	9.62
	6-3/4" Float Sub	1	6-3/4	2-1/4	4-1/2" IF Box x Box	0.42	8.69
	6-3/4" PDM	1	6-3/4	-	4-1/2" Reg B x 4-1/2" IF B	8	8.27
	8-1/2" PDC Bit	1	8-1/2	-	4-1/2" Reg. P	0.27	0.27

Figure 7-1: 8-1/2" Directional Drilling BHA

7.2 6-1/8” Hole Section

BIT SIZE (inch)	6-1/8"	BHA Configuration to C/Out Inside 7" Liner	Collar	OD
Depth In/Out (m)			DC-1	4-3/4"
HOLE INCLINATION	0			
MUD WEIGHT (pcf)	103			
B.FACTOR	0.790			
Weight in Air (lb)	29,642			
Bouyed Weight (lb)	23,410			
Bouyed Weight Below Jar (lb)	16,591			

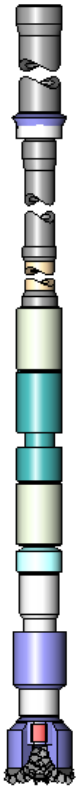
SCHEMATIC	DESCRIPTION	QUANTITY (JT)	OD (in)	ID (in)	CONNECTION	LENGTH (m)	
	5" Drill Pipe	-	5	4.276	4-1/2" IF Pin x Box	-	
	XOS	1	4-3/4	2.25	3-1/2" IF P x 4-1/2" IF B	1	1094.41
	3-1/2" Drill Pipe	90	3-1/2	2-7/16	3-1/2" IF P x B	864	1093.41
	3-1/2" Heavy Weight Drill Pipe	6	3-1/2	2-1/4	3-1/2" IF P x B	54.84	229.41
	4-3/4" Drill Collar	3	4-3/4	2.25	3-1/2" IF P x B	27.42	174.57
	4-3/4" Jar	1	4-3/4	2.25	3-1/2" IF P x B	8	147.15
	4-3/4" Drill Collar	15	4-3/4	2.25	3-1/2" IF P x B	137.1	139.15
	4-3/4" Bit Sub W/ Float Valve	1	4-3/4	2.25	3-1/2" Reg. P x 3-1/2" IF B	0.75	2.05
	Junk Sub	1	4-1/4	1-1/2	3-1/2" Reg. B x P	1	1.3
	6-1/8" MT Bit	1	5-7/8	-	3-1/2" Reg. P x B	0.3	0.3

Figure 7-2: 6 – 1/8” C/Out BHA

BIT SIZE (inch)	6-1/8"	BHA Configuration to Drill 6-1/8" Hole Section (PDC Bit)	Collar	OD
Depth In/Out (m)	3940 - 4150		DC-1	4-3/4"
HOLE INCLINATION	0			
MUD WEIGHT (pcf)	104			
B.FACTOR	0.788			
Weight in Air (lb)	29,642			
Bouyed Weight (lb)	23,350			
Bouyed Weight Below Jar (lb)	16,548			

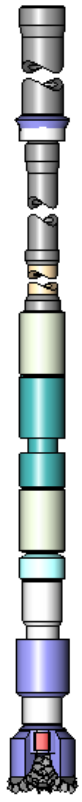
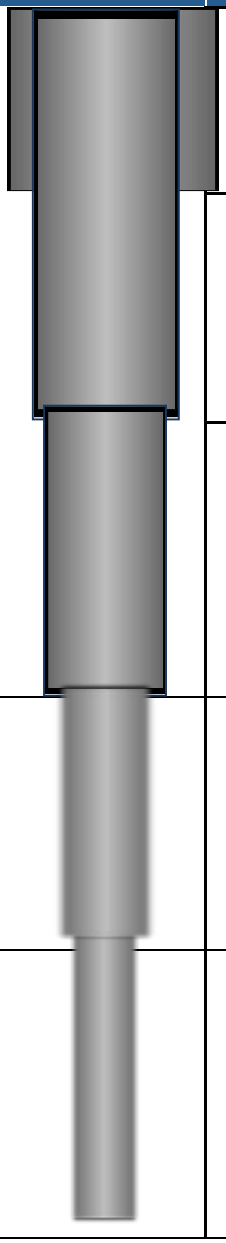
SCHEMATIC	DESCRIPTION	QUANTITY (JT)	OD (in)	ID (in)	CONNECTION	LENGTH (m)	
	5" Drill Pipe	-	5	4.276	4-1/2" IF Pin x Box	-	
	XOS	1	4-3/4	2.25	3-1/2" IF P x 4-1/2" IF B	1	1496.61
	3-1/2" Drill Pipe	132	3-1/2	2-7/16	3-1/2" IF P x B	1267.2	1495.61
	3-1/2" Heavy Weight Drill Pipe	6	3-1/2	2-1/4	3-1/2" IF P x B	54.84	228.41
	4-3/4" Drill Collar	3	4-3/4	2.25	3-1/2" IF P x B	27.42	173.57
	4-3/4" Jar	1	4-3/4	2.25	3-1/2" IF P x B	8	146.15
	4-3/4" Drill Collar	15	4-3/4	2.25	3-1/2" IF P x B	137.1	138.15
	4-3/4" Bit Sub W/ Float Valve	1	4-3/4	2.25	3-1/2" Reg. P x 3-1/2" IF B	0.75	1.05
	6-1/8" PDC Bit	1	6-1/8	-	3-1/2" Reg. P x B	0.3	0.3

Figure 7-3: 6-1/8" Drilling BHA

8. Mud & Cement Program

8.1 Drilling Fluid Summary

Hole Size (in)	Casing Size (in)	Casing Program	Depth (m MD)	Drilling Fluid system	Mud Properties
26”	20”		146 m	**Already Has been Drilled**	
17 ½”	13 3/8”		1441 m	**Already Has been Drilled**	
12 ¼”	9 5/8”		3909 m	**Already Has been Drilled**	
8 1/2”	7” Liner		K.O.P: 3055 m To 3940	**KCL/$CaCO_3$**	MW: 72 - 85^{pcf} PV: ALAP (15 – 25) YP: 12 – 20 lb/100.ft² F.L: < 2.5 cc/30.min PH: 9 – 10 Vis: 40 – 50 sec/qt
6 1/8”	5” Liner		4150 m	**$CaCO_3$/Polymer Mud**	Mw:100 – 106^{pcf} PV: ALAP YP: 14 – 24 lb/100.ft² F.L: < 2.5 cc/30 min PH: 9 – 10 Vis: 45 – 60 sec/qt

8.2 8 – 1/2" Hole Section (3055 – 3940m)

Challenges

- Tight Hole and Hole stability
- Hole instability issues
- Wellbore losses.
- Shale

Interval Description

The 8 1/2"section will be drilled from KOP to an estimated section TD KCl/$CaCO_3$ Polymer mud system necessitating a minimum requirement of 80 to 85^{pcf} based on nearby well data.

In case of having problem with Marl and Shale layer 3 - 5wt. % of KCl, should be added to system to minimize associated problem.

It is necessary to adjust mud weight and mud properties as hole condition dictate. Some tight hole, well flow and low ROP maybe experienced in this hole section. The drilling fluid engineer will advise the drilling supervisor of any extended circulation or sweep requirement. It uses clarified viscosifier to provide viscosity and carrying capacity and starch to reduce mud filtrate.

In the directional and horizontal section required amount of Lubricant such as Bit Lube and Glass beds will be added to active system to minimize torque and drag.

8 1/2" HOLE SIZE	7" Liner
FROM : 3055 MT	TO : 3940 MT
885 meter to drill in Sarvak, kazhdomi, Darian and Gadvan by KCl Polymer mud	
DRILLING FLUID SYSTEM	KCl/$CaCO_3$/Polymer Mud
Key products	Limestone Powder , Salt (NaCl) , KCL , Xanthan gum , PAC LV , Starch LV , Caustic Soda, Soda ash, Biocide
Water type	Drill Water + Waste Water
Solid Control	Shale shakers, Desilters , Desander, Centrifuge

Sarvak	Limestone to argillaceous limestone, the lower part is shale and dolomite, with salt aquifer. Salt water 1.15sg.H_2S
Kazhdomi	Shale interbedded with argillaceous limestone, sandy claystone, claystone, marl, sandy limestone, limestone beds
Daryan	shale interbedded with argillaceous limestone
Gadvan	limestone, claystone, sandy claystone and sandstone

8.2.1.1 Interval Mud Discussion

The water-based KCl Polymer system provides effective inhibition when drilling reactive shale's through below mechanism:

Potassium Chloride provides a source of Potassium ions, which are small enough to fit between the clay platelets without distorting the shale lattice. The Potassium ions are adsorbed on the exchangeable caution sites in the shale lattice and this holds the platelets together, which helps to minimize clay and shale hydration, swelling and dispersion.

A low solids, non-dispersed Salt/KCl/$CaCO_3$ Polymer drilling fluid is selected for drilling this section. This system is used primarily to drill moderately reactive shale's. It can be used effectively at temperatures up to 275˚ F – 300˚

Product Concentration & Function

Salt / KCl / Polymer mud		
Product N.	**Ppb**	**Product Function**
Caustic Soda	0.5 - 1	Alkalinity Control
Soda Ash	1 – 2	Hardness Remover
Salt	Up to 98	non - solid weighting agent
KCl	10. – 18	Shale Inhibitor
Limestone Powder	Up to 113	weighting agent
Starch LV	12 – 14	Fluid loss controller
Xanthan gum	1 – 1.5	Viscosifier
Biocide	As Requirement	Anti-bacteria
MIXING PROCEDURE		
1. PH , chloride , calcium and total hardness of drill water (that water which will be used for mud making) must be check by Mud engineer		
2. Fill the mud tank with Drill water		
3. add Salt & KCL to reach mentioned mud weight		
4. Remove Hardness by add Soda ash		
5. adjust PH & alkalinity by Lime and Caustic Soda		
6. add Starch LV to control fluid loss		
7. add Xanthan gum as Need to reach good rheology		
8. Add limestone powder to reach Mentioned Weight		
9.Check the pH and keep in range by add Lime		
10. Mix Biocide as need		

Required Properties

Fluid Properties Summary		
MUD PARAMETERS		Interval
		IV
Bit diameter	inch	8 1/2”
Density	pcf	80 – 85
Marsh Viscosity	sec	40 – 50
PV	cps	ALAP
Yield Point	lb/100ft²	12 – 20
Gel 10 Sec	lb/100ft²	2 – 4
Gel 10 min	lb/100ft²	3 – 6
API Filtrate	cc/30'	< 2.5
MBT	ppb	<5
pH	-	9 – 10
Total Hardness	mg/l	< 1200
Ca++	mg/l	< 800
LGS (Drilled Solid)	% vol	< 7
KCl	% by Wt.	3 – 5
CL^-	Mg/l	180,000 – 188,000

****Please note that Rate and controlling of PV and solid percentage are depends on the solid control EQ efficiency****

Mud Volume

DRILLING FLUID TYPE			
Mud Density (pcf)	80 - 85	LGS Below (%vol)	6
Bit Size (inch)	8.5	Solid Control Equipment Eff. %	80
Section TD (m)	3940	Hole Wash Out %	15
Casing OD/ID (inch)	13.375 / 12.525	Dilution Factor (bbl/m)	2.2
Water Depth (m)		Meters To Drilled (m)	885

Volume information (Salt KCL Pol Mud)		
Surface System	800	bbl
Cased Hole Volume	1,528	bbl
New Hole Volume	204	bbl
Wash Out Volume	31	bbl
Sub Total	2,562	bbl
Dilution Volume	448	bbl
Sub Total	3,010	bbl
Carried from last section	0	bbl
Volume Received	0	bbl
Sub Total	3,010	bbl
New Vol Required	3,010	bbl
Vol transfer out	0	bbl

Product	Unit Size lb or bbl	Concentration	Expected Consumption
Soda Ash	110	1-2	55
Caustic Soda	55	0.5 - 1	55
Salt (NaCl)	2,200	Up to 98	134
KCL	2,200	10.7 - 18	25
Starch LV	55	12-14	766
Limestone Powder	2,200	up to 113	155
Xanthan Gum	55	1-1.5	82
Biocide	55	As Requirement	

Contingency Materials			
Product Name	Function	Unit Size	QTY
Pipe Lax	Pipe freeing agent	55 Gal	2
De foamer	De foamer	55 Gal	2
Shell fish F,M	Loss control material	55 lb	each size 60 sxs
$CACO_3$ F,M	Loss control material	55 lb	each size 80 sxs
Water Base Mud lubricant	Torque & Drag Reducer/ROP Enhancer	55 Gal	40
Starch HT	Fluid Loss Controller	55 lb	80
Pac-LV	Fluid Loss Controller	55 lb	120
PHPA	Encapsulate	55 lb	20
Glycol	Shale Inhibitor	55 Gal	28
Glass Beads - 500/800 mic	Torque & Drag Reducer	55 lb	120
Sodium Bicarbonate	CMT Contaminiation Remover	55 lb	20

1-Please note all calculation are based on normal hole condition and maximum SCE effieciency. Concentration and usage of material may be change as hole condition dictate.

2-All the calculation based on max material Concentration which mentioned.

8.3 6 – 1/8" Hole Section (3940 – 4150m)

Interval Description

After cementing the 7" inch Liner and confirmation of its proper cementing, the 6 1/8" section will be drilled from liner shoe to an estimated section TD of 4150 m horizontally by Salt/Polymer mud system necessitating a minimum requirement of 100 pcf to 106 pcf based on nearby well data.

It is necessary to adjust mud weight and mud properties as hole condition dictate. Some tight hole, well flow and low ROP maybe experienced in this hole section. The drilling fluid engineer will advise the drilling supervisor of any extended circulation or sweep requirement. It uses clarified viscosifier to provide viscosity and carrying capacity and starch to reduce mud filtrate.

In the directional and horizontal section required amount of Lubricant such as Bit Lube and Glass beds will be added to active system to minimize torque and drag

6 1/8" HOLE SIZE	**5" Liner Size**
FROM : 3940 MT	**TO : 4150 MT**
210 meter to drill in Fahliyan, by $CaCO_3$/Polymer mud	
DRILLING FLUID SYSTEM	**$CaCO_3$/Polymer Mud**
Key products	Limestone Powder, Hematite , Salt (NaCl), Xanthan gum , PAC LV , Starch LV , Caustic Soda, Soda ash, Biocide
Water type	Drill Water + Waste Water
Solid Control	Shale shakers, Desilters , Desander, Centrifuge

Fahlyan:	Limestone interbedded with green gray, Argillaceous Limestone, sticky Clay stone beds.

Product Concentration & Function

Salt /$CaCo_3$/ Polymer mud		
Product N.	**Ppb**	**Product Function**
Caustic Soda	0.5 – 1	Alkalinity Control
Soda Ash	1 – 2	Hardness Remover
Salt	Up to 109	non - solid weighting agent
KCl (Optional)	10.7 – 18	Shale Inhibitor
Limestone Powder	Up to 258	weighting agent
Starch LV	12 – 14	Fluid loss controller
Hematite	Up to 98	weighting agent
Xanthan gum	0.25 – 0.5	Viscosifier
Biocide	As Requirement	Anti-bacteria
MIXING PROCEDURE		
1. PH , chloride , calcium and total hardness of drill water (that water which will be used for mud making) must be check by Mud engineer		
2. Fill the mud tank with Drill water		
3. add Salt to reach mentioned mud weight		
4. Remove Hardness by add Soda ash		
5. adjust PH & alkalinty by Casutic soda		
6. add Starch LV to control fluid loss		
7. add Xanthan gum to reach good rheology		
8. Add Limestone powder to reach Mentioned Weight		
9.Check the pH and keep in range by add caustic soda		
10. Mix Biocide as need		

Required Properties

Fluid Properties Summary		
MUD PARAMETERS		Interval
		II
Bit diameter	inch	6 1/8”
Density	pcf	100 – 106
Marsh Viscosity	sec	45 – 60
PV	cps	ALAP
Yield Point	lb/100ft²	14 – 24
Gel 10 Sec	lb/100ft²	2 – 4
Gel 10 min	lb/100ft²	3 – 7
API Filtrate	cc/30'	< 2.5
pH	-	9 – 10
Ca++	mg/l	< 800
Total Hardness	Mg/l	<1200
MBT	ppb	< 5
LGS	% vol	< 10
CL^-	Mg/l	180,000 – 188,000

****Please note that Rate and controlling of PV and solid percentage are depends on the solid control EQ efficiency****

Mud Volume

DRILLING FLUID TYPE			
Mud Density (pcf)	100 - 106	LGS Below (%vol)	6
Bit Size (inch)	6.125	Solid Control Equipment Eff. %	80
Section TD (m)	4150	Hole Wash Out %	15
Casing OD/ID (inch)	9.625 / 8.835	Dilution Factor (bbl/m)	2
Water Depth (m)		Meters To Drilled (m)	210

Volume information (Salt Pol Mud)		
Surface System	800	bbl
Cased Hole Volume	847	bbl
New Hole Volume	25	bbl
Wash Out Volume	4	bbl
Sub Total	1,676	bbl
Dilution Volume	50	bbl
Sub Total	1,726	bbl
Carried from last section	0	bbl
Volume Received	0	bbl
Sub Total	1,726	bbl
New Vol Required	1,726	bbl
Vol transfer out	0	bbl

Product	Unit Size lb or bbl	Concentration	Expected Consumption
Soda Ash	110	1-2	31
Caustic Soda	55	0.5 - 1	31
Salt (NaCl)	2,200	Up to 109	86
KCL (Optional)	2,200	10.7 - 18	14
Starch LV	55	12-14	439
Limestone Powder	2,200	up to 258	202
Xanthan Gum	55	0.25-0.5	16
Ferobar	3,300	Up to 98	51
Biocide	55	As Requirement	

Contingency Materials			
Product Name	Function	Unit Size	QTY
Pipe Lax	Pipe freeing agent	55 Gal	2
De foamer	De foamer	55 Gal	2
Pac LV	Fluid Loss Controller	55 lb	120
Starch HT	Fluid Loss Controller	55 lb	80
Shell fish F,M	Loss control material	55 lb	each size 60 sxs
$CACO_3$ F,M	Loss control material	55 lb	each size 80 sxs
Water Base Mud lubricant	Torque & Drag Reducer/ROP Enhancer	55 Gal	40
Glass Beads - 500/800 mic	Torque & Drag Reducer	55 lb	120
Sodium Bicarbonate	CMT Contaminiation Remover	55 lb	20

1-Please note all calculation are based on normal hole condition and maximum SCE effieciency. Concentration and usage of material may be change as hole condition dictate.

2-All the calculation based on max material Concentration which mentioned.

8.4 Technical Appendix

Loss Control Procedure

Loss of fluid returns will lower the hydrostatic head of the drilling fluid in the wellbore, thereby inducing a kick. The influx fluid will then flow to the surface or into the zone of lower pressure. Lost circulation can occur in naturally occurring fractured, vuggy, cavernous, sub-normally pressured or pressure depleted formations. Induced losses can occur from mechanical fracturing due to pressure surges while breaking circulation. In all cases of lost circulation, attempts should be made to keep the hole full. The hole can be filled with either light drilling fluid or water. A record of the amount of fluid pumped should be made.

The concentration of LCM will vary according to losses. In the event of seepage losses, treat system with a finer grade of LCM at 20 – 30 Kg/m³. In the event of moderate to high rate of well bore loss, prepare pills at different graded concentrations and spot over thief zone. The concentration may vary between 18 to 70 lb/bbl. In order to avoid LCM plugging MWD tools a circulation sub should be installed in the drill string. The MWD engineer is to advise on the size and concentration of LCM that can be pumped through MWD Tools. Ensure that the drilling displace the LCM pill out of the string prior to shutting down the mud pumps.

Important Note: Please note below concentration and recommended materials may be change with same materials on rig site.

In case of having circulation losses:

Losses less than 10 bbls/hr:

- Active system to be loaded with 10 lb/bbl of Fiber Fine.

Losses 10- 40 bbls/hr:

- Active mud to be loaded with 15- 20 lb/bbl of Fiber Medium.
- Sweep the hole frequently with 50 bbls high viscous pill contains;
- 15 lb/bbl Mica Medium - Corse.
- 10 lb/bbl Walnut Shells Medium -Fine.
- 10 lb/bbl Oyster Shell Fine.

Losses 40 – 100 bbl/hr:

- Stop Drilling.
- Pull out string above the loss zone
- Spot 100 bbls pill contains;
- 30 lb/bbl Fiber coarse.
- 30 lb/bbl Walnut Fine.
- 20 lb/bbl Mica Med.
- Repeat the previous pill minimum of two times.

Losses more than 100 bbls:

- Up to company procedure, may be squeeze 50 bbls of Cement squeeze.

8.4.1.1 Mixing LCM Pill

The amount of LCM that can be added to a mud depends very much on the mud properties. High gels and YP will help keep the material in suspension. Since LCM is generally of low gradient, excess material tends to float on top if too much is added. The following procedure may prove useful for mixing LCM pills and avoiding the problem of flotation.

1. ENSURE the pill has enough viscosity to keep the low gradient material in suspension before addition of LCM. ADD appropriate viscosifier if required.
2. ADD the material in stages e g. 5, 10, 15, 20 etc. ppb.
3. STOP after desired concentration has been reached, or when it is observed that the material cannot be held in suspension
4. If a higher LCM content is desired, LEAVE the pill to soak for a period of time and then TRY ADDING more material.

8.4.1.2 LCM Pill Spotting Procedure

1. After spotting LCM pill, pull the bit above the calculated top of the pill. Wait 30minutes and monitor the annulus.
2. After 30 minutes, attempt to fill the annulus, if not already full. If the hole can be filled, close the BOP and squeeze 10 bbls of LCM pill into the formation without exceeding 200 psi.
3. If the hole remains stable, continue to drill ahead.

4. Should the LCM pill fail to reduce the losses or the hole cannot be filled, then mix and spot another 50 bbls of LCM pill.
5. Attempt to fill the annulus. If successful, close the BOP and squeeze 20 bbls of LCM pill into the formation without exceeding 200 psi. If a squeeze pressure develops, hold the pressure for 1 hour before staging the drill string back to bottom.

Notes:

- All time hole must be full, transfer mud to trip tank and asked driller to fill hole from trip tank.
- All LCM pills must be thoroughly mixed and homogenised before being pumped.
- When pumping an LCM pill through a bit with nozzle smaller than 18/64, DO NOT USED LCM Medium and Coarse sizes.
- Remove filter screens from the drill pipe and if necessary from the mud pump suction lines.
- This is particularly true if the mud density is high, 14 lb/gal or greater. Fine loss circulation material which does not adsorb a lot of water should be used in weighted mud.

Guide Line for Fiel Engineers for Shale Inhibition

Shale Hydration takes two forms in water-sensitive shales: surface and osmotic adsorption. Surface hydration occurs when a small volume of water is strongly adsorbed onto the planar surfaces of the clays which causes little softening or swelling, but can lead to excessive stresses if the swelling is confined.

Osmotic swelling occurs when a large volume of weakly held water is attracted to the clay surfaces by electrostatic forces. Osmotic swelling causes softening and significant swelling as the adjacent clay layers hydrate water and expand. Osmostic swelling does not generate excessive stresses, even when confined, and can be substantially reduced if Extremely difficult drilling conditions have led to more systematic approaches to analysing wellbore instabilities.

One such approach that can be used as a guideline for the mud engineer involves three basic steps:

1. Monitor and analyse the fundamental failure mechanism.
2. Determine and understand the factors governing this failure mechanism.

3. Model the problem considering impact of proposed changes.

8.4.1.3 Monitor & Analyze the Failure Mechanism

Monitoring of data and events on the rig is the first step of the process and involves:

• **Collection, organization and processing of drilling data.** Drilling data such as Rate of Penetration (ROP), Revolutions per Minute (RPM), Weight-on-Bit (WOB), Bottom-Hole Assembly (BHA), torque, over pull Trip In Hole (TIH) and Pull Out Of Hole (POOH), swab and surge, flow rate, mud weight, mud Rehology , etc. should be collected, organized and analyzed on a continual basis and compared to the symptoms of wellbore stability being observed. This will help evaluate performance and establish trends or any deviations and measure the effectiveness of treatments and changes.

• **Formation characterization.** The standard procedure in drilling operations is to apply the knowledge

gained from previous wells to subsequent wells drilled in similar geologic regions. Wellbore-instability

investigation requires evaluation of shales during the drilling process. Since rig site shale testing is limited

due to equipment and time limitations, most evaluations are performed in a properly equipped laboratory.

Laboratory tests are conducted to evaluate shale swelling, dispersibility, cation exchange capacity, hydration capacity, colloidal content, capillary suction time, shale hardness tests and mud compatibility properties.

A number of methods have been used to perform rig site tests. These include tests as simple as the methylene blue test (CEC) measurement of shale; shale hardness measurements; hot-rolled dispersion testing with various mud formulations and additives; and sophisticated analytical measurements of various clay properties. It is particularly important to try and determine the failure mechanism: swelling and softening or brittle spalling. Information from these tests, current shaker observations and mud logger

data are used to evaluate drilling fluid performance, adjust mechanical drilling parameters and making casing design decisions. The primary options to counter shale problems at the rig site are:

- Change the drilling fluid density.
- Change to an oil- or synthetic-base system.
- Change to a more inhibited water-base mud.
- Reduce fluid loss to very low levels and increase filtrate viscosity.
- Utilize pore-plugging additives, emulsified insoluble materials or plastering agents.
- Adjust mechanical drilling parameters.
- **Evaluation and analysis of instability symptoms.**

Instability warning signs such as the presence of excessive cavings on the shakers for a given ROP, fill on connections, tripping difficulties, torque, stuck pipe, etc. indicate that something is wrong down hole. The magnitude or volume of failed rock should be estimated and recorded with the other drilling parameters. The number and location of specific incidents should be recorded, such as pipe

sticking, fill or packing-off, reaming and re-drilling. All of this information can then be organized and correlated with time and depth to detect the location and quantify the severity of critical sections in the well.

8.4.1.4 Determine & Understand Failure Mechanism

All of the monitored and plotted parameters are then evaluated to identify the most probable mechanism causing the instability. The most useful symptoms used to indicate the mechanism responsible for wellbore instability are:

- Cavings analysis.
- Tripping difficulties.
- Drilling conditions.
- Mud system analysis.

- In-situ conditions.

These symptoms should be evaluated based on an interdisciplinary knowledge of the causes of wellbore instability to determine the most likely failure mechanism.

8.4.1.5 Model & Considering Proposed Changes

Once the monitoring and determination steps have produced parameters, organize and analyze this data to obtain qualitative solutions and evaluate proposed changes. One good way to do this is through the use of a decision tree or flowchart approach to evaluate the possible failure mechanism and to identify the most promising remedies. After arriving at a proposed solution, a corrective measure should be applied and

the results used as feedback into the method and decision tree method.

A decision tree or flowchart, such as shown in Figure 9 for evaluating the cause of excessive cavings, provides a logical path for identification of the mechanism and a qualitative solution. Maintaining hole stability while drilling shale sections can be particularly troublesome. Simple solutions do not always exist, but good drilling practices combined with good mud practices are most often successful. Analysis of the mechanical stress/strain relationships has helped to develop the proper density requirements for hole stability.

8.5 Directional Drilling

Drilling Fluid Properties & Selection

It is common knowledge among drilling engineers and drillers that borehole problems can be considerably more severe in a directional hole than in a vertical well. The problems are magnified as the hole angle and horizontal displacement increases. Although most of these reactions are mechanical in nature, much can be done to alleviate them by proper selection and maintenance of the drilling fluid. For example, if a water-base fluid is being used, lubricants can be added to lower the coefficient of friction between the drill string and borehole or casing. Inhibiting salts can be added to reduce the interaction of the drilling fluid with the exposed formations, thereby improving borehole stability.

There are a number of drilling factors influenced by the drilling fluid, but the following are of primary concern in drilling a horizontal well.

- Borehole instability
- Hole cleaning
- Torque and drag
- Formation damage
- Barite sag
- Hydraulics

Borehole Instability

Borehole instability is one of the more costly, but relatively common problem associated with drilling oil and gas wells. Not only can it impede drilling, it can result in loss of the hole. The term normally conjures thoughts of borehole collapse with accompanying stuck pipe. This is one of the more critical reactions of borehole instability, but it can manifest itself in more subtle ways such as hole-fill, bridges, hole enlargement, or hole size reduction.

These reactions lead to other associated problems including increased torque and drag, logging difficulties, poor cement jobs, and increased drilling days, all of which add significantly to the drilling cost. It should be pointed out that the discussions to follow

do not cover hole enlargement problems from weakly consolidated sands, or rubble zones that erode from the hydraulic action of the drilling fluid.

Stabilization

The swelling activity is controlled by adjusting the electrolyte content of the fluid to minimize expansion of the clays. Lime and gypsum-treated fluids have long been used for this purpose, but basically do not satisfy the inhibition needs of the more troublesome shales. The potassium ion is an effective swelling inhibitor and is used extensively for this purpose through the addition of potassium salts. The concentration of potassium utilized ranges from several hundred mg/L to over 60,000 mg/L. The most cost-effective potassium concentration, however, appears to be in the 3000 to 7000 mg/L range. In marine environments, seawater or seawater with additional sodium chloride, are good alternates to potassium. The tendency for the shale to disperse or subdivide into discrete particles appears to be best controlled by the addition of an adsorptive, high molecular weight water-soluble polymer.

One of the more effective polymers for this purpose is the partially hydrolyzed poly acrylamide with molecular weight in excess of 15 million. These anionic polymers adsorb on to the surface of the cuttings and borehole wall to act as a binder or elastic coating to retard dispersion. Another approach to stabilizing the hard brittle shales that contain natural or induced micro fractures is the use of water-dispersible asphalt or asphaltines. The addition of these type materials appear to plug the micro-fractures, or perhaps even surface pore throats to retard further fluid invasion. This action has proven very effective in certain shales and can be applied in most water base fluids.

Hole Cleaning

Cleaning cuttings from a vertical hole has been covered extensively by numerous investigators. Hole cleaning in a vertical hole is essentially a matter of overcoming the slip velocity of the cuttings with sufficient annular velocity and carrying capacity of the drilling fluid. Although any velocity greater than the settling velocity of the largest cutting will theoretically lift all the cuttings to surface eventually, too low an annular velocity will lead to an undesirable high concentration of cuttings in the annulus. Experience has shown that cuttings concentrations more than 5% by volume cause tight hole or stuck pipe when circulation is stopped.

Cuttings Bed Formation

Radial slip velocity increases with increasing angle of inclination, forcing more cuttings to the low side of the annulus, which eventually results in the formation of a cuttings bed. reported that bed formation was dramatic in the 40° to 50° angle with the bed sliding downward against the flow, resulting in a very high cuttings concentration.

The fact that a cutting bed forms is a serious problem, but the instability of the bed in this critical angle of the well makes the problem even more severe. High torque and drag resulting from the high cutting concentration at the lower part of the angle can obviously result in stuck pipe or loss of circulation.

Hole Cleaning Sweeps

In static drill string concluded that high-density sweeps were more effective than high-density high-viscosity sweeps based on field measurements. The higher viscosity of the latter caused flow diversion to the high side of the wellbore. Not surprisingly, high-viscosity sweeps also performed poorly. Also concluded that the best hole cleaning sweeps were high-density low viscosity sweeps, with a density 3 – 4 ppg higher than that of the drilling fluid. the following procedure is recommended:

- Calculate sweep height in the vertical part of the annulus, and added hydrostatic head in the annulus.
- Rotate at 80+ rpm while pumping the sweep, 180+ rpm is preferred. It is not recommended to pump a weighted sweep while sliding.
- Keep pumping until the sweep is out of the hole. If pumping stops while the sweep is in the deviated hole, it falls out of the "jet stream" and re-deposits captured fines.
- Avoid increasing the viscosity of the weighted sweep.
- Plan to run 3 – 4 sweeps to achieve desired results when hole cleaning problems are indicated. One sweep will probably not be sufficient to remedy an existing situation.
- Avoid checking for flow or closing the annulus with the weighted sweep in the drill pipe.
- If desired for preventative maintenance: 1) schedule sweeps every 6 – 8 hours or 500 – 1000 feet (whichever comes first, unless the hole dictates greater frequency); barite/silt build- ups are dependent on time and footage: and 2) pump a sweep while circulating bottom up before tripping, and another once back on bottom.

concluded that high-density sweeps were more effective than high-viscosity sweeps.

They strongly advocated the use of sweep reports, which should be standard practice in any drilling fluids operation.

Short Tripping

Another technique that has proven helpful in controlling problems created by the cuttings bed in the high-angle portion of the well is to make periodic short trips. Tripping the bit through the bed not only assures that the bottom hole assembly and bit can be pulled through it, but the action should agitate the bed to the point that a portion will slide down the hole and subsequently be picked up when circulation is commenced. The time intervals for these short trips will be determined by the torque and drag conditions that exist in any particular well. Lacking specific well history, a general rule of thumb is to short trip at least once for every 500 feet of new hole.

Drill String Rotation

The industry is in total agreement that drill string rotation improves hole cleaning, especially in hole sizes less than 17 ½". The general opinion in the industry is that the optimum drill string rotation rate is approximately 120 rpm. It is believed that the drill string rotation improves hole cleaning by mechanically disturbing the cuttings bed on the low side of the hole. There could also be effects due to skewing the velocity profile in the wellbore and the creation of Taylor Vortices. Encouraged the use of rotary steerable systems.

Turbulance

Bottom hole assembly design becomes important because the flow restrictions of some of the components, such as the measurement-while-drilling tools or positive displacement motors, may not allow sufficient flow to reach turbulence in the annulus. Turbulence can, of course, be induced at some constant flow rate by decreasing the annular area. This may be more difficult to accomplish in a horizontal well because of limitations on the design of drill strings and tool assemblies. An increase in the effective viscosity of the drilling fluid will decrease the Reynolds Number, and thus increase the flow rate required to obtain turbulent flow. Thus, a lower effective viscosity fluid may be desirable in the horizontal well, but attention must also be given to cleaning the larger diameter portion of the upper hole or possible washed-out sections where it is impractical to maintain turbulent flow.

It is interesting to point out a basic fact between static fluid pressure and dynamic fluid pressure. Static fluid pressure is a function of true vertical depth, while dynamic fluid pressure, or ECD, is a function of measured depth. Thus, the ECD will continue to increase, albeit slowly, even after the hole becomes essentially horizontal. The fracture pressure could be reached if the hole was extended to sufficient depth and the fluid density is close to the fracture gradient.

Hole Cleaning Guideline

The single most important factor related to hole cleaning in deviated wells is flow rate. During directional drilling operations drilled cuttings will settle on the low side of the hole and form a stationary bed if insufficient annular fluid velocity is used. The critical flow rate required to prevent cuttings bed formation must be determined .

Typically few hole cleaning problems exist in vertical or horizontal sections. Most problems

associated with hole cleaning are seen on deviated wells and occur in the 40 – 60 degree section where gravity effects can cause cuttings beds to slump down the hole.Typical flow rates to aim for in ERD wells:

Hole Size	Typical Flow Rates
17 ½"	800 gpm minimum. Some rigs achieve 850 – 1000 gpm.
12 ¼"	Aim for 800 gpm (although 800 – 1000 gpm is typically achieved). If 800 gpm is not achievable, ensure tripping procedures are in place for poorly cleaned hole.
8 ½"	Aim for 400 gpm.

When correctly designed, both laminar and turbulent flow regimes will effectively clean a deviated well. In general increasing the viscosity of a fluid in laminar flow will improve hole cleaning as will a reduction of the viscosity of a fluid in turbulent flow. It is important that one or other regime is selected and that the transition zone between the two is avoided.

Generally viscous fluids in laminar flow are preferred because:

- It is possible to achieve higher cleaning capacity.
- Viscous fluids give better transport in the near-vertical sections.
- Viscous fluids have better suspension characteristics when circulation is stopped.
- It is difficult to achieve "turbulent flow" except in small hole sizes.

Experience has shown that good drilling fluid rheological properties are extremely important to hole cleaning when drilling a high angle well. Studies show that the effects of drilling fluid

rheological properties and annular flow regime are mutually dependent. In the laminar flow regime, increasing fluid viscosity will improve hole cleaning and this is particularly effective if the Yield Stress is high. In the turbulent regime, however, reducing fluid viscosity will help removing cuttings. Therefore the drilling fluid rheological properties should be designed to avoid the transitional flow regime. For hole sizes above 8 ½", the annular flow is laminar under most circumstances. Therefore it is necessary to specify a minimum Yield Stress. The optimum level required is best established based on field data and experience.

A higher ROP requires a higher flow rate to clean the hole. It is good practice to drill the hole with a steady ROP and select the required flow rate for hole cleaning accordingly. Before making a connection, the hole should be circulated at the normal flow rate to clear the cuttings from around the BHA. Depending upon the hole angle and length of BHA, a circulation time of 5 to 10 minutes is often required.

Before tripping out, the hole should be circulated at the normal flow rate until the shakers are clean, and the drill pipe be rotated at maximum speed/reciprocated in the meantime. This may require up to 3 bottoms-ups, depending on the hole angle and hole size. Below are listed the recommended number of bottoms-up prior to tripping.

Hole Angle, deg	8 ½"	12 ¼"	17 ½"
0 – 10	1.3	1.3	1.5
10 – 30	1.4	1.4	1.7
30 – 60	1.6	1.8	2.5
60+	1.7	2.0	3.0

Overall friction can become so great in an extended-reach well or horizontal section that the drill string can only be rotated with great difficulty. Needless to say, this condition is conducive to stuck pipe. Obviously, this coefficient of friction will be significantly affected by the lubricating properties of the drilling fluid.

any of the causes of torque and drag are mechanical, while others are related to hole geometry. Several, however, can be directly attributed to the drilling fluid. Since wall contact with the drill string will be extremely high in a horizontal hole, it is imperative that the lubricating characteristics of the fluid be considered.

Formation Damage

The obvious reason for drilling a horizontal well is to expose more of the producing zone to the wellbore for increased production. Thus, the importance of preventing formation damage during drilling and completion of the horizontal well is just as important as it is in its vertical counterpart.

In fact, it may be more important since the increased cost of drilling the horizontal well must show additional production for its justification. A 50% damage or permeability loss in a potential 500 bbl/day horizontal well represents a considerably greater loss in production than a 50% damage in a 100 bbl/day vertical hole through the same reservoir.

Sandstone production is by far the most susceptible reservoir rock to damage or permeability loss during drilling and completion operations. Though formation damage may be caused by poor drilling and completion practices, one of the primary causes is due to exposure to the drilling and completion fluid.

This loss, attributable to the fluid, can be caused by one or more of the following occurrences:

- Particle transport from the drilling fluid into the production zone, subsequently plugging pore throats.
- Fluid filtrate reacting with expandable clays in the rock to decrease pore throat diameters.
- Particle movement within the permeable rock due to dispersion of clays and other minerals from quartz surfaces.
- Wettability changes of the formation from exposure to the drilling fluid filtrate.
- Interaction of fluid filtrate with formation fluids to form water - insoluble precipitants.

The effect of these actions on the permeability of sandstones has been the subject of a considerable investigation. In almost all cases, some damage is caused by the drilling and completion fluid. The primary objective is to limit this damage as much as possible.

Consequently, selection of the drilling and completion fluid becomes an extremely important function in drilling the horizontal well.

Recommended Fluid Parameters

- Velocity – One of the major obstacles to overcome in horizontal and ERD drilling is hole cleaning. In vertical holes, cuttings have room to settle without causing problems. In horizontal wells, cuttings only have to settle 4+ inches to form a cuttings bed. Research and practical experience has shown that cuttings bed formation is analogous to sand dunes on an ocean beach. The bed moves in a wave fashion up the hole. If sufficient annular velocity is not present, the bed thickens until it reaches equilibrium, which usually occurs when turbulent flow conditions exist. With water, turbulent flow is easily obtained since viscosity is in the denominator of the turbulent flow equation, and water has a viscosity of 1. Fluids other than water require more pump output to approach turbulent flow conditions. Practical experience in other industries has proven that materials can be transported in water slurries if the fluid velocity is between 3 and 7 ft/sec.

- Rheological Properties – Ideally one would want a true thixotropic fluid to drill horizontally. In effect, two different conditions exist simultaneously. One occurs in the vertical portion of the hole where larger diameter casing may be set, or wash-outs in the open hole occur. In that case, a fluid having a high Yield Stress, or if the fluid is visco-elastic, the initial gel strength needs to be elevated. The opposite is needed in the horizontal section from a hole-cleaning point of view. Here one wants water for the reasons pointed out earlier. Compromises are the result. If casing is set into the zone of interest, or if formations up-hole are not a problem, then water may be possibly combined with occasional sweeps.

- Surface Apparatus – Special attention must be placed on the solids removal equipment at the surface. Solids removal efficiency must be continually monitored so that unwanted particulate material is not retained in the fluid system that may negatively affect fluid weight or filter cake quality, resulting in lost circulation or increased torque and drag. The cost of problems that may occur due to not doing this correctly may far outweigh the cost of equipment needed.

8.6 Hydrogen Sulphide Gas in Drilling Fluid

Description of Hydrogen Sulphide Gas

Hydrogen sulphide is a colourless gas, with an offensive odour and a sweetish taste. It is soluble in water, alcohol, oils, and many other solvents. It has a specific gravity of 1.1895

with reference to air. It is considered a weak acid, it is toxic to humans and corrosive to metals. Hydrogen sulphide can be dangerous to personnel on the surface as it is extremely toxic to human and even animal life, and is extremely corrosive to most metals as it can cause cracking of drill pipe and tubular goods, and destruction of testing tools and wire lines. When hydrogen sulphide enters the borehole during drilling, completion, or testing for hydrocarbons, it creates several very detrimental problems. These problems are encountered regardless of the source of the hydrogen sulphide. Hydrogen sulphide has created unsolved problems in the oil field.

Origin of Hydroge Sulphide

During oil and gas operations, hydrogen sulphide can be found as a component of formation gases, dissolved in water, hydrocarbons, or even liquid sulphur. Thermal degradation of organic materials and sulphate reducing bacteria (SRB) can create hydrogen sulphide along with other gases.

Problem Cused by Hydroge Sulphide Gas

8.6.1.1 Health & Safety of the Personnel

One of the main problems hydrogen sulphide causes is in respect to the health and safety and safety of the personnel. A maximum of eight hours exposure to concentrations greater than 100 ppm H2S will cause hemorrhage and death. Concentrations above 600 ppm can be fatal in three to five minutes. Highly stressed, high strength steel can fail in a matter of minutes in the presence of 50 ppm H2S. At high pressure, as little as 0.1 ppm H2S can greatly reduce the time to failure of highly stressed, high strength steel. Relying solely on its odour is not a good idea because at concentrations above 100 ppm it deadens a person's sense of smell within a few minutes. The pure gas is heavier than air and can collect in low areas such as pit rooms and accommodation. below Table shows the potential hazards hydrogen sulphide can cause to the personnel from short-term exposures. There are many historical cases to document the ill effects of hydrogen sulphide exposure.

The documented cases reported in Table 2 indicate that hydrogen sulphide is very toxic and it takes only a matter of minutes to affect unprotected workers.

Health effects from short-term exposure to hydrogen sulphide	
Concentration (ppm)	**Health Effect**
0.01 – 0.3	Odour threshold

1 – 20	Offensive odour, possible nausea, tearing of the eyes or headaches with prolonged exposure
20 – 50	Nose, throat and long irritation; digestive upset and loss of appetite; sense of smell starts to become fatigued; acute conjunctivitis may occur (pain, tearing and light sensitivity)
100 – 200	Severe nose, throat and long irritation; ability to smell odour completely disappears.
500	Severe lung irritation, excitement, headache, dizziness, staggering, sudden collapse knockdown), unconsciousness and death within a few hours, loss of memory for the period of exposure (results in permanent brain damage if not rescued immediately).
500 – 1000	Respiratory paralysis, irregular heart beat, collapse and death without rescue.
> 1000	Rapid collapse and death

8.6.1.2 Effects of Hydrogen Sulphide on Metal

Hydrogen sulphide mainly cause corrosion problems to drill strings, transport pipes, storage tanks etc. Hydrogen sulphide causes sulphide stress cracking, hydrogen embrittlement and pitting corrosion in oil and gas operations. The corrosion of iron in the presence of hydrogen sulphide and water is dependent upon the dissociation of the hydrogen-sulphide molecule. Iron is oxidized to the ferrous form at the anode and hydrogen sulphide under- goes a two-step dissociation at the cathode. Tung et al22propose the following equilibrium reactions below.

At the anode:

$$Fe \longrightarrow Fe^{2+} + 2e^-$$

And at the cathode hydrogen is produced and either enters the steel or forms hydrogen gas and bubbles off:

If H2S is present:

$$2H^+ + 2e^- \longrightarrow H^\circ + H^\circ$$

If H2S is not present:

$$H^\circ + H^\circ \longrightarrow H_2$$

Hydrogen enters the steel first by adsorption onto the water to steel interface and then by being absorbed into the steel as hydrogen. The amount of hydrogen entry depends on the corrosion rate of the steel surface and the concentration of anions such as HS- that reduce the tendency to produce hydrogen gas (as shown in second equation above) and promotes the hydrogen (Ho) to enter the steel. pH also has an influence on amount of corrosion damage. Basically, the term "sulphides" used in oil and gas operations includes all three water-soluble species, H2S, HS- and S2-, which coexist in a sulphide-water system. Shown in the ionization chart for hydrogen sulphide in figure 1, it can be observed that molecular H2S predominates in the acidic range, where the pH is below 7. In the range of 7 to 12, monovalent HS- predominates and above pH 12, divalent S2- predominates.

The illustration in figure 1 indicates that the most important practice is that when H2S enters in alkaline mud, it reacts to form an alkaline sulphide, most often sodium sulphide.17 While elevated pH is a good way to keep H2S from causing problems under certain conditions, it does not remove it from the fluid and any drop in pH can create a significant hazard. This then brought about the advent of sulphide scavengers in the history of oil and gas operations. The only safe method for the total removal of hydrogen sulphide or soluble sulphides is with a sulphide scavenger.

8.6.1.1 Effects of Hydrogen Sulphide on Drilling Fluids

The first positive identification of soluble sulphide in the drilling fluids must be taken as a possible indication of H2S entering the fluid from some source. In the simplest case an increase in mud weight may be sufficient to control minor influxes. If levels persist chemical treatment will be required. This can take two forms; neutralization and scavenging. The two terms must not be confused. Neutralization of H2S in drilling fluid by caustic soda or lime is a common field approach. This alkalinity maintenance method has proved successful when small amounts of H2S enter the drilling fluid, for example, from drilling up pore spaces containing the gas.

This is not a suitable approach to the control on influxes of Hydrogen Sulphide. The alkalinity neutralization of H2S is instantly reversible by reduction in pH. Should the pH of the fluid decrease due to further influxes of H2S, Co2 or brine flows the sulphids may come out of solution as hydrogen sulphide gas. To ensure that such influxes are handled safely, it is essential to employ a sulphide scavenger. This term refers to any additive which is capable of reacting with one or more of the sulphide species and as a result, converting them to a more inert form. To be truly effective an irreversible and complete reaction must occur. As a mutual equilibrium exist between the three sulphide species in

solution (H2S, HS- and S2-) irreversible and complete removal of one serves to removal all three.

To remove the problem it is necessary to precipitate the sulphide as an insoluble inorganic sulphide. Zinc or iron compounds are usually chosen because of the relative inertness of the sulphids formed. Zinc source (Zinc oxide or carbonate) has in the past been the preferred treatment. However, continuous treatment with Zinc carbonate will produce significant undesirable carbonate concentration (Carbonate contamination) which can adversely effect on drilling fluid rehology and fluid loss. See the below table:

Contaminant	FV	PV	YP	Gels	FL	pH	Pm	Pf	Mf	Cl^-	Ca^{2+}
Carbonate	↑		↑	↑	↗	↓	↓	↗→	↓		↓
H_2S	↑		↑	↑	↑	↓	↓	↓	↓		↗

Table 8-1 Quick reference for recognizing carbonate and H2S contamination

As a precaution against Hydrogen Sulphide influxes into the drilling fluid system should be pre- treated with Zinc Source. Zinc oxide can be used as an alternative, provided that the possible environmental payloads associated with its use are appreciated. pH and alkalinity tests (Pf/Mf) are a good first indicator of Hydrogen Sulphide influx. However, the use of scavenger is used up. It can be seen from this that while a scavenger increase safety levels, it also makes detection of small Hydrogen Sulphide influxes difficult by conventional methods. Measurement of the concentration of scavenger in the mud system is possible but the analysis method is not safe for use on the rig.

The most effective environmentally acceptable scavengers for Hydrogen Sulphide in water base muds are Zinc sources like Zinc Carbonate or Oxide, primarily because of their solubility which leads to very fast reaction rates compared to other scavengers.

Zinc Sources clearly reacts very much faster than the other scavengers. This is due to the fact that the iron in solution ensures a fast reaction with the sulphides and the chemistry in this case produces elemental sulphur as the reaction product. Elemental sulphur is completely insoluble in water. The iron is reduced from Fe+3 to Fe+2 in the process and some of this will be regenerated to produce Fe+3 when the mud comes in to contact with the air. Any regeneration of Fe+3 will aid in reducing product consumption without affecting is scavenging properties.

Recommended treatment levels of Zinc sources are as follows:

pre- emptive treatment is recommended at 1.0 lb/bbl treatment of a mud system in the event of gas influx should be made on the basis that 2.0 lb/bb; will treat approximately 100 ppm Hydrogen Sulphide.

8.6.1.1 Testing, Reporting, Treatment of Hydrogen Sulphide

The active mud system must be regularly analyzed and monitored for the presence of active soluble Sulphide with the Garrett Gas train. This test must be carried out at the same time as each complete mud check, i.e.: three times per day while drilling.

Alkalinity checks should be carried out intermittently in between these full checks and any significant reduction in the pH and Mf/Pf should be followed up by testing for the presence of active soluble sulphides with the Garrett Gas Train.

For routine checks lead acetate paper can be used to give a qualitative check. Should this test be positive then it must be followed by a qualitative analysis with a Dragger tube.

The following actions must be taken after the initial positive test result:

1. Inform the Drilling Supervisor
2. Start adding H2S Scavenger to the active mud system. Active soluble sulphides can be analysed and monitored because of their characteristic of readily regenerating hydrogen Sulphide gas in reaction with mild acid. By adding citric acid to whole mud sample in a Garrett gas train detection of Hydrogen Sulphide on lead acetate paper indicates the presence of active Sulphides in the mud. When this qualitative test proves positive a dragger tube should be used to give qualitative results.

However it must be remembered that pre-treating the mud with MILGARD Fe means that no Sulphides will be delectable until all the scavenger has reacted, i.e.: it is impossible to detect small influxes of Hydrogen Sulphide in a system pre- treating with MILGARD Fe .

8.6.1.1 Contingency Stock to be Kept on Hand on the Rig

The following Contingency stock of treatment chemicals must be kept on hand on the rig over and above the normal mud maintenance material requirements.

Zinc Carbonate or Oxide 120 Sxs 25 kg

Lime 200 Sxs 25 kg

As above mentioned Hydrogen Sulphide is a poisonous and dangerous acidic gas encountered in many formations. It can quickly deaden senses and can be fatal, even at low concentration. Hydrogen Sulphide is characterized by its typical “rotten egg” smell.

For safety reasons, it should be neutralized immediately with caustic soda or lime to increase pH to >11.5 to form sulphide (S2-), then precipitated with a source of Zinc.

Neutralization with caustic soda:

$2\ NaOH + H_2S\ 2Na^+ + S^{2-} + 2H_2O$ (at pH>11.5)

Neutralization with maintaining excess lime:

$2\ Ca(OH)2 + H_2S\ Ca^{2+} + S^{2-} + 2H_2O$ (at pH>11.5)

Converting Hydrogen Sulphide into just sulphide by increasing pH is not a permanent reaction. If the pH were drop into the acid region, sulphide will be converting back into the poisonous H2S form. For this reasons H2S should always be precipitated by adding a source of Zinc such as Zinc Oxide.

Removal by precipitation with treatments of (ZnO):

$H_2S + ZnO\ ZnS + H_2O$

if H2S gas detected in mud following action should be taken to:

3. Immediately rise up the pH to at least 11.5 to 12 with caustic soda.
4. Buffer the pH with lime.
5. Begin treatment with Zinc oxide to remove sulphide (S2-) from the system. (1 lb/bbl treatment of Zinc oxide removes approximately 1100 mg/lit of sulphide. No more than 2 lb/bbl of Zinc oxide should be used to pre-treat the system for H2S)

If it appears that H2S bleeding in (flowing) form the formation, the mud density should be increased to shut off the flow of gas into the well-bore.

8.7 Health, Safety & Enviroment

Drilling Engineer Rules & Responsibilities

- Attend Operators/Contractors safety meetings and advise on all HSE matters pertaining to our products. The requirements for Drilling Fluids Engineers is not only to attend but to contribute to Safety meetings onsite is also important, including giving

- Presentations on the fluids and chemicals being used.

- Give toolbox / pre-tour talks on chemical safety. ƒ Take part in risk assessments relevant to fluids, in particular for the first use of new systems e.g. SBM / OBM etc.,
- Follow all procedures related to HSE and follow all well site directives issued by the operator and / or drilling contractor.
- Ensure correct and updated safety posters are in place in the mud and sack rooms or mixing areas for land rigs.
- Ensure MSDS are up to date and easy to locate.
- Use engineering controls and personal protective equipment, as
- Appropriate.
- Use rig specific HSE Observation system e.g. STOP.

Hazard & Risk Assessment

The use of chemicals in the workplace presents hazards and risks to personnel involved in their handling and application. In order to minimize these hazards risk assessments are performed and HSE control measures and management systems are established to achieve the following:-

- ➢ Identify hazards
- Safely manage those hazards.
- ➢ Identify risks
- Where possible, eliminate those risks through control / engineering measures e.g. ventilation and collection of dust.
- Where not possible, manage those risks through processes including the use of PPE.
- Provide training and awareness systems designed to achieve the above and promote continuous improvement.

Hazardous Materials

- The effect on a person of a hazardous material depends on:
- The nature of the hazardous material.
- The site of the action.
- The amount of the hazardous material involved (“dose”).

- The reaction of the individual ("susceptibility").

Hazardous Material Effects

8.7.1.1 Local Effects

- Skin and eye irritation & burns.
- Skin defatting leading to dermatitis.

8.7.1.2 Systemic Effects

- Central nervous system (headaches, nausea, dizziness).
- Cardiovascular system (CO poisoning).
- Sensitisation (allergy) and asthma.
- Teratogenic and carcinogenic.

Chemical hazardous effects may be :

- Acute - effects lasting minutes, hours or days e.g. irritation i.e. generally short term recoverable effects.
- Chronic - effects lasting weeks, months or years e.g. occupational asthma – generally long and possibly permanent effects.
- Effects may be reversible or irreversible.

Personal Protective Equipment (PPE)



- Hard Hat



- Gloves, long rubber gloves for handling hazardous materials.



- Eye Protection, glasses, goggles or full face mask (as appropriate).



- Coveralls, i.e. long sleeved, fire resistant, to cover as much body skin as possible, rubber apron when handling hazardous material.



- Boots – recommended rubber or treated leather.



- Dust mask, particulate filter mask, respirator (as appropriate).

PPE should be checked before use each time and examined on a regular basis if not in regular use. Remember – PPE requires care and maintenance.

Material Safety Data Sheet (MSDS)

An assessment is made of the physical and health hazards of each chemical supplied by our company. This information is included in a Material Safety Data Sheet (MSDS) and, in part, on container labels.

Material Safety Data Sheets contain the following information:

1. Product Identification
2. Composition, information on ingredients
3. Hazard identification
4. First aid measures
5. Fire fighting measures
6. Accidental release measures
7. Handling and storage measures
8. Exposure controls, personal protection
9. Physical and chemical properties
10. Stability and reactivity

Ensure that the following practices are followed regarding MSDS information at the workplace:

- Provide active up-to-date MSDS files covering all drilling fluid chemicals on location either on CD or paper copies. This must also include the laboratory testing chemicals.
- Distribute the MSDS files to the Well site Manager/ OIM, Medic (or designate) and sack storage / mixing areas.
- Update the MSDS file list whenever a new item is received.

Mixing Guideline

8.7.1.3 Pre – Job

- Select the chemicals to be mixed.
- Review MSDS (Material Safety Data Sheets) .
- Review the wall charts.
- Obtain appropriate tools, e.g. barrel pump.
- Inspect the condition of the chemicals to be mixed.
- Obtain appropriate PPE and WEAR IT.
- Ensure mixing personnel have clear written instructions.
- Perform a Job Hazard Analysis for any new chemicals or personnel.

8.7.1.4 Mixing

- Check that hopper is running and that the correct lines valves and pits have been selected.
- Ensure sufficient extraction and ventilation in hopper area.
- Ensure that sacks and drums are conveniently positioned and use correct lifting procedures.
- Be aware of any forklift operations.
- Clean up spills as soon as possible.
- Close the hopper any time chemicals are not being mixed.

8.7.1.5 Clean – Up

- Inform derrick man, pump man or supervisor that job is complete.
- Clean up mixing area.
- Dispose correctly of empty sacks, drums and pallet waste, e.g. Banding, wood & plastic wrapping in the correct manner.
- Ensure forklift is parked in designated area with forks lowered.

8.8 Completion Section

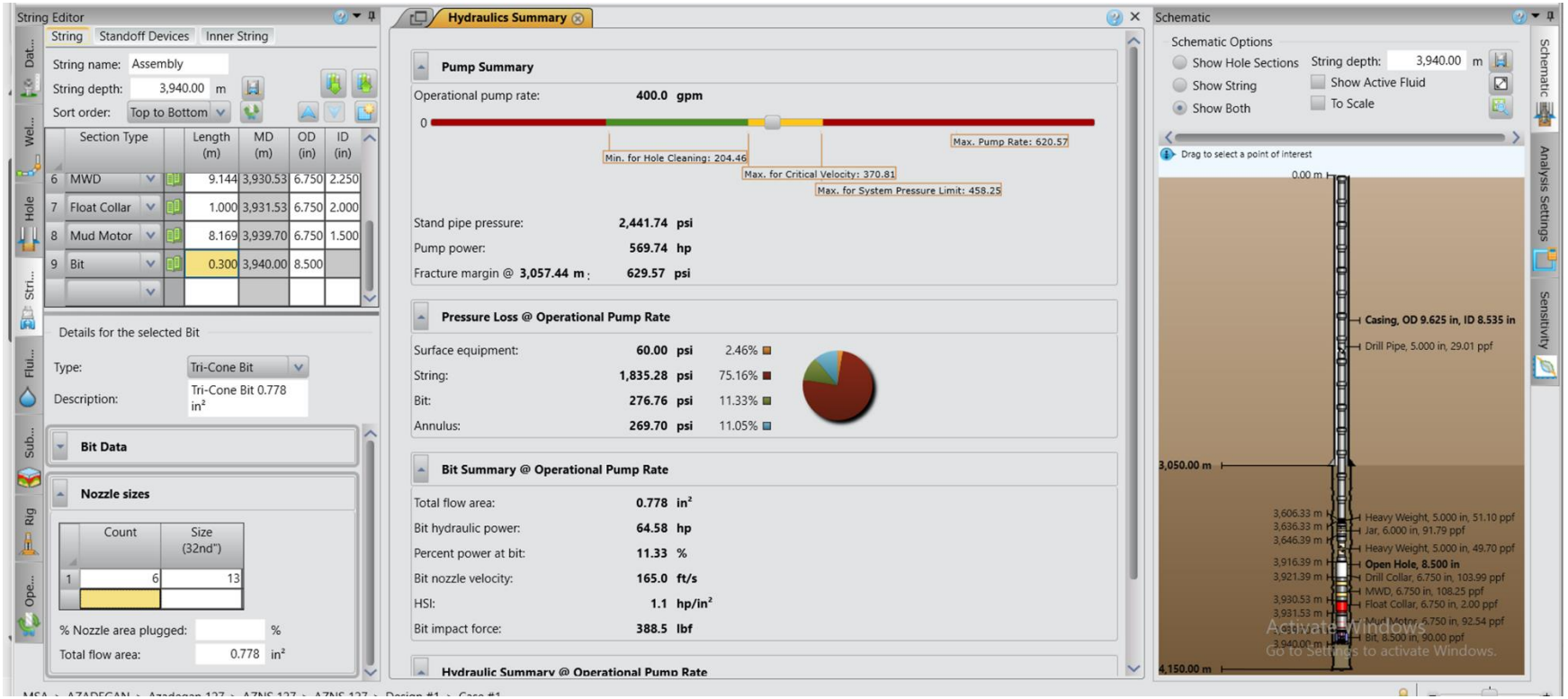
Completion Fluid program will be issued in due time.

8.9 Cement Program

Cement program will be issued in due time.

9. Bit & Hydraulic Program

9.1 8 – ½" Hole Section

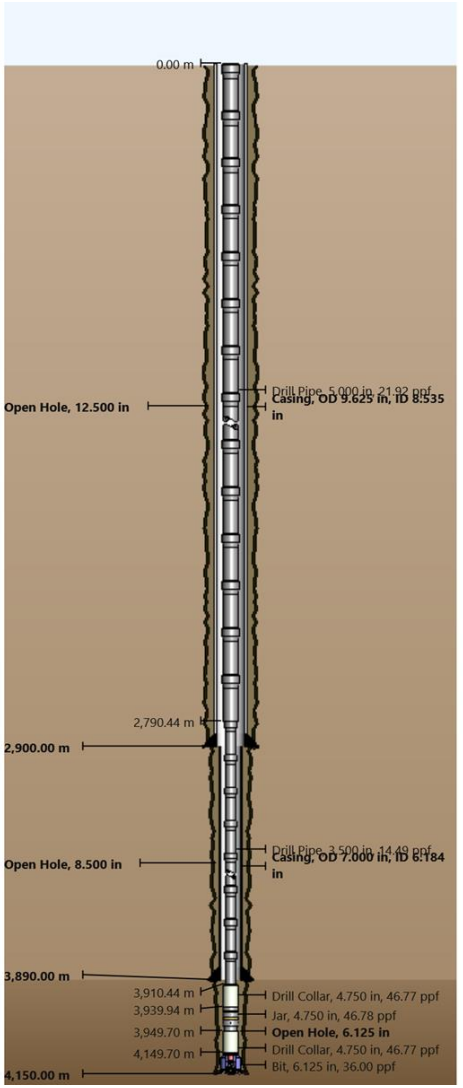


9.1 6 – 1/8” Hole Section

Pressure Loss @ Operational Pump Rate			
Surface equipment:	150.00	psi	5.13%
String:	1,906.57	psi	65.25%
Bit:	393.28	psi	13.46%
Annulus:	472.18	psi	16.16%

Bit Summary @ Operational Pump Rate		
Total flow area:	0.451	in^2
Bit hydraulic power:	57.35	hp
Percent power at bit:	13.46	%
Bit nozzle velocity:	177.8	ft/s
HSI:	1.9	hp/in^2
Bit impact force:	320.1	lbf

Hydraulic Summary @ Operational Pump Rate		
Mud weight:	13.90	ppg
ECD at shoe:	14.40	ppg
ECD at bit:	14.57	ppg



10. Casing Program

Casing/Liner Characterize Overview

Table 10-1: Casing/Liner Characterize Overview

Casing Size (in)	Depth (m)		Weight (PPF)	Grade	Connection	Burst (psi)	Collapse (psi)	A.S.F	Triaxial	Formation Remark
	From	To								
20”	0	146	94	K-55	BTC	Already run and set				AJ
13-3/8”	0	1441	68	N-80	BTC	Already run and set				GS
9-5/8”	0	3909	53.5	T-95	TSH Blue TPCQ	Already run and set				GD
7” ST	2955	3940	29	L-80	Premium Gas Tight Con. TPCQ	**8160**	**7030**	-	-	GD
7” 1st T.B	1355	2955	29	L-80	Premium Gas Tight Con. TPCQ	**8160**	**7030**			
7” 2nd T.B	100	1355	29	L-80	Premium Gas Tight Con. TPCQ	**8160**	**7030**			
5”	3840	4150	18	L-80	Premium Gas Tight Con. VFJ	**8290**	**7250**	-	-	FH

11. BOPs, Wellhead and Completion Schematic

10.1 BOPs Specification and Configuration

Table 11-1: AZNS-127 Well BOP Specification

Casing String	Wellhead	BOP	Hole Size
13-3/8" Intermediate casing	Section "B" Casing Head Spool 20-3/4", 3K BTM x 13-5/8", 5K TOP	13-5/8", 5K Annular BOP 13-5/8", 10K Double Rams (Pipe Ram & Shear Ram) 13-5/8", 10K Single Ram 13-5/8", 10K Drilling Spool 13-5/8", 10K Spacer Spool Adapter Flange 13-5/8", 5K BTM x 10K Top	12-1/4″
9-5/8" Production casing	Section "C" Tubing Head Spool 13-5/8", 5K BTM x 11", 10K TOP	13-5/8", 5K Annular BOP 13-5/8", 10K Double Rams (Pipe Ram & Shear Ram) 13-5/8", 10K Single Ram 13-5/8", 10K Drilling Spool 13-5/8", 10K Spacer Spool Adaptor Flange 11", 10K BTM x 13-5/8" 10K TOP	8-1/2"

BOPs Arrangement

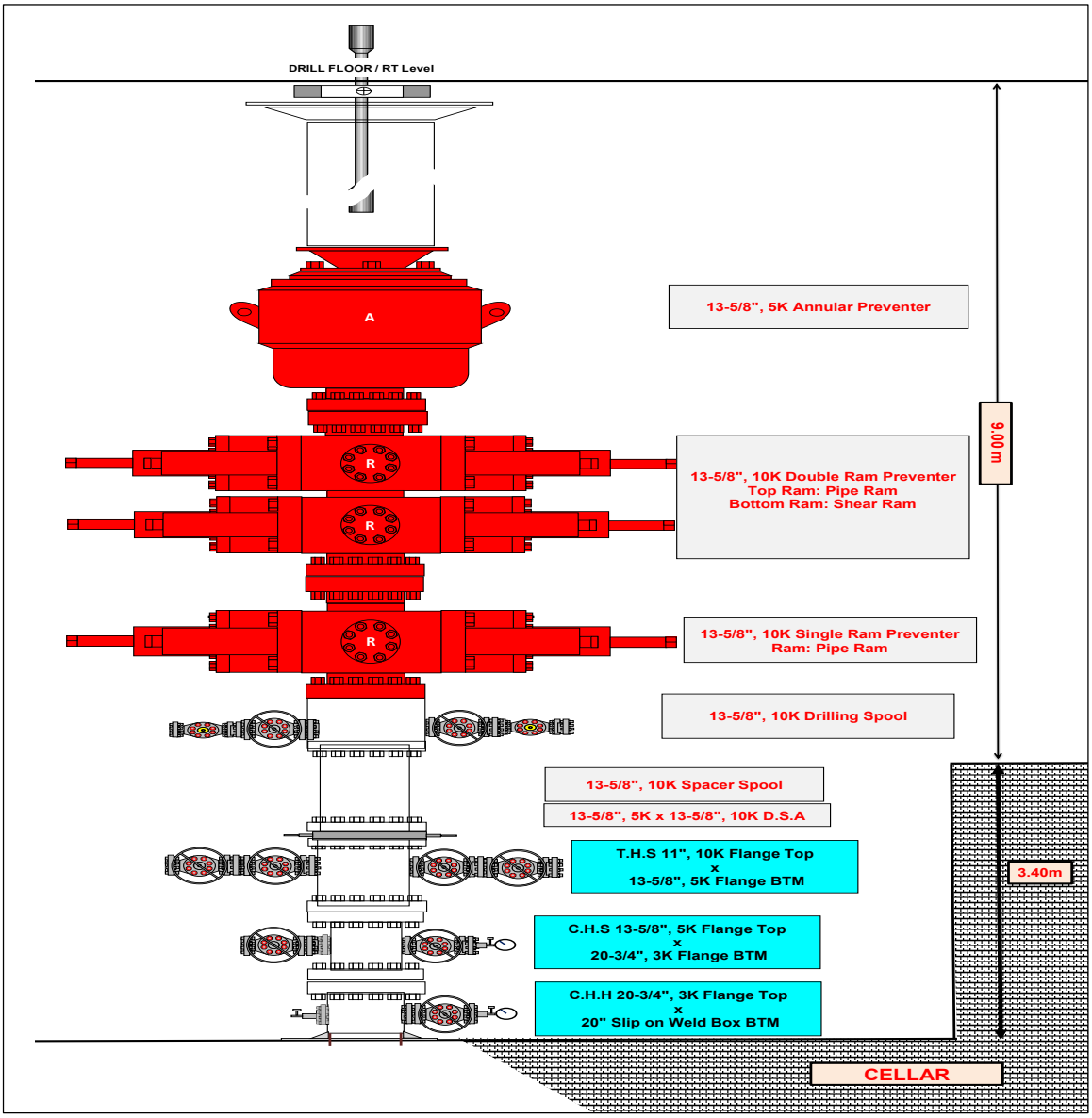


Figure 11-1: BOPs

Kill and Choke Manifold Arrangement

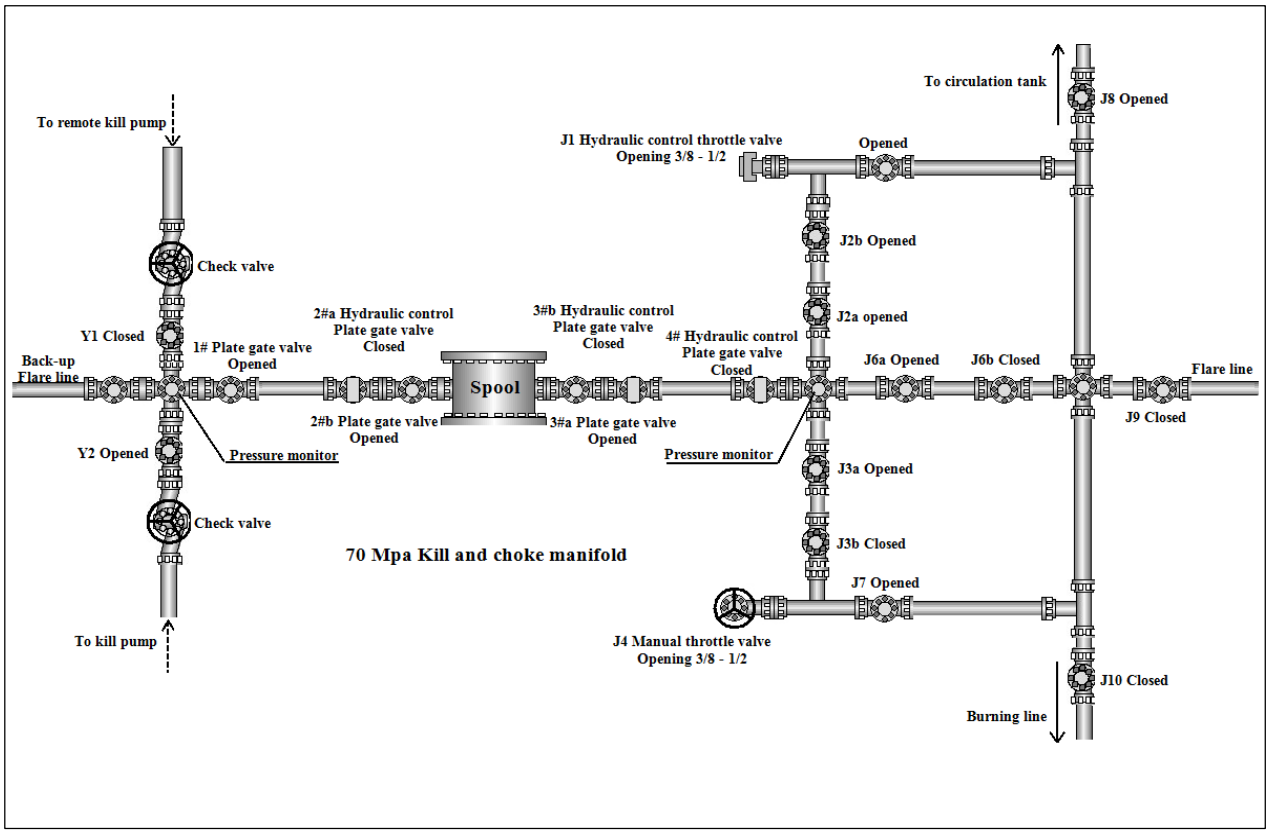


Figure 11-2: Kill and Choke Manifold Arrangement

10.2 Wellhead

Wellhead Material Selection Criteria

A. Selection of material class of Wellhead & X-mas Tree:

Table 11-2: H2S & CO2 Content in AZNS Offset Wells

Well No	Reservoir formation	H2S from PVT (mol%)	CO2 from PVT (mol%)
AZN2	Sarvak	0.13	6.71
AZN4	Sarvak	0.06	5.44
	Kuzhdumi	0	2.62
	Gadvan		2.64
AZN6	Gadvan		2.47

Considering the information above, API 6A 19th & NACE MR-0175 (2003) /ISO 1516, and Cl-content which is being most concerned, the South AZADEGA X-mas tree material and the tubing hanger should be FF0.5 class. As to the casing head, it's barely exposed to resident fluid, or only for quite a short time, when it comes to desert working condition, according to API 6A19th & NACE-MR-0175 (2003) /ISO 1516, the tubing spool should be EE0.5, the casing head material class should be DD0.5 (ANSI 4130) , stainless steel cladding in ring groove

B. Selection of PSL (product specification level):

According to API 6A 19th, The final PSL for wellhead and X-mas tree are as below:

Table 11-3: PSL for Wellhead and X-mas Tree

Item	Specification
Casing Head Housing assembly	PSL-1
Casing spool assembly	PSL-1
Tubing spool assembly	PSL-3
Tubing hanger	PSL-3G
X-Tree assembly	PSL-3G

C. Selection of PR (Performance Requirement)

According to API Specification 6A 19th edition, performance requirements are specific and unique to the product in the as-shipped condition. All products shall be designed to perform according to the requirements of API 6A clauses 4 and clauses 10 while in the pressure and temperature ranges and corrosion environment. Other requirements include load capability, cycles, and operating force or torque. There are two performance requirement levels: PR1 and PR2. Valves operating as safety valves shall be of the PR2 performance level and meet the requirements of Annex I in API 6A For SA wells, the guideline for PR selection is that all parts not contacting well fluids in normal production period are PR1, all parts contacting well fluids are PR2 as follow, Casing head housing and casing head spool assembly - PR1. Tubing head spool, tubing hanger and X-mas tree - PR2.

D. Temperature Class

Environment temp:
Well temp: 0.0244×4100+27.384=127℃
According to API 6A 19th, temp should be U (-18℃-121℃)

Table 11-4: Temperature Class for Wellhead and X-mas Tree in AZNS Oilfield

API TEMP. CLASSIFICATION	OPERATING RANGE ◦C
U	-18 TO 121

E. Designed Pressure

Base on wells data of this project, and API spec 6A 19TH, considering the planning casing program, Wellhead and X-mas tree system for the project is designed as followed:

Table 11-5: Wellhead and X-mas tree Designed Pressure in AZNS Oilfield

Item Well type	Casing Head Housing	Casing Spool	Tubing Spool	X-mas Tree
Savark	20-3/4" 3K	13-5/8" 5 K × 20 3/4" 3K	11" 5K × 13 5/8" 5 K	5K
Gadvan / Fahliyan	20-3/4" 3K	13-5/8" 5 K × 20 3/4" 3K	11" 10K × 13 5/8" 5 K	10K
Water disposal wells	20-3/4" 3K	13-5/8" 5 K × 20 3/4" 3K	11" 5K × 13 5/8" 5K	5K

Item	Specification
Casing Head Housing assembly	MC:DD-0.5 3000PSI PSL-2 PR-2, TC: U
Casing Spool assembly	MC:DD-0.5 5000PSI PSL-2 PR2, TC: U
Tubing Spool assembly	MC: FF-05 5000PSI PSL-3, PR-2, TC: U (Savark & Water disposal wells)
	MC: FF-0.5 10000PSI PSL-3G, PR-2, TC: U (K&G&F wells)
Tubing hanger	MC: FF 0.5 disposal wells 5000PSI PSL-3G, PR2, TC: U (Savark & water)
	MC: FF 0.5 10000PSI PSL-3G, PR2, TC: U (K&G&F wells)
X-Tree assembly	MC: FF 0.5 5000PSI PSL-3G, PR-2, TC: U (Savark & Water disposal wells)
	MC: FF 0.5 10000PSI PSL-3G, PR-2, TC: U (K&G&F wells)

Wellhead Schematic

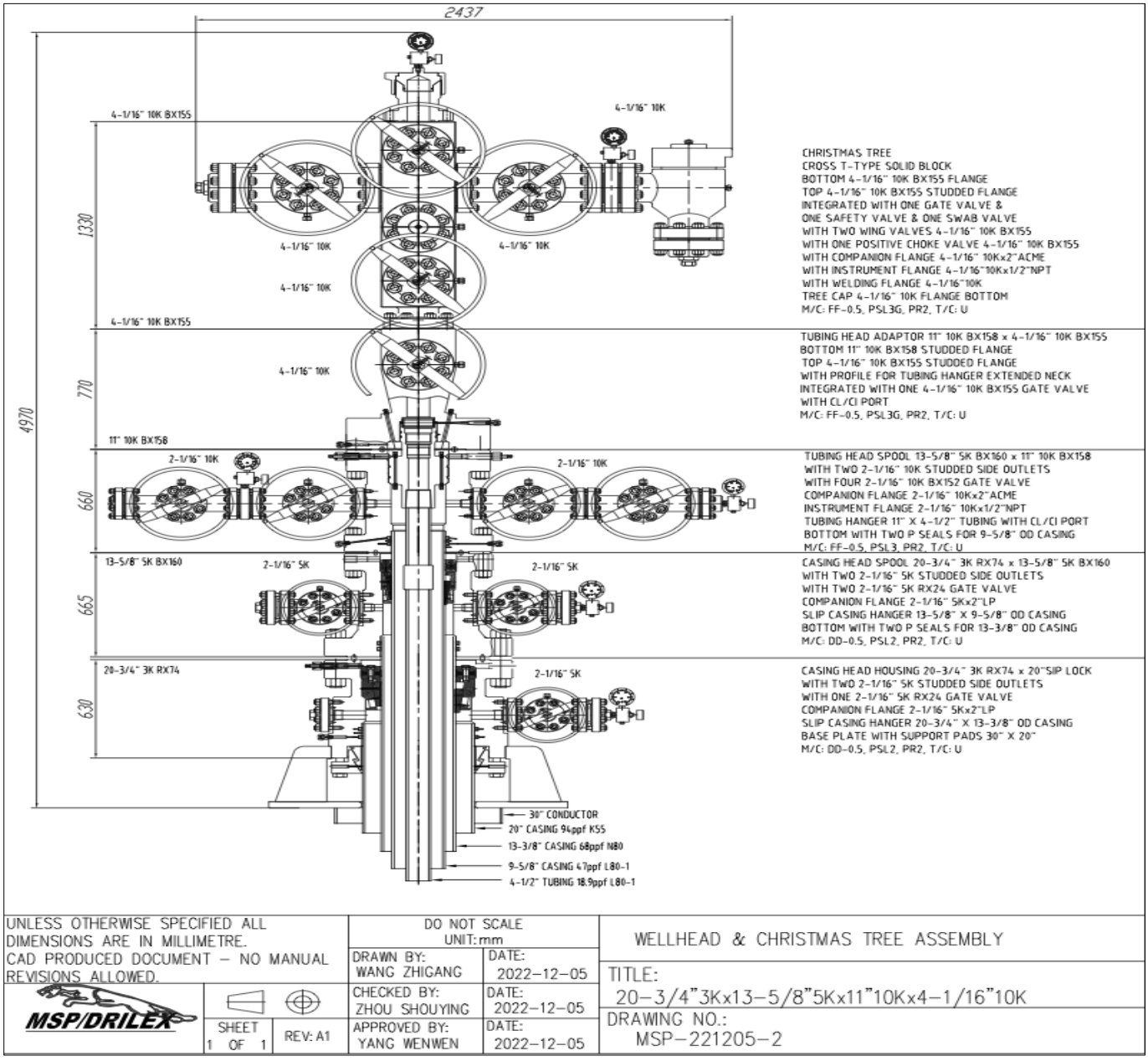


Figure 11-3: AZNS-127 Well Head Schematic

Wellhead Specification

Table 11-6: AZNS-127 Wellhead Specification

20-3/4" x 13-5/8" x 11" x 4-1/16" x 10,000 psi WP for Fahliyan wells	
Section A	**Casing head housing** for 20" (94lb/ft) OD casing x 20 3/4" 3,000 psi API flange **has hold down bolts for wear bushing,** with 2 x 2- 1/16" 3000 psi studded side outlets, test ports and seals. One side outlet has **one** 2 1/16"x 3,000 psi manual gate valve, 2" LP threaded companion flange and 2" bull plug threaded for ½" NPT and fitted with 0-3,000 psi gauge and needle valve and, the other side to have 2 1/16" x 3,000 psi x 2" LP threaded flange and 2" bull plug. Casing slip and seal assembly self-energizing to hang 13 3/8" (68lb/ft) OD casing. This section is installed on 20" casing stub by way of **slip lock** with landing plate on 30" conductor pipe. **Material DD-0.5, PSL-2, PR-2, Temp U. Height: 630mm** Material: Forged alloy steel 4130 Automatic casing slip material: Low alloy steel 4130
Section B	**Casing head spool** with 13 5/8" 5,000 psi top flange x 20 3/4" 3,000 psi bottom flange has **hold down bolts for wear bushing** and bottom preparation to seal on 13 3/8" OD casing with test ports to test annular space and casing seals. One side outlet has **one** 2 1/16"x 5,000psi manual gate valve, 2" LP threaded companion flange and 2" bull plug on one side and, the other side has **one** 2 1/16"x 5,000psi manual gate valve, 2" LP threaded companion flange, 2" bull plug threaded for ½" NPT and fitted with 0-5,000psi gauge and needle valve. Casing slip and seal assembly self-energizing to hang 9 5/8" (47lb/ft) OD casing. **Material DD-0.5, PSL-2, PR-2, Temp U. Height: 665mm** Material: Forged low alloy steel 4130 13 5/8"x 9 5/8" automatic casing slip material: Low alloy steel 4130
Section C	**Tubing head spool** with 11" 10,000 psi top flange x 13 5/8" 5,000 psi bottom flange has **hold down bolts for wear bushing** and bottom preparation to seal on 9 5/8" OD casing with test ports to test annular space and casing seals C/W T.D.Screws with 410SS material Side outlets have 1.900" sharp vee VR threads on both sides, to have **two** 2 1/16"x 10,000psi manual gate valve, 2"ACME 10K female thread companion flange, 2" bull plug threaded for ½"NPT and fitted with 0-10,000psi gauge and needle valve on one side and, the other side to have the first 2 1/16"x 10,000psi manual gate valve, 2 1/16" 10,000 psi instrument flange installed with 0ne ½"NPT and fitted with 0-10,000psi gauge and needle valve, and the second 2 1/16" x 10,000 psi manual gate valve, 2" ACME 10K female thread companion flange and 2" bull plug.(manual G.valve to be non-rising stem w/back seat capability). Clear bore for passing 8-1/2" BHA. The 11" x 4 1/2" **tubing hanger** has 4 1/2" 18.9ppf premium conn. Top & bottom thread one continuous 1/4" NPT control line exit for SSSV, two channels for **3/8"** chemical injection line exit, one 1/4" line exit for DHPG. 4" type "H" BPV thread preparation, one metal seal & two S-SEALS to pack off the upper neck and two S-SEAL to pack off the body of hanger, and tonging preparation on the lower neck. Tubing hanger shall have factory-torqued double pin tubing pup joint 4 1/2" 18.9 ppf, L80, New Vam approximate length 1600mm to facilitate makeup. **Material FF-0.5, PSL-3, PR-2 with exception to tubing hanger: FF 0.5, PSL-3G, PR-2, Temp U. Height: 850mm**

	Material of body: Forged low alloy steel 410SS G. valves, material of seats/gates & stem: Forged alloy steel AISI 410SS Seats & gates coated by tungsten carbide by HVOF method.
Section D	**Tubing head adaptor (w/one integral manual G. valve)** 11" 10,000psi studded flange x 4 1/16" 10,000psi full bore manual gate valve with bottom preparation to seal on tubing hanger (one metal seal & 2 S. seal) and with test ports to test annular space and seals, top prepared with one continuous 1/4" NPT control line exit, one 1/4" line exit for DHPG and two channels for **3/8"** chemical injection line exit. (Manual G. valve to be non-rising stem w/back seat capability) The T-TYPE **Christmas tree** consists of the following: (1) **Solid block valve (T-Type):** 4 1/16" 10000psi bottom flange for tubing adaptor. 4 1/16" 10000psi lower manual gate valve, 4 1/16" 10000psi upper gate valve with SSV w/ hydraulic actuator, including wire line cutting function & lock open cap, (2) 4 1/16" 10000psi manual swab valve. With two 4 1/16" 10000psi studded outside for wing valves. With one 4 1/16" 10000psi studded top flange for tree cap. (3) one wing valve 4 1/16" 10,000 psi full-bore manual gate valve, one **instrument flange** with one 1/2" NPT ports c/w one 0-10,000 psi pressure gauge and needle valve installed and one 4 1/16"10,000psi Cage type positive choke with 90 deg outlet.(With 2 no of each Wrench): positive choke with 2 sets of bean sizes 8, 12, 16, 20, 24, 28, 32, 36, 40, 44, 48, 52, 56, 60, 64, 128/64 inch and with welding flange prepared for 4" pipe(specification to be confirmed). (4) The other wing with one 4 1/16" 10,000psi full-bore manual gate valve and one companion flange w/2" ACME 10K female thread. (5) Quick union type Tree Cap above, with 4 1/16" 10,000psi flange at bottom, ACME thread union on top, including 1/2"NPT port for needle valve and pressure gauge. **Material FF-0.5, PSL-3G, PR-2, Temp U. Height of integral tubing head adaptor and gate valve=800mm**. Material of body: 410ss G. valves, material of seats/gates & stem: Forged alloy steel, AISI 410SS Seats & gates coated by tungsten carbide by HVOF method.

11. Waste Management Program

11.1 general Guidelines

- Prior to Spud-in, inspect complete Drilling Waste Management system prior start rig spud date. Make sure all equipment
- installation and prepare to work.
- Test all solid control equipment prior to spud to be able to use efficiently.
- Make an inventory of Dewatering chemical and cutting stabilization. Make sure to order more chemical if required by keeping in
- view the lead-time.
- Check the quality of treated water and should have enough reserve volume of water for transfer to the rig tanks.
- Check the total Hardness of treated water and treat accordingly.
- Calibrate the HACH test kit prior to using it.
- Confirm MSDS/TDS of all chemicals is available at rig site.
- Planning / Daily Monitoring and Reporting.
- Before to request chemical, confirmed by operation manager.

Daily

- DWM Report (to full distribution list)
- Water treatment transfer Report (Included properties and volume) if required.
- Meeting with the Drilling Supervisor to discuss properties, problems, De-Watering, Fixation recommendations and order anticipated chemicals or Equipment in a timely manner.
- Call project manager on a daily basis to discuss water properties, cutting stabilize, problems, and recommendations to the system.

Weekly

- Check Materials inventory and send to operation manager.
- Equipment’s check List.
- Final Volume of weekly Water treated and cutting fixation.
- SOC
- Purchase Request

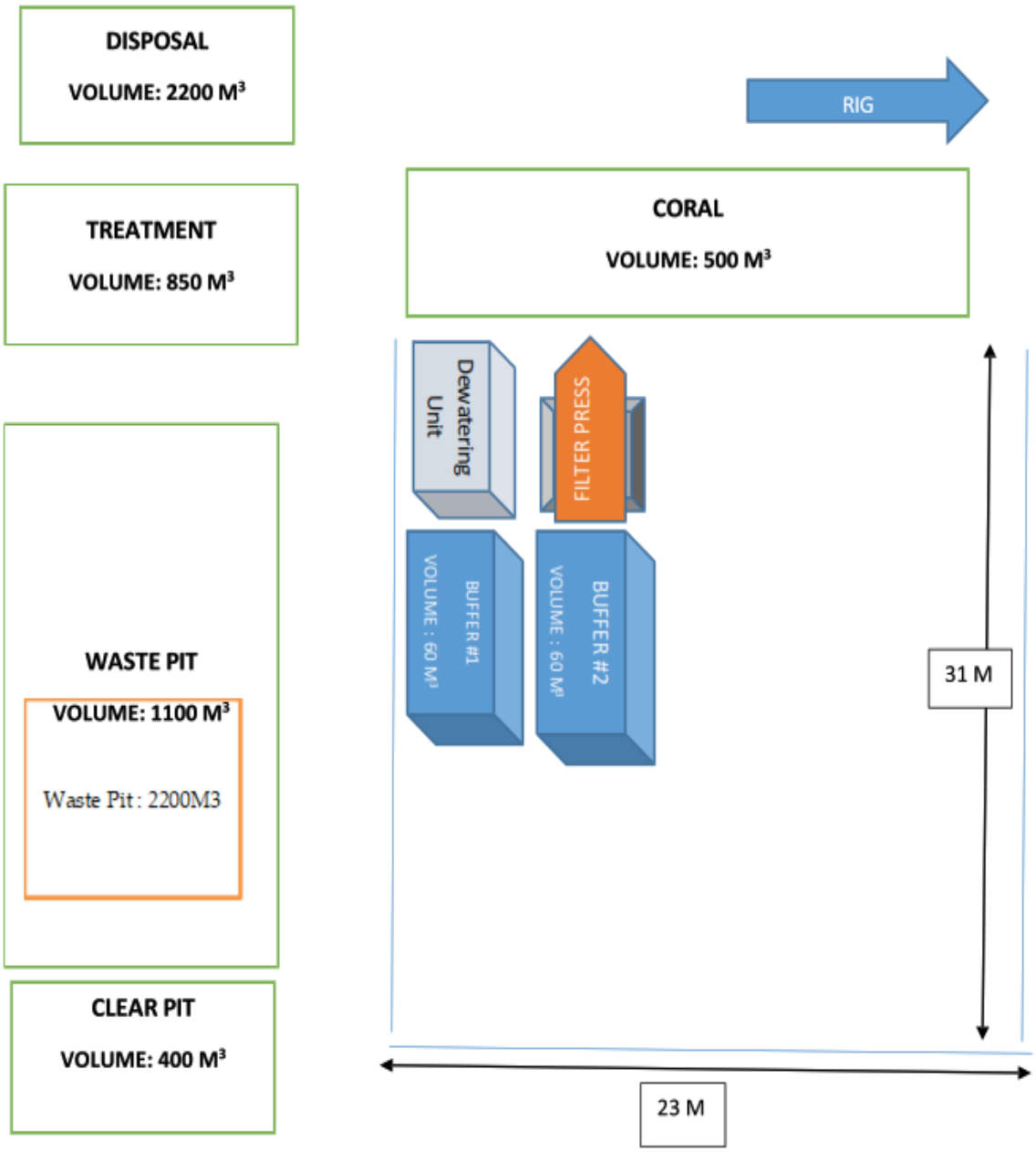
End of Section

- Interval summary (discussion and cost analysis) within 2 days of completion
- Update EOWR standard format

End of Well

- Full EOWR will be submitted to operator within 2 weeks of well completion.
- In drilling operations, good communications are the best tool of successful operation. So active communications are required between SPHK DFS & operator representative at location, keep good communication with driller and Drilling waste management supervisor for any indication of Dump Drilling fluid and keeping volume for them and the end give them De-Water solution for preparing Drilling fluid.
- Cleaning all the pit and location without any stain of the oil
- Read the DWM program very carefully and act according, if any doubt, please check with operation manager at any time (24/7).
- Always plan ahead regarding drilling operation & get all Chemical and keep ready all the equipment for using on location in time to avoid any rig time loss.
- The DWM Engineers are fully responsible for managing the inventory and other Equipment and chemicals.
- It is the DWM engineer's duty to monitor the shakers' screen and works and give advice to the drilling team for best performance.
- DWM Engineer should report the descriptions of the cuttings on DWMR in terms of type, volume and properties.
- Similarly report and evaluate the effectiveness of the sweeps on the Daily DWM Reports. Describe the observations in terms of shape, size and quantity of cuttings received at shakers after returns of each sweep. Pay attention on lead-time prior to order of any chemicals.
- Ensure that a member of the rig crew monitors the shale shakers at all times while drilling and circulating. This is vital for a number of reasons including monitoring mud-flow over the shakers and continuous observation of the nature of shale cuttings, i.e., early detection of caving.

11.2 Design Plan



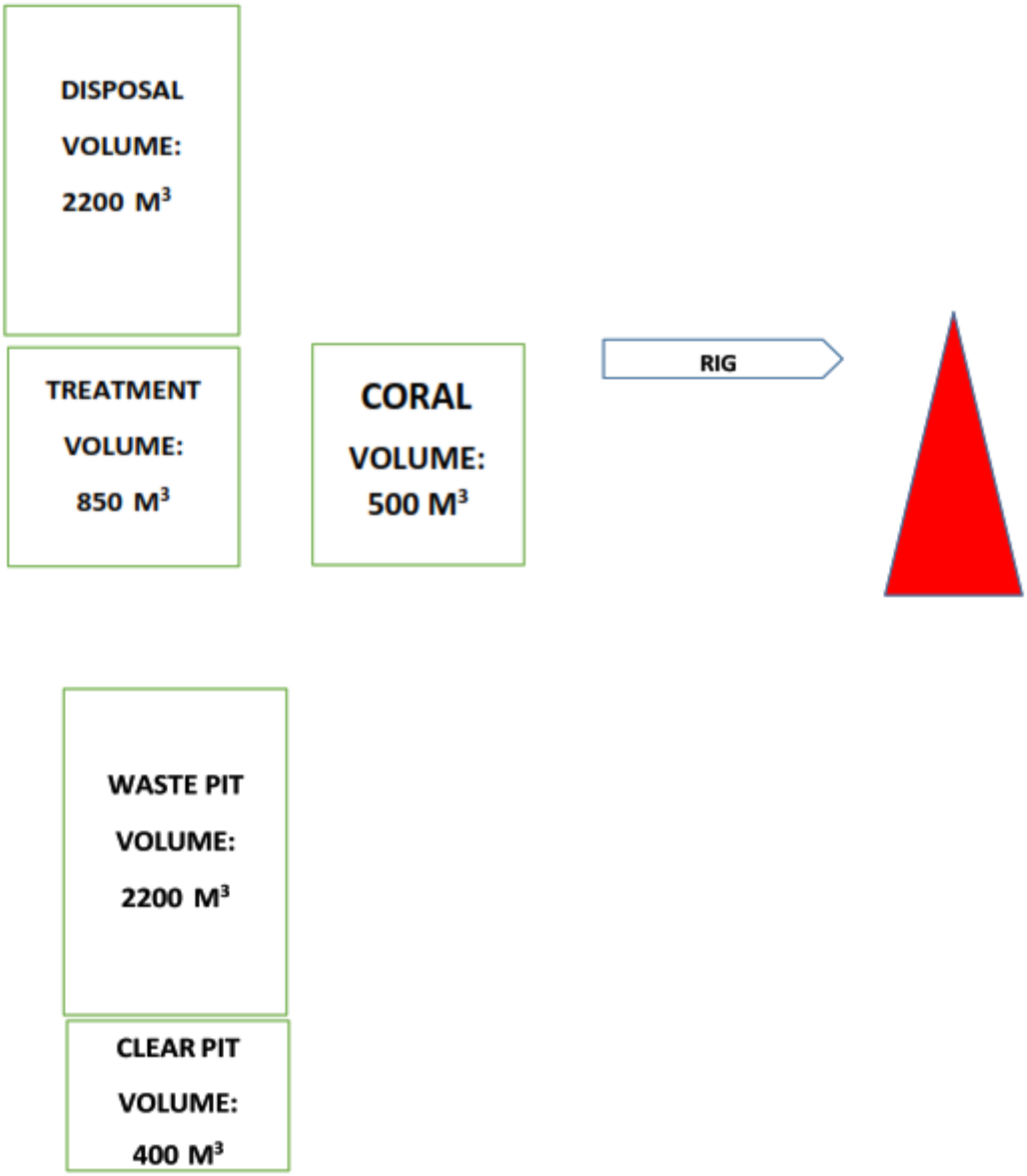


Figure 11-1 _ Drilling waste management Desig Plan

11.3 Expected Challenges

8 – 1/2" Hole Section from 3055m to 3940m

11.3.1.1 Challenges

- Hole instability issues
- Wellbore losses.
- Shale
- Tight hole and well instability

11.3.1.2 Recommendation

- It is mandatory to monitor closely the rheological properties in order to ensure adequate hole cleaning, run Hydraulic calculation to ensure that mud carrying capacity is enough to meet this critical goal.
- Mud can be treated with additional shale inhibitor if deemed necessary to control shale reactivity and washouts.
- Optimize the procedure to fight the losses (see the guide at the end of this mud program) and be ready with adequate LCM material to mix LCM pill as quick as possible.
- The previouse mud in hole must be dump, due to have enoght surface volume, keep 300 bbl heavy mud in mud pits as kill mud. The concentration of KCl in the system is insufficient; shale cuttings will be soft and mushy. If the concentration of KCl is sufficient, they will retain their integrity. As drilling commences the potassium ion becomes depleted and must be replenished. The rate of depletion provides an indication of how reactive the shales are. While drilling gumbo, the appearance of soft mushy cuttings at surface is a good indication that the potassium concentration should be increased. Other indicators include a drilled solids increase, a CEC increase and the occurrence of tight hole or mud rings.
- Since this system is very solids-sensitive, it is best to mix the system from scratch instead of converting an existing drilling fluid (containing drill solids) to a potassium chloride system.
- This system is very sensitive to solids and hardness contamination so the first step in mixing this system is to treat the hardness.
- Due to necessary of directional drilling, it is recommended to pump (at the first 20 bbl Low Vis pill follow by 20 bbl Hi Vis Pill should be pumped every stand) for clean the hole. For better understanding of hole cleaning procedure based on hole angels please refer to technical appendix.

- Due to face with torque and drag, it is recommended to mix water base mud lubricant 1 – 3% vol reduce torque and drag while drilling horizontal or vertical.
- In the directional drilling section, it is recommended to use Glass beads 2 – 3 ppb as due to have torque and drag.
- The mud weight in this interval is critical for successful drilling, hole condition will dictate the density. Mud weight will be increased by adding $CaCO_3$ F.

11.3.1.3 Interval Description

The 8 1/2“section will be drilled from KOP to an estimated section TD KCl/Caco3 Polymer mud system necessitating a minimum requirement of 80^{pcf} to 85^{pcf} based on nearby well data.

In case of having problem with Marl and Shale layer 3 - 5wt. % of KCl, should be added to system to minimize associated problem.

It is necessary to adjust mud weight and mud properties as hole condition dictate. Some tight hole, well flow and low ROP maybe experienced in this hole section. The drilling fluid engineer will advise the drilling supervisor of any extended circulation or sweep requirement. It uses clarified viscosifier to provide viscosity and carrying capacity and starch to reduce mud filtrate.

In the directional and horizontal section required amount of Lubricant such as Bit Lube and Glass beds will be added to active system to minimize torque and drag.

11.3.1.4 Product Concentration & Function

Salt / KCL / Polymer mud		
Product N.	**Ppb**	**Product Function**
Caustic Soda	0.5 - 1	Alkalinity Control
Soda Ash	1 – 2	Hardness Remover
Salt	Up to 98	non - solid weighting agent
KCL	10.7 - 18	Shale Inhibitor
Limestone Powder	Up to 113	weighting agent
Starch LV	12 – 14	Fluid loss controller
Xanthan gum	1-1.5	Viscosifier
Biocide	As Requirement	Anti-bacteria

MIXING PROCEDURE

1. PH, chloride, calcium and total hardness of drill water (that water which will be used for mud making) must be check by Mud engineer

2. Fill the mud tank with Drill water
3. add KCL as inhibitor
4. add Salt to reach mentioned mud weight
5. Remove Hardness by add Soda ash
6. adjust PH & alkalinity by Caustic soda
7. add Starch LV to control fluid loss
8. add Xanthan gum to reach good rheology
9. Add Limestone powder to reach Mentioned Weight
10. Check the pH and keep in range by add caustic soda
11. Mix Biocide as need

Table 11-1 Product Concentration & Function

11.3.1.5 Required Properties

Fluid Properties Summary		
MUD PARAMETERS		Interval
		IV
Bit diameter	inch	8 1/2"
Density	pcf	80 – 85
Marsh Viscosity	sec	40 – 50
PV	cps	ALAP
Yield Point	lb/100ft²	12 – 20
Gel 10 Sec	lb/100ft²	2 – 4
Gel 10 min	lb/100ft²	3 – 6
API Filtrate	cc/30'	< 2.5
pH	-	9 – 10
Ca++	mg/l	< 800
Total Hardness	mg/l	<1200
MBT	ppb	< 5
LGS (drilled Solids)	% vol	< 7
KCl	% by wt	3 - 5
CL^-	Mg/l	180000-188000

Table 11-2 _ Required Properties

11.3.1.6 Drilling Fluid Volume & Material Estimation

Interval			Drilling Hole 8.5"		
Interval Size Inch			8.5"		
Previous Casing Size Inch			9 5/8"		
Casing ID			8.835		
From M			3050		
To M			3940		
Interval Length M			885		
Mud Type			Salt KCL Pol Mud		
Mud Weight PCF			80-85		
Hole Volume bbls			1732		
Cased Volume			1528		
Formation Volume			204		
Dilution Factors SCE Effic. %	LGS%	Hole Enlargment %	18	6	15
Surface Volume bbl			800		
EX. volume (BBL)			479		
Total Volume bbl			1732		
Total volume of drill water for this Section			3011		

Products	Unit	Ppb for Mixing	Ppb for Treating	Expected Consump. For mixing	Expected Consump. For Treating	Total
Citric Acid	25Kg/Sx	As needed	As needed	As needed	As needed	As needed
Caustic Soda	25 Kg/sx	As Needed	As Needed	As Needed	As Needed	As Needed
Soda Ash	50 Kg/sx	As Needed	As Needed	As Needed	As Needed	As Needed
PAC	25 Kg/sx	0.06-0.14	0.06	7	4	11
Polymer Anionic	25 Kg/sx	0.04-0.08	0.04	4	2	6
Cement	Ton	5-15	6	15	16	31

Table 11-3 _ Drilling Fluid Volume & Material Estimation

6 – 1/8" Hole Section from 3940m to 4150m

11.3.2.1 Challenges

- Wellbore losses
- Possible HT environment
- Oil/Water Influx

11.3.2.2 Recommendation

- Hole cleaning is highly related to low shear rate parameters, so an adequate low shear rate yield point should have maintained to ensure hole cleaning. Run Hydraulic calculation to ensure that mud carrying capacity is enough to meet this critical goal.
- The previous mud will be transfer to this section.
- A suitable LCM material for reservoir should be available on location to combat down hole losses if occurred (see the guide at the end of this mud program).
- Since this system is very solids-sensitive, it is best to mix the system from scratch instead of converting an existing drilling fluid (containing drill solids) to a potassium chloride system.
- This system is very sensitive to solids and hardness contamination so the first step in mixing this system is to treat the hardness.
- Due to necessity of directional drilling, it is recommended to pump (at the first 20 bbl Low Vis pill follow by 20 bbl Hi Vis Pill should be pumped every stand) for clean the hole. For better understanding of hole cleaning procedure based on hole angels please refer to technical appendix.
- Due to face with torque and drag, it is recommended to mix WATER BASE MUD Lubricant 1 – 3% vol reduce torque and drag while drilling horizontal or vertical
- In the directional drilling section, it is recommended to use Glass beads 2 – 3 ppb as due to have torque and drag.
- The mud weight in this interval is critical for successful drilling, hole condition will dictate the density. Mud weight will be increased by adding $CaCO^3$ F.

11.3.2.3 Interval Description

After cementing the 7" inch Liner and confirmation of its proper cementing, the 6 1/8" section will be drilled from liner shoe to an estimated section TD of 4150 m horizontally by Salt/Polymer mud system necessitating a minimum requirement of 100^{pcf} to 106^{pcf} based on nearby well data.

It is necessary to adjust mud weight and mud properties as hole condition dictate. Some tight hole, well flow and low ROP maybe experienced in this hole section. The drilling fluid engineer will advise the drilling supervisor of any extended circulation or sweep requirement. It uses clarified viscosifier to provide viscosity and carrying capacity and starch to reduce mud filtrate.

11.3.2.4 Product Concentration & Function

Salt /$CaCo_3$/ Polymer mud		
Product N.	**Ppb**	**Product Function**
Caustic Soda	0.5 - 1	Alkalinity Control
Soda Ash	1 – 2	Hardness Remover
Salt	Up t0 109	non - solid weighting agent
KCL (Optional)	10.7 - 18	Shale Inhibitor
Limestone Powder	Up to 258	weighting agent
Starch LV	12-14	Fluid loss controller
Hematite	Up to 98	weighting agent
Xanthan gum	0.25-0.5	Viscosifier
Biocide	As Requirement	Anti-bacteria
MIXING PROCEDURE		
1. PH, chloride, calcium and total hardness of drill water (that water which will be used for mud making) must be check by Mud engineer		
2. Fill the mud tank with Drill water		
3. add Salt to reach mentioned mud weight		
4. Remove Hardness by add Soda ash		
5. adjust PH & alkalinty by Casutic soda		
6. add Starch LV to control fluid loss		
7. add Xanthan gum to reach good rheology		
8. Add Limestone powder & Hematite to reach Mentioned Weight		
9.Check the pH and keep in range by add caustic soda		
10. Mix Biocide as need		

Table 11-4 Product Concentration & Function

11.3.2.5 Required Properties

Fluid Properties Summary		
MUD PARAMETERS		Interval
		V
Bit diameter	inch	8 1/2"
Density	pcf	100-106
Marsh Viscosity	sec	45 – 60
PV	cps	ALAP
Yield Point	lb/100ft²	14 – 24
Gel 10 Sec	lb/100ft²	2-4
Gel 10 min	lb/100ft²	3-7
API Filtrate	cc/30'	< 2.5
pH	-	9 – 10
Ca++	mg/l	< 800
Total Hardness	Mg/l	<1200
LGS	%	<10
MBT	ppb	< 5
CL^-	Mg/l	180000-188000

Table 11-5 _ Required Properties

11.3.2.6 Drilling Fluid Volume & Materials Estimation

Interval			Drilling Hole 6.125"		
Interval Size Inch			6.125"		
Previous Casing Size Inch			7"		
Casing ID			6.835		
From M			3940		
To M			4150		
Interval Length M			210		
Mud Type			Salt Pol Mud		
Mud Weight PCF			100-106		
Hole Volume bbls			872		
Cased Volume			847		
Formation Volume			25		
Dilution Factors SCE Effic. %	LGS%	Hole Enlargment %	18	6	15
Surface Volume bbl			800		
EX. volume (BBL)			54		
Total Volume bbl			1672		
Total volume of drill water for this Section			1726		

Products	Unit	Ppb for Mixing	Ppb for Treating	Expected Consump. For mixing	Expected Consump. For Treating	Total
Citric Acid	25Kg/Sx	As needed	As needed	As needed	As needed	As needed
Caustic Soda	25 Kg/sx	As Needed	As Needed	As Needed	As Needed	As Needed
Soda Ash	50 Kg/sx	As Needed	As Needed	As Needed	As Needed	As Needed
PAC	25 Kg/sx	0.06-0.14	0.06	4	2	6
Polymer Anionic	25 Kg/sx	0.04-0.08	0.04	2	1	3
Cement	Ton	5-15	6	8	9	17

Table 11-6 _ Drilling Fluid Volume & Material Estimation

11.4 Treatment Properties

Properties range	
Turbidity	<50 NTU
Suspended solid	<80 mg/lit
Total Hardness	<800 mg/lit
Aluminum	< 3 mg/Lit
Color	Clearly
Calcium	<600 mg/lit
pH	7-9

11.5 Volume for Stabilize & Dewater

Mud total volume bbl	1700
Theoretical Drilling Cutting bbl	125
Shaker out Cutting (Drilling Cutting with hole enlargement) bbl	0
Surface Water bbl (per day)	40-50
Dumping Approximately Mud (Base on solid estimation)	41 bbl cutting of mud + 90bbl of pit cleaning = 131 bbl
Cement and sodium silicate (Estimation volume increased)	33
Excess volume (Fixation volume of Cellar, Leakage and Rig repairing) by estimation BBL	0
Total Volume of fixation (BBL)	410
Total Volume of fixation cutting M^3	66
Total Volume	1700

11.6 Contingency Stock

Product	Unit size, lbs / Gal	Total Units
Caustic Soda	55LB	5
Soda ash	55 LB	5
PAC	55 LB	10
Polymer anionic	55 LB	10
Lime	55 LB	40
Citric Acid	55 LB	5
Cement	2200LB	20
Sodium silicate	55 Gal	14

Flocculation unit

This unit, has the bellow specifications:

- Fully automated system,
- Full 316 stainless steel material,
- Eqipped with high pressure level and pH sensors,
- Equipped with digital flow meter and totalizer to monitor flow rate, and
- Touch panel technology to control prose.



Figure 11-2 _ Flocculation unit.

400^{bbl} Dewatering unit

This tank provides more than 400 bbls capacity to store contaminated water for using in dewatering process and ability to adding chemicals such lime from hopper. Two Mixer 25 hp and Jet System give ability to holding any type of mud for hours without any stealing.



Figure 11-3 _ Conditional tank.

400^{bbl} Storage tank

Each tank provides more than 400 bbls capacity to store clear water for recycling.

High Performance Filter Unit

Filtration plays an important role in the natural treatment of waste water as it percolates through the soil. It is also a major part of most water treatment. Groundwater that has been softened, or treated through iron and manganese oxidation, requires filtration to remove floc created by coagulation or oxidation processes. Since surface water is subject to run-off and does not undergo natural filtration.

it must be filtered to remove particles and impurities. Filtration can be compared to a sieve or micro- strainer that traps suspended material between the grains of filter media. However, since most suspended particles can easily pass through the spaces between grains of the filter media, straining is the least important process in filtration. Filtration primarily depends on a combination of complex physical and chemical mechanisms, the most important being adsorption. Adsorption is the process of particles sticking onto the surface of the individual filter grains or onto the previously deposited materials. Forces that attract and hold particles to the grains are the same as those that work in coagulation and flocculation. In fact, coagulation and flocculation may occur in the filter bed, especially if coagulation and flocculation before filtration was not properly controlled. Incomplete coagulation can cause serious problems in filter operation.



Figure 11-4 _ High Performance Filter Unit.

Centrifugal pump

This pump can be utilized in different situation. 3 x 4 pump can be used for transferring processed water back to rig water tanks for reuse.



Figure 11-5 _ Centrifugal pump.

Submersible pump

These pumps designed to clear and waste water from water pits and tanks to designated areas or tanks. They are very easy to handle and can be very useful in case of transferring from bottom of tanks or pits.



Figure 11-6 _ Submersible pump.

200 bar wash down guns

These wash guns minimized water consumption and can be used on the shakers to unplug screens or washing al the area on the rig from rig floor to mud pits.



Figure 11-7_ 200 bar wash down guns.

Fluid transfer diaphragm pump

This pump works with air pressure and has the responsibility of transferring mud and contaminated water.
Three inches diaphragm pump transfers mud and waste from cellar to coral to keep cellar empty.



Figure 11-8_ 3" Diaphragm pump.

Two inches diaphragm pumps transfer waste water and mud between different spots on location. They also can be utilizing to transfer mud from mud tanks dead volume to treatment tanks.



Figure 11-9_ Diaphragm pump

These pumps designed to transfer clear and waste water from water pits and tanks to designated areas or tanks. They are very easy to handle and can be very useful in case of transferring from bottom of tanks or pits.

DWM laboratory cabin

Lab equipment for DWM is consisting of:

- Retort kit,
- API Filtrate test kit,
- Testing reagents and equipment for Chloride, Calcium, Hardness and PH, and
- Hatch Water Test kit





Figure 11-10_ Hatch Water Test kit

This kits analys all conserned parameters, including aluminum, iron, nitrat, nitrite, zinc, phosphate, sulfate, sulfide, copper and cyanide concentration.

Storage container





Figure 11-11 _ Storage container.

Loader



Figure 11-12 _ Loader.

12. Health, Safety and Environment

12.1 General Quality, Health, Safety & Environmental Requirements

- PPL/PEDEC's committed to conducting operations in a manner which minimizes the risks to the health and safety of those involved in, and affected by, such operations. PPL/PEDEC's also committed to conducting its operations in a manner which minimizes the impact on natural and 'built' environments in the area of operations.
- This section sets out minimum Quality, Health, Safety and Environmental (QHSE) requirements for the WORK that MSA and subcontractors will be required to comply with whilst conducting operations for PPL/PEDEC.
- PPL/PEDEC may perform audits and assessments of MSA's and its subcontractors' performance under the CONTRACT. PPL/PEDEC reserves the right without prior notice to conduct quality, health, safety, security and environmental audits of the operations.
- MSA shall at all times be responsible for performing the WORK in a manner so as to protect against loss of life, injury to people, damage to property and the environment. The detailed manner means and methods of performing the WORK shall be under the control and direction of MSA.
- Prior to the start of Operations, certain QHSE processes must be completed. These QHSE processes include developing the final QHSE plan format and frequency of any QHSE audits, permitting and licensing and PEDEC's approvals. MSA shall ensure that sufficient resources are dedicated to this task to ensure completion prior to mobilization.
- PPL/PEDEC may at any time stop the WORK or any portion of the WORK without penalty to PPL/PEDE Cif the WORK is not conducted in accordance with the Quality, Health, Safety or Environmental guidelines specified in this CONTRACT or by the laws of the appropriate jurisdiction. PPL/PEDEC reserves the right to require MSA to remove any person from the operations in the case of non-compliance with QHSE procedures. PPL/PEDEC may also prohibit the use of equipment and plant in cases where that equipment or plant does not meet the required standards or is deemed unsafe.
- MSA shall ensure that activities associated with this CONTRACT are conducted in compliance with applicable national and local HSE laws and regulations, MSA's standards and PEDEC's requirements and IADC documents.

- Limits of Authority
- PEDEC's Representative(s) do not have the authority to relax health, safety or environment standards. If they judge the operation to be inadequate with regards to any aspect of quality, health, safety, security or the environment, then the operation may be immediately suspended.
- MSA shall be required to develop a project-specific QHSE plan to manage the WORK and associated risks. This plan shall, as a minimum, address topics included in the list detailed in Item 7. Where existing MSA QHSE publications detail the topics listed, it is sufficient to either include them verbatim, or provide reference to where they may be found in MSA's QHSE Manuals. The final project-specific QHSE plan shall satisfy MSA standards, PPL/PEDEC requirements and applicable HSE laws and regulations.
- MSA should establish an effective HSE system and implement it strictly.
- MSA must adhere to the standards and regulations of Health, Safe and Environment constituted by the local government.
- MSA must strictly adhere to the procedures / standards and regulations of Health, Safety and Environment promulgated by NIOC.
- MSA should comply with all applicable local government regulations including guidelines and recommended practices and any other Iranian authorities having jurisdiction.
- All personnel should respect the local culture, traditions and social values of the indigenous population of the Iran and South Azadegan region.
- MSA shall hire qualified and experienced worker and familiar his personnel with hazards especially with H2S and Mine and UXO (unexploded ordinance) according to related procedure.
- MSA shall conduct proper training about UXO (unexploded ordinance) and mine awareness to all personnel

12.2 Health & Safety Management

As per national/local and client (PPL/PEDEC) standards and MSA/OEOC HSE plan. Following are minimum requirement for the AZNS project:

- MSA/OEOC shall take all necessary safety measures in relation to the WORK to be provided and shall conduct itself and its work force in such a way as to comply at all times with the provisions of the national and local health and safety regulations pertinent to the contracted Scope of WORK.

- MSA/OEOC shall also comply with all and any of PPL/PEDEC'S safety standards and requirements as particularly applicable to the Operating Area and/or as may be imposed by PPL/PEDEC from time to time.
- MSA/OEOC and its subcontractors shall take all necessary steps to provide a safe and healthy working environment for employees, PPL/PEDEC's personnel and third parties in the performance of the WORK.
- MSA/OEOC shall operate a management system that ensures:
- MSA/OEOC's and sub contractors' personnel operate safe systems of work at all times for the performance of the WORK.
- Hazards associated with the work are identified and appropriate actions are taken to prevent accidents.
- Risks to safety and health of personnel involved in the WORK are identified and assessed, and any actions and measurements are implemented to either eliminate or reduce these risks to a level that is considered as low as reasonably practicable, given the nature of the WORK.
- Personnel are aware of the identified hazards and risks to their safety and health and to the safety and health of others. The detailed safe working procedures established by MSA/OEOC to ensure that the WORK will be conducted with due regard to the QHSE aspects, shall be detailed or referenced in the QHSE plan. Copies of the source documents shall be provided to PPL/PEDEC if the procedures are referenced only.

QHSE Representative:

- MSA/OEOC shall provide a dedicated, on-crew QHSE Representative. The primary responsibility of the QHSE Representative shall include, but not be limited to, the maintenance and monitoring of the implementation of HSE guidelines and procedures and where required provide HSE awareness training for the crew. Such activities shall be approved and actively supported by MSA/OEOC at corporate management level.

Training and Competence:

- MSA/OEOC shall ensure that its employees and those of its subcontractors are trained and competent to perform the WORK in a safe and environmentally responsible manner. Such training and demonstration of competence will include but not be limited to vocational, safety, emergency, environmental and other training required by regulatory agencies and IADC/OGP guidelines. MSA/OEOC's senior staff shall have received training in safety management.

Accident/Incident Reporting:

- MSA/OEOC shall report all incidents. Any hazardous incident/near miss involving PPL/PEDEC, MSA/OEOC's, or any third-party personnel, plant or equipment, shall be reported to PPL/PEDEC, whether or not injury to personnel or damage/loss to plant or equipment resulted. The following incidents shall be verbally reported immediately to the on-site PPL/PEDEC Representative(s): damage to property or equipment belonging to PPL/PEDEC, third parties, MSA/OEOC or its subcontractors; injury to personnel of PPL/PEDEC, third parties, subcontractors or MSA/OEOC; near misses which may have had serious potential consequences; environmental damage; media attention. An initial written report, containing factual information only, shall be prepared by MSA/OEOC and delivered to the PPL/PEDEC Representative within twenty-four (24) hours of the incident.
- PPL/PEDEC reserves the right to participate in the investigation of any incident arising out of performance of the WORK.
- MSA/OEOC and all subcontractors shall have an accident and incident reporting system, details of which shall be included in the QHSE plan.
- MSA/OEOC shall maintain and forward to PPL/PEDEC, a monthly summary of its safety performance.
- MSA/OEOC shall prepare and forward to PPL/PEDEC, a weekly report stating all incidents and near misses that occurred during that week.
- All MSA/OEOC and subcontractor personnel and PPL/PEDEC Representative shall attend safety meetings on a regular basis. PPL/PEDEC Representative shall be given a copy of the minutes of the meetings.

Safety Equipment & Personal Protective Equipment (PPE) requirements:

- MSA/OEOC shall ensure that appropriate PPE compliant to national standards is supplied to MSA/OEOC's personnel involved in the WORK. PPL/PEDEC intends to use water-based mud and hence MSA/OEOC's Personnel that may be exposed to this must be provided with appropriate PPE.
- MSA/OEOC shall be solely liable for all costs in equipping its own and all other personnel on contract to, or under the control of, MSA/OEOC with the required safety equipment. MSA/OEOC will provide to all MSA/OEOC's Personnel appropriate PPE throughout the WORK. Any defective PPE will be replaced without delay. Personnel without the correct PPE will not be allowed to continue work, until replacement PPE has been issued. If any person is disallowed to work owing to inadequate PPE, then PPL/PEDEC will deduct his day- rate from MSA/OEOC's invoice.

- MSA/OEOC shall detail in the QHSE plan the PPE proposed for issue to the crew. This proposal should include the arrangements for replacement (due to wear and tear and damage) of this equipment.
- This well is committed to develop for oil in a responsible manner that promotes employees' and local inhabitants' health.
- A trained Paramedic and proper MEDIVAC equipment and ambulance shall be stationed on the drilling rig during the drilling campaign.
- An emergency medical evacuation plan shall be coordinated, and shall keep in touch with emergency response centers manned by qualified doctor 24-hours per day in Tehran or Ahwaz city.
- Medical emergency report and investigation shall be performed as soon as possible in accordance with Medical Emergency Procedures.
- MSA/OEOC shall, as a minimum, conduct the WORK and provide personnel, equipment, facilities and services consistent with good oil field practices and OGP Guidelines. An assessment of the Contract Area with regard to "Environmental and Occupational Health Risks" and "Local Medical Support" shall be provided by MSA/OEOC in the QHSE plan. This assessment and the resulting "Health Risk Assessment Score" will be discussed with PPL/PEDEC, and a final assessment will be agreed.
- To ensure the fitness of those personnel engaged to undertake the WORK, MSA/OEOC shall conduct periodic medical examinations of its staff and ensure that its subcontractors' personnel are also subject to periodic medical examination. As a minimum this shall include pre-hire medical examinations of temporary local personnel and bi-annual medical examinations of its permanent and expatriate staff. Evidence should be made available to PPL/PEDEC to verify an appropriate program of immunizations consistent with the recommendations of the World Health Organization for the area of operations.
- MSA/OEOC shall be responsible for the cost of medical examinations as required to provide PPL/PEDEC with medical certificates which confirm the fitness of MSA/OEOC's assigned personnel for the WORK. Each drilling rig shall be equipped with a hospital, doctor and an ambulance type-2 level.

Drug Abuse and Control:

- MSA/OEOC warrants that it shall obtain valid written consent of all Personnel to a search of property of such Personnel when PPL/PEDEC suspects that such Personnel have in their possession drugs and/or other prohibited substances or any other item likely to prejudice safety or security. PPL/PEDE will provide a representative to attend the search.

- MSA/OEOC warrants that it shall obtain valid written consent of all Personnel to medical testing for the presence of drugs or other prohibited substances.
- MSA/OEOC further agrees that any Personnel who refuse to provide such consent or consents or withdraws such consent or consents shall at PPL/PEDEC's option be removed from the WORK and shall not, at PPL/PEDEC's option, be assigned to any future services to be provided by MSA/OEOC for PPL/PEDEC.
- MSA/OEOC further agrees that it shall immediately notify PPL/PEDEC in the event that MSA/OEOC becomes aware that any of its Personnel has been found in possession of drugs and/or any other prohibited substance. MSA/OEOC is required to notify PPL/PEDEC in the event that any Personnel provided for the WORK have previously been found in possession, or have abused drugs and/or other substances prohibited at Location. The acceptance of such Personnel for the WORK shall be at the sole discretion of PPL/PEDEC.

Camp Standards:

- MSA/OEOC shall abide by all statutory Iranian governmental health and safety regulations, and maintain acceptable standards for cleanliness and hygiene for quarters, camps and the Operating Area(s) in general. MSA/OEOC shall detail the standards and facilities proposed for its camp in the QHSE plan.

Emergency Preparedness:

- MSA/OEOC and its subcontractors shall have emergency preparedness and response plans plus procedures available at all times throughout the duration of the CONTRACT.
- MSA/OEOC shall take all necessary measures to ensure that:

I. Potential emergency situations have been identified and assessed;
II. An emergency response plan, including clear responsibilities, is established;
III. MSA/OEOC's management, employees and third parties are aware of response procedures and responsibilities;
IV. Regular drills and exercises are carried out to test the emergency response organization and procedures and that any resultant issues are properly addressed.

MSA/OEOC shall notify PPL/PEDEC immediately with respect to any emergency situation and shall provide regular updates on emergency planning procedures, tests etc.

MSA/OEOC shall detail in the QHSE plan a list of potential contingencies for which emergency response procedures are planned. The QHSE plan should also detail a schedule of proposed exercises and drills to verify the

effectiveness of the proposed arrangements in place. Development and implementation of the above plan shall incorporate any PPL/PEDEC assigned personnel to the project.

12.3 Environmental Management

- COMPANY shall comply with any and all laws, rules and regulations of governmental agencies having jurisdiction, which now exist or may be promulgated during the term of the CONTRACT, relating to the control and prevention of damage to the environment.
- Without prejudice to above, COMPANY shall ensure that all personnel use the highest prevailing standards of care and diligence to prevent unlawful disposal of all waste oil, water and other waste materials or fluids which may accumulate or be caused by COMPANY's performance of the CONTRACT. COMPANY shall not dispose of any objects, scrap, waste, rubbish, equipment, fluids or fuel into the surrounding area or allow anything to be stored loose where it can be driven into surrounding areas by wind. COMPANY shall take all reasonable precautions to prevent pollution of any nature or kind from its performance of the WORK. COMPANY shall have contingency plans and procedures in place for such events.
- In the event of any disposal prohibited by Item 3.2 above, whether by dumping, spilling, emission or otherwise arising from COMPANY's performance or non-performance of the CONTRACT, COMPANY shall be fully liable for and bear the costs of removal of any such item(s) dumped, spilled, emitted or otherwise disposed regardless of the sole, concurrent or contributory negligence or breach of duty (statutory or otherwise) of PPL/PEDEC. In the event of any such disposal, COMPANY shall immediately undertake all reasonable remedial and protective measures required in such circumstances.
- COMPANY shall notify PPL/PEDEC immediately with respect to any pollution, loss, damage, claim or demand (or occurrence which may give rise to the same) resulting from the CONTRACT. COMPANY shall report to PPL/PEDEC any incidents of non-compliance with legislative and regulatory environmental requirements that occur during the performance of the WORK. Without relieving COMPANY of any of its obligations under Item 3.2, if COMPANY does not promptly proceed to perform its obligations under this Item 3, PPL/PEDEC may take part to any degree it deems necessary in the control and removal of any pollution, dumping, spillage or contamination which is the responsibility of COMPANY under this Item 3.2, or which is due to any act or omission of COMPANY. COMPANY shall reimburse PPL/PEDEC for costs arising from such actions of PPL/PEDEC upon receipt of invoice from PPL/PEDEC. PPL/PEDEC shall have the right to deduct such costs from any sum due or becoming due to COMPANY under the CONTRACT.

- COMPANY shall operate a management system that is able to demonstrate compliance with environmental laws, rules and regulations applicable to its activities, products and services.
- MSA/OEOC and its subcontractors shall take all necessary steps to ensure that the environmental impacts of their work are responsibly managed and shall take all necessary measures to ensure that:
- COMPANY's personnel are aware of the environmental impact of the COMPANY's operations and are sufficiently trained and competent to perform the WORK.
- COMPANY's personnel operate systems, which will ensure that significant environmental impacts of WORK are identified and managed in accordance with a commitment to legal compliance and continual improvement.
- The environmental risks of operations associated with the WORK are evaluated and appropriate actions are taken to prevent accidents and reduce pollution, and that contingency plans are in place in the event of an incident.
- COMPANY will have adequate processes for monitoring the environmental performance of their WORK.
- COMPANY will, as a minimum, conduct the WORK and provide personnel, equipment, facilities and services consistent with IADC Environmental Guidelines. COMPANY's proposals for environmental protection should be detailed in the QHSE Plan. This should include, but not be limited to:

- procedures and precautions to minimize damage to crops and vegetation, the local ecology and disturbance of wildlife and farm animals;
- recycling and waste disposal;
- Clean-up and restoration of seismic lines, Rig sites access roads and trails, fixed sites occupied during the survey.

- An ecological team shall be set up by COMPANY, with the purpose of collecting all waste, rubbish, food scraps, office waste and other discarded material from camps, created by the WORK. Such activities shall be performed on a daily basis. Such material shall be appropriately disposed of, according to PPL/PEDEC approved methods or existing methods established by COMPANY, which are proven to be acceptable to PPL/PEDEC. Well location and camp site shall be delivered to PPL/PEDEC without any waste, rubbish, food scraps, office waste and other discarded material. PPL/PEDEC has the right to hold Completion certificate till cleaning the well location and camp site by COMPANY.
- The access road and well location construction shall be accorded with construction program of well location. It minimizes any impacts on the local residents, prevent all soil contamination, prevent all ground water contamination, and prevent all surface water contamination. Cutting and waste water pit shall be lining and drainage system shall be good and to avoid farm and season canal contamination.
- Management regulation for transportation truck, drilling operation machine and tools shall be carried out strictly to avoid damage farm and trees, etc.

- Well site shall be flat and clean. All the equipment, materials, drilling tools shall be put in order.
- Control the rig noise to decrease the influence to surroundings while drilling at the inhabitation area.
- COMPANY shall perform on site segregation of wastes prior to transfer to the transfer area and/or delivery to the transportation COMPANY in accordance with the following waste hazard classification.
- Non-Hazardous Wastes (Re-usable, Recyclable and Recoverable Wastes)
- Ferrous and non-ferrous metals, plastics, and wood wastes, PETs and potentially paper and glass
- Materials shall be separately segregated and provision may be made for size reduction and containerization equipment.
- The COMPANY/Generators are encouraged to seek for recycle/re-use alternatives based on the Local available facilities, for final disposal of these materials.
- It should be noted that landfill disposal of metal waste materials and wood wastes shall be prohibited. For wood wastes re-use of and/or salvage of commercial timber, or deliver/release to
- local community are recommended.

Hazardous Wastes:

- Hazardous Waste, such as clinic waste, oil waste, waste produced by painting, insulation works shall be duly segregated.

Facilities of environment protection supplied to well site:

- Mud purifying circulation system
- Cooling water spraying and circulating system for drilling pump
- Special recovery tank for deposed oil
- Reserve house for valuable mud chemicals
- Automatic mud weighting supply system
- BOP stack is equipped according to drilling design
- Define a separation area for gathering all waste according to waste management procedure.
- Requirement of environment protection on location Sewage pit Area of the sewage pit: 1500 m2
- Prior to spud mud pit and sewage pit must comply with the standard. Install cooling water spraying and circulating system for drilling pump and sewage circulating system for cleaning rig floor and equipment.
- Install BOP stack and pressure test shall be operated according to the drilling design to avoid blowout.
- Install oil supply lines and check whether there is any leakage.

Management of environment protection on location:

- In drilling operation keep mud and sewage from overflowing. If it happens, clean up timely.

- Mud chemicals shall be stored according to the standard and shall not be scattered around well site. If mud chemicals scatter while loading or unloading, clean it up at once. The sack of mud chemicals shall be recovered. Do not dispose it on site.
- Enhance water management at well location and do not wash equipment with fresh water. Stop water flowing for long time. Overhead tanks shall be utilized.
- Oil management shall be carefully at the well site. No oil dirt on site, no dripping and leaking at oil tank and manifold system area.
- After the well is completed, reinforce the sewage pit， solidify the disposed mud and have environment procedure fulfilled.
- Install fuel supply lines and keep them in good condition.
- All the drilling fluid materials shall be recovered and shall not be left at the well site after completion.
- The environmental protection personnel from COMPANY and operator side shall check and accept the well site environment according to the regulation after completion. Only well site environment is qualified, the well can be delivered.
- Transition Area for Hazardous Waste shall also be provided by each COMPANY in his allocated area,
- PPL/PEDEC shall approve each area before use as well. The transition area of hazardous waste shall be:

- Paved, fenced, marked and illuminated
- Equipped with appropriate Fire Extinguisher and other emergency response equipment (e.g., Eye-Wash and Emergency Shower, appropriate First Aid kits, neutralizer substances etc.).
- In compliance with the requirements of handling, transportation and storage of hazardous waste (based on usage chemical MSDS).
- Configured to accommodate a safe physical separation of incompatible waste materials (e.g., acids, and alkalis, organic materials and strong oxidizing agents).
- Capable enough to provide storage for the entire range of hazardous wastes generated during the work.
- The area shall be provided with Spill Containment and means of preventing run off to and from the area.
- All waste containers to be stored shall be appropriately labeled.
- Hazardous Waste shall be transferred in a timely manner, within a maximum period of 48 hours, for any period in excess of this permission shall be granted from PPL/PEDEC.

- Standard of utilization and discharging of wastes on location
- Conduct drilling operation according to the design, drilling mud is needed before spud.
- Sewage that cannot be recovered shall be put into the pit, do not discharge
- Make full use of waste water. Water in mud pump and water brake must be utilized by circulating. Water that is used to wash drill floor, shall be recovered after deposition. The recovery rate of waste water shall reach 85%.

- Disposed oil shall be recovered entirely. Recovery rate shall be 100%. Dumping of waste oil shall be prohibited.
- In drilling operation, non-chrome thinner high temperature mud system shall be used often if engineering is qualified.
- While drilling in special area adopt mud creation inhabitation techniques to decrease the waste mud creation.
- After completion, all the mud chemicals, bentonite, clays shall be recovered and shall not be left at the well site. Mud in purifying tank shall also be recovered after completion.
- In case of any flammable or noxious gas encountered, steps shall be taken at once to prevent kicks and blowout. Gas shall be guided into ignition equipment to burn it away.
- Burning material that may result in serious smog or horrible smell at the well site is not allowed. For cement or barite that may result in granular dust pollution, operation shall be performed in a closed system.
- For abandoned well, filled with cement plugs to seal the oil, gas or water formation.
- COMPANY/Generators are required to transport all generated wastes to the location/facilities designated, and licensed by authorities and/or approved/verified by the PPL/PEDEC (as appropriate).
- All wastes shall be transferred by approved Sub- COMPANY (which could be a 3rd party Sub-COMPANY of the COMPANY/Generator itself). The Waste Transporter and any other waste management service COMPANY shall be licensed in accordance with local regulations and are in compliance with such regulations. PPL/PEDEC shall verify and approve the Waste Transportation COMPANY by checking licenses and qualifications, audit and inspections, verification of human resources and received trainings etc.
- The transportation COMPANY shall ensure that all wastes are transported in a safe and responsible manner.

12.4 Security

As per PPL/PEDEC as main client and MSA/OEOC agreed plan.

- Security for the WORK shall be arranged and administered by the MSA/OEOC. MSA/OEOC shall be fully advised in writing covering all aspects of security upon. Company shall liaise with PPL/PEDEC QHSE to develop operational and contingency plans prior to Mobilization.
- The Security guards shall be appropriately housed and provided for. Transport for the Security guards to perform field activities shall be made available according to standard Iranian Oilfield procedures.

- All MSA/OEOC's personnel involved in the WORK shall possess the necessary security clearance documentation required by the Iranian authorities to work within the Operating Area.

12.5 Quality

1. Ensure that relevant experiences from the appraisal and production wells (if any) are captured and included in the operation programs /procedures.
2. Check lists to be prepared for all equipment, available for PPL/PEDEC prior to start of operations.
3. Procedures planned for all operations, available for PPL/PEDEC prior to start operations.
4. High focus and follow up of KPIs (Key Performance Indicators) by NIOC procedures.
5. Drilling risk assessment.
6. All goods and equipment shall be checked as per engineering design and inspection certificates, PPL/PEDEC approval shall be obtained before running any goods in hole.
7. Routine reports shall be submitted to PPL/PEDEC and any commented shall be corrected and informed engineering team.
   - Rig Specification shall be submitted to PPL/PEDEC any commented shall be corrected and informed engineering team.
   - Rig Layout shall be submitted to PPL/PEDEC any commented shall be corrected and informed engineering team.
   - Rig certification (COF) and record shall be submitted to PPL/PEDEC any commented shall be corrected and informed engineering team.
   - Apply PPL/PEDEC inspection work instructions.
   - Drilling String inspection must be done before and during operation based on client request and its certificates and record shall be submitted to PPL/PEDEC.
   - Pre-Spud check list must be verified by PPL/PEDEC before operation.
   - NDT certificates and record shall be submitted to PPL/PEDEC.

12.6 Meetings

HSE meetings shall be convened on a monthly basis with an agenda to discuss the general issues about HSE, any change in the contractor, in the nature and risk of the activities, HSE training program, safety analysis of the activities, as well as investigation and analysis of the accidents. The Contractor's site manager and HSE Representative

shall attend such meetings and report to the Client's site manager and HSE representative.

Weekly HSE meeting shall be conducted with the presence of MSA/OEOC HSE & Rig Contractor HSE Supervisor.

12.7 List of Minimum Required Contents

As per below tables:

Table 12-1: Minimum Onshore Rig HSE Requirements

Minimum Onshore Rig HSE Requirements	
ITEM/DESCRIPTION	**Specification**
General Safety Equipment	
Muster station sign board	2
Warning Sign (H2S Danger, Danger zone...)	As needed.
MSDS For all Chemicals	As needed.
T- Card Board	1 Panel in entrance of Rig & 1 Panel in Camp
Wind Socks	3
Fall Arrestor	Minimum 6 sets with different length Install in (Monkey board, 2 for Monkey Board ladders, Crown, In mast for hanging block, Under Rig floor)
Bug Blower (Explosion proof)	Minimum 3 no. Set in (Rig Floor, Cellar area, Spare)
Explosion Proof Torches	Minimum 6 (Company man, Tool pusher, HSE, Medic, Maintenance crew, Mud Crew)
Rescue Equipment	As needed.
Security Well Marker	By MSA/OEOC HSE Concern
Derrick man's escape system I.e. REM chute	By MSA/OEOC HSE Concern

Emergency shower with eye wash to be situated at assigned positions	Minimum 3 sets (Rig floor, chemical matters storage and handling areas)
Portable eye wash set	Minimum 5 sets
Life Ring	2
Spill Kit	3
EX Radio	At least 7
First Aid Kits	Rig Floor, Medic room, Camp Boss, Client Office, Store, Camp Galley
Rig Site Training Materials	Required
Personnel Protective Equipment	
Coverall+ Safety Helmet + Safety boots for all personnel on location and for visitors.	As needed.
Safety Eye protection & Ear Protections (Ear muff & Ear Plug)	As needed.
Anti-Acid (Apron, Gloves, Boot, Face Shield, chemical suits) for all operational personnel	As needed.
Fire Fighting Equipment	
Fire extinguishers: "Powder-50 kg"	7(In Rig site) 4(In Camp site)
Fire extinguishers: "Powder-12 kg" for all fire potential areas.	By MSA/OEOC HSE Concern
Fire extinguishers "CO2- 9 KG" For all electrical fire potential areas.	By MSA/OEOC HSE Concern
Fire extinguishers "Foam-45 Liter" For fire potential areas where required.	4 No. wheeled (Set in Diesel tanks, Engine generators in Rig site and Camp site)
Fire extinguishers "ABC 20 lbs" for all fire potential areas where required.	By MSA/OEOC HSE Concern
Mobile Foam Package (Fiberglass Body-Z4 Inductor) + 1000 Liter FFFP 3% Foam + S4 Low expansion Nuzzle (B.S)	By MSA/OEOC HSE Concern
Fire hoses with (B.S) couplings placed around the location as per firefighting plan.	By MSA/OEOC HSE Concern
Fire blankets for galley, Welding & Cutting operations	As needed.
Fire Fighting Suits	Minimum 4 sets in each Rig site and Camp site

Water pump diesel, capable of charging the fire main in case of deep well failure.	1(Rig site)1(Camp site)
Medical Equipment	
Minimum requirements for clinic and ambulance and relative medical equipment in accordance with fallowing requirements: 1. IOGP publication, Standards for local medical support, Report No: 6.44/222 2. Managing health for field operations in oil and gas activities, IOGP Report Number 343 (Revision – Version 1/ October 2011)	Required
Stretcher	Scoop Stretcher (1No.) Basket Stretcher with lifting belts and hooks) (1No.) Long Back Board with spider straps and immobilizer) (1No.) Army Stretcher(2No.)
Ambulance	1No. (B Type)

Table 12-2: Minimum Rig-less HSE Requirements for South Azadegan Project

Minimum Rig-less HSE Requirements for South Azadegan Project	
ITEM/DESCRIPTION	**Specification**
General Safety Equipment	
Eye wash & safety Shower to be situated at assigned position	(1) chemical storage and handling area
Muster station Sign board	2No.
Warning Sign (H2S Danger, Danger zone, ...)	Required
MSDS For all Chemicals	Required
T- Card Board	Required
First Aid Kits	Hospital, All offices
Wind Socks	2No.
Air Fans (Explosion proof)	1No.
Explosion Proof Torches	Required
Rescue Equipment	Required
Spill kit	3
EX Radio	At least 6

Security Well Marker	Required
Rig Site Training Materials	Required
Personnel Protective Equipment	
Coverall + Safety Helmet + boots for all personnel on location and for visitors.	Required
Safety Eye protection & Ear Protections (Ear Plug)	Required
Anti-Acid (Apron, Gloves, Boot, Face Shield, chemical suits) for all operational personnel	Required
Fire Fighting Equipment	
fire extinguishers: "Powder-50 kg"	At least 3No.
Fire extinguishers: "Powder-12 kg" for all fire potential areas.	Required
Fire extinguishers "CO2- 9 KG" For all electrical fire potential areas.	Required
Fire extinguishers "ABC 20 lbs" for all fire potential areas where required.	Required
Fire hoses with (B.S) couplings placed around the location as per firefighting plan.	Required
Fire blankets for galley, Welding & Cutting operations	Required
Fire truck (10000 Liter Water-1000 Liter Foam) Or (Firefighting systems)	1No.
Fire Fighting Suits	As per fire fighters' number
***The following equipment recommended meanwhile the risk can be retained in case that the company risk assessment considering the risk is so far as reasonably foreseeable. The measures and precautions also need to be as low as reasonably practicable.**	
Medical Equipment	
Minimum requirements for clinic and ambulance and relative medical equipment in accordance with fallowing requirements: 1. IOGP publication, Standards for local medical support, Report No: 6.44/222 2. Managing health for field operations in oil and gas activities, IOGP Report Number 343 (Revision – Version 1/ October 2011)	Required
Stretcher	Scoop Stretcher (1No.) Back Board (1No.) Army Stretcher (2No.)
Ambulance	1(B Type)

12.8 Well Control

1. Installation Requirements of BOP Stack and Control System:

- The distance between BOP control lines and relief lines shall be not less than 1m. Steel Gap Bridge should be placed in such a way to protect lines from damage by vehicle crossing. Sundries shall not be packed on the control line shield, on which welding operations are forbidden.
- The air supply shall be separated from that of driller's console and directly connected to the main air supply system, and shall be equipped with liquid/gas separators. Excessive bending and pinching on air hoses are forbidden.
- Electric power to BOP control system shall be directly from the main switch of the power board, and controlled by independent switches.
- Accumulators shall be in good condition, charged at its rated value, and ready for work at any time.

2. Installation Requirements of Choke Lines and Kill Lines, and Relief Lines:

a) Beside the drilling spool, one manual and one HCR gate valves on the side of chock manifold and two manual gate valves on the side of kill manifold shall be equipped, and the manual gate valve close to the spool shall be always open.

b) The high-pressure co-flex hose shall be installed as straight as possible and fixed by anchor after installation.

c) Relief lines (burning line and bleed line):

- No field welding is permitted on relief lines and the connecting style of relief lines must be flanged.
- The arrangement of relief lines shall take wind directions, dwellings, roads, oil tank areas, power lines and other facilities into consideration.
- While the two lines go in the same trend, the distance between the lines shall be at least 0.3m, and the two lines shall be fixed independently.
- The outlets of bleed line connecting with choke manifold shall be in a safe area which is at least 300m. The bleed line on the kill manifold side connecting with drilling spool shall be prepared and connected when necessary (about 70m from the wellhead).
- At every 10m~15m, turning points, and outlets, the lines shall be supported and fixed by cement foundations with anchor bolts or ground anchors. Hanging

portions shall be supported firmly. If crossing the stumbling blocks such as brooks or ponds is needed, metal bridging supports are necessary.

- Pre-buried anchor bolts for cement foundations shall not be less than 20mm or 3/4" in diameter, and greater than 0.5m in length.

3. Pressure test value and requirements:

a) A pressure test shall be conducted successfully where there is no pressure decline in 5 minutes for a low-pressure test and a minimum of 10 minutes for a high-pressure test.

b) Prior to each high-pressure test, a low-pressure test with 200-300 psi shall be carried out to verify sealing capability at the circumstance of low pressure and avoid washing out sealing elements.

c) High pressure test should be to the rated working pressure of ram preventers or the rated working pressure of the wellhead, whichever is the lowest, or to the rating required in DRILLING ROGRAM. Variable bore rams should be initially pressure tested on the largest and smallest OD pipe sizes that may be used during the OPERATION. Annular preventers, with a joint of drill pipe installed, should be tested to 70% of their working pressure. The high test for Kelly cocks, kill lines and choke lines should be to their rated working pressures, each part in the choke manifold is tested respectively to its rated working pressures, the relief lines be pressure tested to no less than 1,450psi (10Mpa).

d) When casing cup tester is used, the above test pressure should not exceed 80% of the internal yield pressure (burst pressure) of the exposed casing.

e) The control system of BOP shall be pressure tested to 3,000psi (21Mpa) for its reliability before putting into use.

f) Except that the BOP control system is tested with hydraulic oil, other well control equipment shall be pressure tested by clear water.

g) Testing tools

- All required sizes of the plug type testers, and cup type testers should be available for pressure testing purpose.
- When the cup type tester(s) is used for pressure test, the connection of the cup testers has to be checked, to ensure it will stand the load during the high pressures test. For surface casing, the cup tester is to be set in middle of the first casing joint in the top, so the first casing connection (between wellhead and

casing head housing) below the well head is to be tested too while testing BOP and wellhead.

4. Pressure test frequency:

All ram type preventers, annular BOP, valves, choke and kill lines, choke manifold, Kelly valves and safety valves should be pressure tested at the following frequencies:

- Prior to installation where possible.
- After installation of wellhead and BOP assembly and prior to drilling.
- After the disconnection or repair of any pressure containment seal in the BOP stack, choke line, or choke manifold, but limited to the affected component.
- Prior to drilling into a confirmed or suspected high-pressure zone.
- At any time requested by the COMPANY REPRESENTATIVE.
- In any case, every one month or during the first trip after the one-month interval with a maximum interval of 45 days. This frequency is not required for shear rams. As a minimum, the shear rams are to be tested prior to resume drilling operation after 9-5/8" casing string has been set and adequate WOC.

5. Work Requirement of Well Control Equipment:

- With shut-in casing pressure lower than 1000psi, while snubbing with annular preventer, drill tools shall have 18-deg shoulder tool joints, and the snubbing velocity shall not be greater than 0.2m/s.
- After shut off the annular preventer or ram preventer, with the shut-in casing pressure not greater than 2000psi, drill tools are allowed to pull up and down with the velocity less than 0.2m/s, however, rotating or tool joint passing through are forbidden.
- While drilling through pay zone and thereafter, BOP function (close-open) test for the BOPs shall be performed regularly.
- Kill manifolds shall not be used for drilling fluid fill-in.
- The maximum allowed shut-in casing pressure shall be remarkably indicated on choke manifolds.
- All gate valves of well control manifolds shall be tagged with serial numbers, and be indicated with on/Off.

- In case shear rams were used, after shearing the drill pipe with shear rams, retrieve the upper sheared drill pipe, lower a drill pipe joint with TIW valve installed into the hole and close pipe rams on it to prevent the possibility of shear ram's failure.

6. Management of Well Control Equipment:

- Well control equipment used by the drilling crew shall be managed and operated by designated personnel with clear post responsibilities.
- Well site shall have specified component storeroom and rubber element storeroom with air conditioners respectively. The temperature of storerooms shall meet storage requirements of components and rubber elements.
- At least 1 set is available as back-up for every type ram assembly in the well site. Ram preventer side door seals and ram rubber, every type gasket and bolt etc. should be enough for backup.

7. Killing Method and Shut-in Method:

- Driller's method is recommended for well killing.
- The hard Shut-in method is recommended for shutting in the well in Fahliyan reservoir.

8. Well Control during drilling:

a) Shut in Procedure in While on Bottom

- Position Kelly (if in use) and tool joint so that tool joints are clear of sealing elements.
- Shut off pumps, check for flow.
- If flow is noted, close the annular preventer and open remote-controlled choke line valve (HCR).
- If the well has been closed using the annular preventer, check space out (pipe rams vs. toll joint relative position) and close pipe rams and ram locks
- Bleed off pressure between pipe rams and annular preventer (if possible).
- Record shut-in drill pipe and casing pressures.
- Record volume gain and mark pits.
- Weigh mud in suction pits.
- Check blowout preventer and manifold for leaks.
- Check flowline and choke exhaust for flow.
- Check accumulator pressure.

b) Shut in Procedure in While Tripping

- If a flow is detected, install full opening safety valve and close same.
- Position tool joints clear of the sealing elements and close annular preventer and open remote-controlled choke line valve (HCR).
- Make up grey valve (dart type inside BOP on full open safety valve). Then Make up Kelly or top drive (insert a pup joint or single between safety valve and top drive) and open safety valve.
- Record casing pressure.
- Check blowout preventer, stack, and manifold for leaks.
- Check valves for correct position.
- Check accumulator pressure.
- Begin stripping to bottom.

c) Shut in Procedure for Empty Wells

- Try to RIH with drill pipe as more as possible if hole condition permits, and the drill pipe must be installed a Kelly cock or safety valve.
- Close the annular preventer and open remote-controlled choke line valve (HCR).
- If the well has been closed using the annular preventer, check space out (pipe rams vs. tool joint relative position) and close pipe rams and ram locks
- Bleed off pressure between pipe rams and annular preventer (jf possible).
- Record casing pressure

9. Well Control During Running Casing:

The well control while running casing is important and critical, best practices shall be followed, including but not limited to:

- Change out pipe rams to casing rams prior to running casing into hydrocarbon bearing formation, if possible.
- The crossover between casing and drill pipe has to be available, and good practice is to make it up with a joint of DP and keep on V-door.

10. Shut in Procedure During Logging:

Cable shall be pull out of the hole immediately when overflow is found during logging. If pit gain exceeds the specified amount, the cable shall be cut immediately and treat the overflow in accordance with the shut-in procedure for empty wells. Continuing to pull the cable through the closed annular preventer is not recommended.

11. Well Control During Testing:

- No difference to while DRILLING OPERATION, since either there is testing string in the well (same as Shut-in procedure while on bottom), or tripping the string. (Same as shut-in procedure while tripping).
- Likely there is some crude in the testing string, and precaution has to be taken while circulating the crude out of hole during killing operations to ensure it is properly handled without causing environmental impact.
- To minimize the formation damage, the killing fluids used shall be weighted brine if possible, however in well control case when it occurs, killing well safely will take priority rather than protecting formation from damage on selection of killing fluids.

12. Well Control During Stimulation:

- Similar to during testing.
- For any residual or alive acid to be circulated out during killing operations, measures have to be taken to treat the returned acid properly to avoid personnel from getting hurt, and without causing any environmental damage.

13. Well Control Drills:

Drills shall be conducted only when hole conditions and other rig activities permit. Drills should be scheduled so that crews can practice drills while on bottom, while tripping in and while out of the hole. Each operating shift shall perform at least one BOP drill under different situations per month.

14. Basic Requirement of Well Control:

- For 13-5/8”, 10K BOP, BOP should be schemed as follows: upper rams 5”, middle rams are shear rams, lower rams is variable rams (2-7/8” to 5”). For easily connecting, armored fire-resistant 10000psi and ID 4-1/16" flexible hoses are recommended as kill/choke lines which connected with BOP directly.
- The BOP stack and choke manifold, kill line and choke line, HCR and manual gate valves, inside BOP (including kelly cock) have to be H2S trimmed and stand the working temperature from -10 degrees C (in case of doing N2 for DST) to the maximum up to 100 deg C (estimated highest temperature condition).
- The BOP equipment shall be visually inspected, function tested, and hydrostatically pressure tested during commissioning and Acceptance of Drilling Unit prior to Commencement Date, then upon installation on a wellhead, prior to drilling any hole section with a potential hydrocarbon bearing reservoir, after breaking any BOP connection (such as after installation of an

additional wellhead section after running casing), and at least every 14-21 days while the BOP is installed on the Well.

- CONTRACTOR shall submit the current well control procedures manual of CONTRACTOR to COMPANY for review and approval by COMPANY. Valid IWCF Well Control Certificates (or IADC equivalent) must be held by all tool pushers, Drillers (optional) at Drilling Supervisor level and Drillers and Assistant Drillers at Driller level, as further specified in CONTRACTOR Personnel HSE Requirements.
- The accumulator must be of sufficient capacity so that it is possible to close one (1) ram and the bag, open a ram and the bag, re-close a ram and the bag and retain 1200psi accumulator pressure.
- The "OPEN" and "CLOSED" operating positions of BOP activators of the rams, annular, and remote actuated valves must be clearly identified on the BOP master control panel and remote panels, in English in addition to the native language of the majority of the senior drilling crew.
- Shear-ram BOP activator must have a flap-cover to ensure only pre-meditated and deliberated activation and to prevent accidental WORK.
- All lines and connections in the choke and kill manifold, standpipe manifold, choke and kill lines and high-pressure mud lines should be welded hard-piping with weld-on FMC Weco "hammer-lug" 1502 unions (or Western equivalent, such as Anson) or weld-on API flanges. The well control system should not include any threaded screw-on hammer-lug unions, flanges, or other threaded screw-on connections.
- All adapter or spacer spools between the wellhead and BOP must be manufactured in accordance with API 6A, and must be complete with all required studs, nuts, and ring gasket seals for both lower and upper connections; minimum spare parts should include two (2) OEM spare studs complete with nuts and two spare OEM ring gasket seals for both bottom and top connections.
- There should be a wide-bore gas vent line from the "poor-boy" degasser; the line and line outlet must be securely anchored, the line outlet must be a safe distance from the well cellar, rig floor, choke manifold, and mud logging unit.
- Well Control Equipment Welding: All welds in the well control system must be NDT inspected and records maintained of all welding and weld inspections, including choke and kill manifold, standpipe manifold, mud pump discharge manifold, choke and kill lines.
- Well Control Rig Floor Information: The following must be posted on the rig floor (in the Driller's shack / "dog-house"): Well Shut-In Procedure, Well Information Sheet (showing casing and hole sizes, depths, and such like), Kill Sheet (including most recent slow circulation rate pressures, formation integrity

leak-off test information and MAASP, present mud weight, present drill string and well volumes, and such like).

- The derrick substructure clear height (rotary table to ground level) needs to be enough for fitting in above BOP configuration (including a tubing head and a bell nipple). This need to be double checked in rig tender process. According to the calculation of height of BOP stack, the substructure clear height is at least 9 m.
- In the case if drilling 6-1/8" hole becomes necessary, or using 3-1/2" drill pipe for the cleanup run (including drill 7" shoe track), then taper drilling string (5" DP + 3-1/2" DP) is to be run, so a pair variable ram be installed in the single ram preventer. The defect installing variable ram in single ram preventer is that the ram may be damaged and the string fall in hole bottom due to the ram hanging up too much weight when the drill pipe be cut. If the pipe ram be installed in the single ram preventer and the variable ram be installed in the double ram preventer, it would be very dangerous due to the variable ram failure in frequently using.

12.9 H2S Monitoring and Prevention Measure

As per PPL/PEDEC SOW South Azadegan oilfield contains H_2S bearing formations, precautions shall be taken in place especially in 12-1/4” & 8-1/2” hole section (for typical Fahliyan vertical well) while drilling, completion and well testing operation.
H_2S can be released any time while drilling, completion and well testing operation, therefore:

- The Rig shall be prepared for H_2S (Training, procedures and protective gear).
- Make sure everyone on the Rig should be made aware of the hazard presented by H_2S during their induction when they first arrive on the rig.
- The mud shall be treated to protect the drill pipe and the casing.
- All training program shall be recorded and reported.
- H_2S emergency drill must be conducted frequently.
- Adequate sensors and alarms must be installed for the scope of monitoring the H_2S releases.

13. H2S Monitoring & Control

NOTE: Final plan for H2S Monitoring and Control will be issued in due time by Service company instruction and Rig Contractor standards based on Oilfield safety and NIOC standards.
Minimum H2S Requirements for Rig site will be as following:

Table 13-1: Minimum Rig H2S Requirements for South Azadegan Project

Minimum Rig H_2S Requirements for South Azadegan Project	
Gas Detection Equipment	
H_2S Service cabin	1 No.
H_2S Unit location point	Comply with MSA/OEOC HSE
Main Monitoring Panel	1 No. - 8 channels
Max. number of channels	8 No.
Number of H_2S channels	4 No.
Measuring range	0 - 100
Adjustable alarm siren	Minimum 4 set (Min 120db voice load)
Measuring range	0-25 PPM
Adjustable alarm setting levels	2 No. (low 10 PPM and high 15 PPM)
Gas concentration reading displays	As sensors head number
Electrical Classification	Required
Certification	Required
Monitor panel installation site	H2S Unit
Automatic switch-on back up battery (min. 12 hr. power supply)	1
SO_2 Equipment	
Portable Single Detector	2 No.
H_2S Sensors	
H_2S Sensor installation points	Comply with MSA/OEOC HSE
Electrical Classification	For hazardous area
Certification	Required

Spare sensor head	2 No.
Response gas concentration	10 ppm low & 15 ppm High
LEL Sensors	
LEL Sensor installation points	Comply with MSA/OEOC HSE
Electrical Classification	For hazardous area
Certification	Required
Response gas concentration	20 % low & 40% High
Spare sensor head	Required
LEL Sensor installation points	Comply with MSA/OEOC HSE
Electrical Cables (connecting sensors with monitor panels)	
Suitable for hazardous area	as much as required
Electrical Classification	as much as required
Visual-Acoustic H_2S Alarms	
Low level visual alarms	Yellow flashing lights
High level visual alarms	Steady red lights
Low level acoustic alarms	Intermittent horn
High level acoustic alarms	Continuous horn
Above items suitable for hazardous area	Required
Number of alarm stations	5 No.
Alarm points location	comply with MSA/OEOC HSE
Calibration kit	At least 2 No.
PORTABLE H_2S MONITORING SYSTEM	
Hand pump colorimetric tubes H_2S detectors	1 No.
H_2S colorimetric tubes ranges	Req., comply with MSA/OEOC HSE
H_2S colorimetric tubes quantity	Req., comply with MSA/OEOC HSE
Personal electronic H2S detectors	Min. 6 No
Multi Gas Portable Detectors	
Quantity	4 (two main two spare)

Measuring range	0 – 100 ppm
Alarm setting levels	two
Certification	Required
Calibration kit	Required
Electrical Classification	For hazardous area
Certification	Required
Respiratory Protection Equipment	
Air Compressors	
Quantity	2 No. (Electrical- Diesel)
Approved for air breathing	Required
Gross weight	Required
Flow rate capacity	480 lit./min each
Charging pressure	3200
Filtration and purifier device	Specify approval
Certification	As per EN-132
Alarm for contaminating gases	Required
Auto shut down for contaminating gases	Required
Air compressor driving system	Electrical
Suitable for refilling bottles of air breathing apparatuses	Required
Air test kit	At least 1 No.
FIXED CASCADE AIR BREATHING SYSTEM	
The fixed cascade air breathing system complies with code type	Specify
Certification details	Required
No. of racks	At least 3*8
Dimensions (LxWxH) each rack	required
Bottles certified as pressure vessel	Required
Total air storage capacity	50 Liter

At max. working pressure of	3300 Psi
Air Pressure regulator	Req., victory Type
Safety/relief valve	Required
Low pressure alarm device	Required
Stainless steel piping/lines	Required
Air Breathing Delivery Stations	
Number of stations	At least 7 No.
Stations locations	Comply with MSA/OEOC HSE
Each Fixed Station Consisting of:	
Outlet manifold	4-man manifold 3 No & 6-man manifold 2 No
Connection outlets	Required
Hose lines	20(25 feet)20(15 feet)
PORTABLE AIR BREATHING APPARATUSES	
Self-Contained Breathing apparatuses (SCBA 45 min)	25
EEBA (20 min.)	20
Mask type	Full face (positive pressure)
Compliance with specifications	Comply w/ EN 137
Approval Authority	required
Air bottle working pressure	required
Air bottle certificate	required
Air regulator	required
Low Pressure audible alarm	Required
Suitable to be quick connected to cascade outlet manifold	Required
Stowage suitcase	Required
SCBA Radio Communication	Required
Spare Bottles for SCBA	At least 3
BREATHING APPARATUSES location points	comply with MSA/OEOC HSE

14. Requirement

14.1 8-1/2”” Hole Section

Table 14-1: 8-1/2" Hole Section Requirement

Item No.	Supplier	8-1/2” Hole Section Requirement Description	Quantity	Unit
1	MSA/OEOC	8-1/2" Insert Bit, 4-1/2" Reg Pin, IADC Code: 517	1	Ea.
2	MSA/OEOC	8-1/2” PDC Bit, 4-1/2” Reg Pin, Matrix Body 16mm, 6 Blades, IADC Code:M433	1	Ea
3	MSA/OEOC	6-1/8” Bottom Bit, 3-1/2” Reg Pin, IADC Code: 537	1	Ea
4	MSA/OEOC	Nozzle (14/32”) W/ O-Ring & Snap Ring for 8-1/2” Tri-Cone Bit	2	Set
5	MSA/OEOC	Nozzle (15/32”) W/ O-Ring & Snap Ring for 8-1/2” Tri-Cone Bit	2	Set
6	MSA/OEOC	Nozzle (13/32”) for 8-1/2” PDC Bit W/ O-Ring	2	Set
7	MSA/OEOC	Nozzle (14/32”) for 8-1/2” PDC Bit W/ O-Ring	2	Set
8	MSA/OEOC	Nozzle (15/32”) W/ O-Ring & Snap Ring for 6-1/8” Tri-Cone Bit	2	Set
9	MSA/OEOC	Nozzle (16/32”) W/ O-Ring & Snap Ring for 6-1/8” Tri-Cone Bit	2	Set
10	Rig Cont.	Nozzle Installation & Removal Tool for Tri-Cone Bit	1	Ea.
11	MSA/OEOC	Nozzle Wrench for 8-1/2” PDC Bit	1	Ea.
12	Rig Cont.	8-1/2” Tri-Cone Bit Breaker	1	Ea.
13	MSA/OEOC	8-1/2” PDC Bit Breaker	1	Ea.

Item No.	Supplier	8-1/2” Hole Section Requirement Description	Quantity	Unit
14	MSA/OEOC	Nozzle Wrench for 6-1/8” PDC Bit	1	Ea.
15	Rig Cont.	6-1/8” Tri-Cone Bit Breaker	1	Ea.
16	Rig Cont.	8-1/2” Ring Gauge	1	Ea.
17	Rig Cont.	6-1/8” Ring Gauge	1	Ea.
18	Rig Cont.	8-1/2” Junk Sub, 4-1/2” Reg Pin-Box	1	Ea.
19	Rig Cont.	6-1/8” Junk Sub, 3-1/2” Reg Pin-Box	1	Ea.
20	Rig Cont.	6-1/8" Bit Sub, 4-1/2” Reg Box-Box	1	Ea.
21	Rig Cont.	4-3/4” Bit Sub, 3-1/2” Reg Box-Box	1	Ea.
22	Rig Cont.	6-1/2" Spiral Drill Collar, 4-1/2” Reg Pin-Box	12	Jt.
23	Rig Cont.	8-1/2" String Stabilizer (IBS, Spiral) 4-1/2” Reg Pin-Box	1	Ea.
24	Rig Cont.	6-1/2” Drilling Jar, 4-1/2” Reg Pin-Box	1	Ea.
25	Rig Cont.	XOS, 4-1/2” Reg Pin x 4-1/2” IF Box	1	Ea.
26	Rig Cont.	4-3/4” Spiral Drill Collar, 3-1/2” IF Pin-Box	18	Jt.
27	Rig Cont.	4-3/4” Drilling Jar, 3-1/2” IF Pin-Box	1	Ea.
28	Rig Cont.	XOS, 3-1/2” IF Pin x 4-1/2” IF Box	1	Ea.
29	Rig Cont.	5” HWDP, 49.7 PPF, 4-1/2” IF Pin-Box	9	Jt.
30	Rig Cont.	5” DP, G-105/S-135, 19.5 PPF, 4-1/2” IF Pin-Box	As Required/1000	Mt

Item No.	Supplier	8-1/2" Hole Section Requirement Description	Quantity	Unit
31	Rig Cont.	3-1/2" HWDP, 23.2 PPF, 3-1/2" IF Pin-Box	6	Jt.
32	Rig Cont.	3-1/2" DP, G-105, 13.3 PPF, 3-1/2" IF Pin-Box	± 400	Mt
33	Rig Cont.	9-5/8" CSG Scraper (43.5-53.5#), 4-1/2" Reg Pin-Box	1	Ea.
34	Rig Cont.	7" CSG Scraper (32-35#), 3-1/2" Reg Pin-Box	1	Ea.
35	Rig Cont.	Float Valve for 6-1/2" Bit Sub	2	Ea.
36	Rig Cont.	Float Valve for 4-3/4" Bit Sub	2	Ea.
37	Rig Cont.	Pipe Screen for inside 5" DP	3	Ea.
38	MSA/OEOC	X.O.S 7" , 29 ppf, 110SS, Premium Gas Tight Connection pin * 4-1/2" IF Box	1	Ea.
39	MSA/OEOC	Cement "G", 110 lb./sx.	As Req.	
40	MSA/OEOC	Cement Additives	As Req.	
41	MSA/OEOC	7" Float Shoe, L-80, 29#, Premium Gas Tight Con.	1	Ea.
42	MSA/OEOC	7" Float Collar, L-80, 29#, Premium Gas Tight Con.	1	Ea.
43	MSA/OEOC	7" LDC, L-80, 29#, Premium Gas Tight Con.	1	Ea.
44	PEDEC	7" Liner, L-80, 29#, Premium Gas Tight Con.	3840	Mt

Item No.	Supplier	8-1/2” Hole Section Requirement Description	Quantity	Unit
45	MSA/OEOC	Hydraulic Set, Rotating Liner Hanger, 7", 29-32 ppf, L-80, Premium Gas Tight Conn., Tandem Cone, with Integral/Split Weight Set Top Packer, 10 ft PBR, Load Capacity 2,000 KN, Packer Seal Capacity 10K, Temp. 150 ºC, with Liner Setting Sleeve c/w DPOB and Setting Tool thread, to Set Inside 9-5/8", 47-53.5 ppf Casing.	1	Set
46	Service Co.	7” Liner Hanger Running Tool	1	Ea.
47	MSA/OEOC	Liner Wiper Plug for 7” Liner, L-80, 29#, Premium Gas Tight Con.	1	Ea.
48	Service Co.	Polish Mill & Dress Mill for 7” Liner, L-80, 29#, Premium Gas Tight Con. PBR	1	Ea.
49	MSA/OEOC	5” DP, G-105/S-135, 19.5 PPF Pump Down Plug (Dart)	1	Ea.
50	MSA/OEOC	7" Coupling, L-80, 29#, Premium Gas Tight Con.	3	Ea.
51	MSA/OEOC	API Drift for 7” Liner, L-80, 29#, Premium Gas Tight Con.	1	Ea.

Item No.	Supplier	8-1/2" Hole Section Requirement Description	Quantity	Unit
52	MSA/OEOC	8-1/2" x 7" Bow Type Centralizer	As Req.	Ea.
53	MSA/OEOC	8-1/2" x 7" Rotating Type Centralizer C/W Nail and Pins	As Req.	Ea.
54	MSA/OEOC	9-5/8" x 7" Rigid Type Centralizer c/w Nails, Non-Welded	As Req.	Ea.
55	MSA/OEOC	7" Stop Collar c/w Spiral Nails	As Req.	Ea.
56	MSA/OEOC	API CSG Thread Lock Compound (Baker Lock)	As Req.	Can.
57	MSA/OEOC	API CSG Thread Sealing Compound (Dope)	As Req.	DR.
58	Service Co.	Drilling Fluid Service (Material, Lab. & Engineering)	As Required	
59	Service Co.	Mud Logging Service	As Required	
60	Service Co.	Waste Management Service	As Required	
61	Service Co.	Directional Drilling Service (Only MWD unit, MWD Tools, MWD Engineer)	As Required	
62	Service Co.	Fishing service	As Required	
63	Service Co.	H2S Service	As Required	
64	Service Co.	Casing Running Service	As Required	

Item No.	Supplier	8-1/2" Hole Section Requirement Description	Quantity	Unit
65	Service Co.	Cementing Service (Material, Pump Truck, Silo & Engineering)	As Required	
66	Service Co.	Well Head Service	As Required	
67	Service Co.	Liner Hanger Service	As Required	
68	Service Co.	Wireline Logging Service (Logging and Perforation)	As Required	
69	Service Co	Dry Testing & DST/TCP Service	As Required	
70	Service Co	Coiled Tubing Service	As Required	
71	Service Co	Stimulation & N2 Lifting Services	As Required	
72	Service Co	Completion Service	As Required	
73	Service Co	Slick line Service	As Required	

- **Qty & Spec to be confirmed with Contact's SOW.**
- **In case of any shortage or discrepancy, the Contract's SOW will be reference.**

14.2 6-1/8” Hole Section

Table 14-2: 6-1/8" Hole Section Requirement

Item No.	Supplier	6-1/2” Hole Section Requirement Description	Quantity	Unit
1	MSA/OEOC	6-1/8” PDC Bit, 3-1/2” Reg Pin, Matrix Body 16mm, 6 Blades, IADC Code: M433	1	Ea.
2	MSA/OEOC	6-1/8” Tri Cone Bit, 3-1/2” Reg Pin, IADC Code: 537	1	Ea.
3	MSA/OEOC	Nozzle (15/32”) W/ O-Ring & Snap Ring for 6-1/8” Tri-Cone Bit	2	Set.
4	MSA/OEOC	Nozzle (16/32”) W/ O-Ring & Snap Ring for 6-1/8” Tri-Cone Bit	2	Set.
5	MSA/OEOC	Nozzle (13/32”) for 6-1/8” PDC Bit W/ O-Ring	2	Set.
6	MSA/OEOC	Nozzle (14/32”) for 6-1/8” PDC Bit W/ O-Ring	2	Set.
7	MSA/OEOC	Nozzle Wrench for 6-1/8” PDC Bit	12	Ea.
8	MSA/OEOC	6-1/8” PDC Bit Breaker	1	Ea.
9	Rig Cont.	6-1/8” Ring Gauge	1	Ea.
10	Rig Cont.	4-3/4” Junk Sub, 3-1/2” Reg Pin-Box	1	Ea.
11	Rig Cont.	6-1/8" Bit Sub, 4-1/2” Reg Box-Box	1	Ea.
12	Rig Cont.	4-3/4" Spiral Drill Collar, 4-1/2” Reg Pin-Box	12	Jt.
13	Rig Cont.	4-3/4” Drilling Jar, 4-1/2” Reg Pin-Box	1	Ea.
14	Rig Cont.	XOS, 4-1/2” Reg Pin x 4-1/2” IF Box	1	Ea.
15	Rig Cont.	5” HWDP, 49.7 PPF, 4-1/2” IF Pin-Box	9	Jt.
16	Rig Cont.	5” DP, G-105/S-135, 19.5 PPF, 4-1/2” IF Pin-Box	As Required/1000	Mt

Item No.	Supplier	6-1/2" Hole Section Requirement Description	Quantity	Unit
17	Rig Cont.	3-1/2" HWDP, 23.2 PPF, 3-1/2" IF Pin-Box	6	Jt.
18	Rig Cont.	3-1/2" DP, G-105, 13.3 PPF, 3-1/2" IF Pin-Box	± 400	Mt
19	Rig Cont.	7" CSG Scraper (29-32#), 3-1/2" Reg Pin-Box	1	Ea.
20	Rig Cont.	Float Valve for 6-1/8" Bit Sub	2	Ea.
21	Rig Cont.	Float Valve for 4-3/4" Bit Sub	2	Ea.
22	Rig Cont.	Pipe Screen for inside 5" DP	3	Ea.
23	MSA/OEOC	X.O.S 5" , 15 ppf, L-80, Premium Gas Tight Connection pin * 4-1/2" IF Box	1	Ea.
24	MSA/OEOC	Cement "G", 110 lb./sx.	As Req.	
25	MSA/OEOC	Cement Additives	As Req.	
26	MSA/OEOC	5" Float Shoe, L-80, 15#, Premium Gas Tight Con.	1	Ea.
27	MSA/OEOC	5" Float Collar, L-80, 15#, Premium Gas Tight Con.	1	Ea.
28	MSA/OEOC	5" LDC, L-80, 15#, Premium Gas Tight Con.	1	Ea.
29	PEDEC	5" Liner, L-80, 15#, Premium Gas Tight Con.	330	Mt
30	MSA/OEOC	Hydraulic Set, Rotating Liner Hanger, 5", 15 ppf, L-80, Premium Gas Tight Conn., Tandem Cone, with Integral/Split Weight Set Top Packer, 10 ft PBR, Load Capacity 2,000 KN, Packer Seal Capacity 10K, Temp. 150 ºC, with Liner Setting Sleeve c/w DPOB and Setting Tool thread, to Set Inside 7", 29 ppf Casing.	1	Set
31	Service Co.	5" Liner Hanger Running Tool	1	Ea.
32	MSA/OEOC	Liner Wiper Plug for 5" Liner, L-80, 15#, Premium Gas Tight Con.	1	Ea.

Item No.	Supplier	6-1/2" Hole Section Requirement Description	Quantity	Unit
33	Service Co.	Polish Mill for 5" Liner, L-80, 15#, Premium Gas Tight Con. PBR	1	Ea.
34	MSA/OEOC	5" DP, G-105/S-135, 19.5 PPF Pump Down Plug (Dart)	1	Ea.
35	MSA/OEOC	API Drift for 5" Liner, L-80, 15#, Premium Gas Tight Con.	1	Ea.
36	MSA/OEOC	6-1/8" x 5" Bow Type Centralizer	As Req.	Ea.
37	MSA/OEOC	6-1/8" x 5" Rotating Type Centralizer C/W Nail and Pins	As Req.	Ea.
38	MSA/OEOC	Seal Nipple 7" 29 ppf, L-80	2	Set as Req.
39	MSA/OEOC	7" x 5" Rigid Type Centralizer c/w Nails, Non-Welded	As Req.	Ea.
40	MSA/OEOC	5" Stop Collar c/w Spiral Nails	As Req.	Ea.
41	MSA/OEOC	API CSG Thread Lock Compound (Baker Lock)	As Req.	Can.
42	MSA/OEOC	API CSG Thread Sealing Compound (Dope)	As Req.	DR.
43	Service Co.	Drilling Fluid Service (Material, Lab. & Engineering)	As Required	
44	Service Co.	Mud Logging Service	As Required	
45	Service Co.	Waste Management Service	As Required	
46	Service Co.	Directional Drilling Service (Only MWD unit, MWD Tools, MWD Engineer)	As Required	
47	Service Co.	Fishing service	As Required	
48	Service Co.	H2S Service	As Required	
49	Service Co.	Casing Running Service	As Required	

Item No.	Supplier	6-1/2" Hole Section Requirement Description	Quantity	Unit
50	Service Co.	Cementing Service (Material, Pump Truck, Silo & Engineering)	As Required	
51	Service Co.	Well Head Service	As Required	
52	Service Co.	Liner Hanger Service	As Required	
53	Service Co.	Wireline Logging Service (Logging and Perforation)	As Required	
54	Service Co	Dry Testing & DST/TCP Service	As Required	
55	Service Co	Coiled Tubing Service	As Required	
56	Service Co	Stimulation & N2 Lifting Services	As Required	
57	Service Co	Completion Service	As Required	
58	Service Co	Slick line Service	As Required	

- **Qty & Spec to be confirmed with Contact's SOW.**
- **In case of any shortage or discrepancy, the Contract's SOW will be reference.**

15. Comments

Drilling Engineering							
١	در جدول سایز حفره به ١/٨ ۶ " اصلاح شود.	١٠	CM	اصلاح شد			
٢	Well montage: مشخصات آستری ۵" پنداژ و ترد طبق انبار کارفرما اصلاح گردد.	٢٠	CM	اصلاح شد			
٣	Casing & Liner Depth before Side Track-2 نوع tread استری ٧" از TPG2 به TPCQ اصلاح گردد	٢١	CM	اصلاح شد			
٤	در جدول عمق لبه آستری ۵" با صفحه ٢٠ یکسان سازی گردد. عمق ٣٨٤٠ صحیح است.	٢٣	CM	اصلاح شد			
٥	آیتم ٣١ اشتباه نوشتاری اصلاح گردد. • P/U & M/U 7” liner hanger eiyh top packer	٣٥	CM	اصلاح شد			

	• TIH 7” liner on						
٦	ردیف ۲۱، عمق لبه آستری جدید +- ۱۳۵۵ ذکر گردد	۳۹	CM	**اصلاح شد**			
۷	حجم گل ارائه گردد.	۸۲-۷۷	CM	**مشخصات گل و احجام مورد نیاز اضافه گردید.**			
۸	برنامه سیال بخش تکمیل چاه ارائه گردد.	۱۱۱	CM	**برنامه سیال بخش تکمیل چاه در برنامه تکمیل چاه که در زمان مقتضی ارائه می-گردد، به صورت کامل قید خواهد شد.**			
۹	برنامه هیدرولیک برای حفره ۱/۸ ۶ " ارائه گردد.	۱۱۲	CM				
۸	عمق لبه آستری ۷" با دیگر صفحات برنامه یکسان سازی گردد (۲۹۸۸)	۱۱۳	CM	**اصلاح شد**			
۹	نوع متریال و گرید آستری ۷" و ۵ " طبق موجودی انبار کارفرما آپدیت شود.	۱۱۳	CM	**اصلاح شد**			
۱۰	فشار مخزن آپدیت گردد.		CM	**اطلاعات به روز فشار مخزن از سوی کارفرما دریافت نگردیده است.**			

				در برنامه سیال ارائه شده، استفاده از باریت قد نگردیده است و با توجه به فشار مخزن فهلیان و وزن گل مورد نیاز و محدودیت‌های استفاده از گل limestone در وزن‌های ۱۰۰ تا ۱۰۶، استفاده از فروبار اجتناب ناپذیر است.		
۱۱	در حفره مخزنی ۱/۸ ۶" فروبار و باریت استفاده نگردد.		CM			

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